

HYPERRIDGE: LEARNING ARCHIVE GEOMETRY FOR PERSONALIZED ROUTING

Anonymous authors

Paper under double-blind review

ABSTRACT

Personalizing a language-model agent requires learning which prompt, skill, or policy works for each user from limited feedback. The agent observes the outcome of its chosen candidate, not how the alternatives would have performed. We model these user-dependent choices as a contextual bandit, balancing the quality of current decisions with the cost of trying uncertain candidates. We ask whether statistical relationships among candidates can reduce this exploration cost. We introduce HyperRidge, which retains a separate reward predictor for each candidate but uses a shared covariance to regularize their parameters and quantify uncertainty. Our analysis gives finite-time exploration bounds for supplied covariance and for covariance estimated from independent calibration data, accounting for estimation uncertainty and calibration cost. Correct covariance reduces exploration cost by 46–59% on planted problems. However, comparisons across nine synthetic and proxy environments and a 600-case real-LLM study show that estimating a full covariance from selected feedback does not reliably outperform simpler regularization. These results separate the potential benefit of shared structure from the difficulty of estimating it under selective feedback. They show why covariance estimation error and its data cost must be included when assessing whether sharing information improves personalized routing.

1 INTRODUCTION

LLM agents can choose among prompts, skills, tools, and workflows rather than use one fixed policy. We call this collection an *archive* and each deployable entry a *candidate*. Personalization methods learn user profiles and preference-conditioned behavior (Salemi et al., 2023; Zhao et al., 2025; Jin et al., 2026); prompt and agent evolution methods search for stronger candidates (Fernando et al., 2023; Guo et al., 2024; Zhou et al., 2024; Agrawal et al., 2025); and routing methods select a model or policy for each request (Ong et al., 2024; Hu et al., 2024; Li, 2025; Tsai & Tran, 2026).

Consider a support agent with formal and casual policies for refund and shipping requests. The router chooses one policy and observes only that policy’s reward. Trying an uncertain policy can therefore reduce reward, and adding a revised policy introduces another candidate to learn. Yet these candidates need not vary independently: several may differ along shared directions such as tone or level of detail. Learning which directions matter could help the router regularize a new candidate without treating its reward model as identical to those already in the archive.

Our focus is the statistical relation among the candidate reward models. Ordinary disjoint linear bandits estimate each model separately (Li et al., 2010; Chu et al., 2011). Sharing one predictor instead would remove candidate-specific differences. HyperRidge shares the regularization geometry while retaining those differences.

These observations lead to the question studied in this paper:

When can shared archive geometry reduce the exploration cost of personalized routing, and how can that geometry be learned?

Each candidate a has a parameter vector θ_a mapping observed features to expected reward. Ordinary ridge applies the same spherical penalty to every vector (Abbasi-Yadkori et al., 2011; Chu et al., 2011). HyperRidge instead uses a shared covariance Ω : directions with little parameter variation

are penalized more strongly, while directions with substantial variation retain more uncertainty. Candidate-specific observations still determine each predictor. What is shared is the geometry of estimation, not a common reward coefficient or the unobserved reward of another candidate.

We analyze supplied geometry and covariance estimated from independent calibration data, accounting for uncertainty and the cost of calibration. These guarantees require fixed geometry during routing; they do not cover periodic covariance updates from adaptively selected feedback.

The experiments follow the same distinction. Planted low-rank problems test the effect of supplying correct geometry. Matched identity, scalar, diagonal, and full-covariance controls test what periodic estimation adds. The complete method is then evaluated on nine synthetic and proxy environments and on real LLM answers to held-out preference questions. Known geometry helps in the planted study, but full covariance does not consistently outperform simpler adaptive regularizers, and the periodic estimator underperforms ordinary ridge in the real-LLM study. Base-router ablations appear last as component analysis rather than evidence for successful covariance learning.

Contributions. The contributions are:

1. **Shared-geometry routing.** HyperRidge uses a common covariance to regularize separate candidate reward models and their UCB widths.
2. **Known and calibrated geometry guarantees.** We bound exploration under explicit observation, diversity, and confidence conditions. A reward identity gives a downstream sufficient condition for improving over fixed deployment.
3. **Controlled evaluation of estimated geometry.** Planted problems, matched covariance controls, and a real-LLM study distinguish the benefit of correct geometry from the limitations of the periodic estimator.

2 PROBLEM SETUP

2.1 CANDIDATES AND FEEDBACK

Candidates may differ in prompts, skills, tools, or policies. Managed agent platforms motivate this setting (Microsoft, 2026; Amazon Web Services, 2026); no platform-specific implementation is assumed. An optimizer such as GEPA (Agrawal et al., 2025) may propose candidates, while the router decides which available candidate to deploy.

2.2 OBSERVATION MODEL AND EXPLORATION TARGET

A user $U \sim \Pi$ is drawn from a population $\mathcal{U}$ and followed for T rounds. The deterministic initial archive $A_1 = \{a^{(1)}, \dots, a^{(K)}\}$ is shared across users. The archive $A_t(U)$ may grow along that user’s trajectory, with at most $K_{\max}$ candidates appearing by the horizon. Population statements average over U and the online randomness.

The observable history $\mathcal{O}_{t-1}$ contains the user, available candidate features, earlier actions and rewards, and the current archive. The archive and action are $\mathcal{O}_{t-1}$ -measurable. The analysis filtration $\mathcal{F}_{t-1} \supseteq \mathcal{O}_{t-1}$ additionally contains latent candidate parameters and proposer randomness once generated. These additional variables are used in proofs, not given to the router.

The router selects $a_t(U) \in A_t(U)$ and observes the reward

$$r_t = \mu(U, a_t(U)) + \xi_t, \quad (1)$$

where $\mu(u, a) \in [0, 1]$ is the unknown mean reward and $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$. Only the selected candidate’s reward is observed.

The comparison target is the best available mean reward, $m_t(U) = \max_{a \in A_t(U)} \mu(U, a)$. The one-round exploration cost is $L_t(U) = m_t(U) - \mu(U, a_t(U))$, and the principal theoretical target is $\sum_{t=1}^T \mathbb{E}[L_t(U)]$. We write $\bar{L}_T = T^{-1} \sum_{t=1}^T \mathbb{E}[L_t(U)]$ for its average. The oracle value is an analytical comparator, not an input to the HyperRidge routing score. Algorithm 1 also permits a varying user stream; the population reward consequence in Section 4 uses the fixed-user trajectory defined here.

3 HYPERRIDGE METHOD

HyperRidge fits one reward model per candidate and uses a common positive definite matrix to determine regularization and uncertainty. We present the estimator and periodic procedure first, then define its controlled variants.

3.1 MODEL

For the HyperRidge theory, write the bounded mean reward as

$$\mu(u, a) = b(u) + x(u, a)^\top \theta_a, \quad (2)$$

where $x(u, a) \in \mathbb{R}^d$ is the user-candidate feature vector and the known baseline $b(u)$ is common to every candidate offered to user u . We assume $\mu(u, a) \in [0, 1]$. A convenient compatible model is $b(u) = 1/2$ and $x(u, a)^\top \theta_a \in [-1/2, 1/2]$.

Instead of estimating each θ_a with ordinary ridge regression, assume the candidate parameters are random effects:

$$\mathbb{E}[\theta_a] = 0, \quad \mathbb{E}[\theta_a \theta_a^\top] = \Omega, \quad (3)$$

where $\Omega \in \mathbb{R}^{d \times d}$ is a hyper-covariance matrix shared across archive candidates. The common baseline makes this zero-mean random-effects model compatible with nonnegative rewards. It also cancels from every candidate comparison.

The goal is to estimate Ω from the archive and use it to regularize routing.

3.2 HYPERRIDGE ESTIMATOR

For candidate a , define

$$X_{a,t} = [x_s^\top]_{s < t: a_s = a}, \quad y_{a,t} = (r_s - b(u_s))_{s < t: a_s = a}. \quad (4)$$

Given an estimate $\hat{\Omega}_t$, define the generalized ridge estimator

$$\hat{\theta}_{a,t} = \arg \min_{\theta \in \mathbb{R}^d} \left\{ \|y_{a,t} - X_{a,t} \theta\|_2^2 + \lambda \theta^\top \hat{\Omega}_t^{-1} \theta \right\}. \quad (5)$$

Equivalently,

$$\hat{\theta}_{a,t} = \left(X_{a,t}^\top X_{a,t} + \lambda \hat{\Omega}_t^{-1} \right)^{-1} X_{a,t}^\top y_{a,t}. \quad (6)$$

Define the generalized design matrix

$$V_{a,t}^{\text{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \hat{\Omega}_t^{-1}. \quad (7)$$

Then the HyperRidge UCB score is

$$\text{UCB}_t^{\text{hyper}}(u, a) = x(u, a)^\top \hat{\theta}_{a,t} + \alpha \sqrt{x(u, a)^\top \left(V_{a,t}^{\text{hyper}} \right)^{-1} x(u, a)}. \quad (8)$$

The selected candidate is

$$a_t(u) \in \arg \max_{a \in A_t(u)} \text{UCB}_t^{\text{hyper}}(u, a). \quad (9)$$

3.3 HRP: ROUTING WITH PERIODIC COVARIANCE UPDATES

Algorithm 1 summarizes the resulting routing procedure. Every τ rounds, it estimates a shared second-moment matrix from the current candidate models and adds ϵI_d to keep it invertible. Between updates, the router uses that matrix in $V_{a,t}^{\text{hyper}}$ and updates the selected candidate’s observations.

The response vector contains observed centered rewards $r_s - b(u_s)$, not unknown means. A shared router can collect rows from different users; the fixed-user reward guarantee specializes this procedure to $u_t \equiv U$.

A refresh aggregates the candidate estimates into a $d \times d$ matrix and changes the regularizer used in subsequent fits. The update uses shrunk point estimates, including zero estimates for unobserved candidates; it does not add a posterior-uncertainty correction.

Archive growth enters through the eligible set $A_t(u_t)$. The covariance estimator does not itself propose candidates. The simulator’s separate oracle-assisted gate and proposer are described with HRB in Appendix M; the real-LLM protocol builds and freezes its archive during development.

Variants. **HyperRidge-Base (HRB)** uses ordinary isotropic ridge; **HyperRidge-Periodic (HRP)** periodically estimates a shared covariance; **HyperRidge-Calibrated (HRC)** estimates covariance on independent calibration candidates and then freezes it; **HyperRidge-Diag (HRD)** retains only diagonal covariance; **HyperRidge-Scalar (HRS)** uses a scalar multiple of the identity; and **HyperRidge-Known (HRK)** uses the supplied true covariance. HRP is the periodic method in Algorithm 1. HRK is a known-geometry reference. The guarantees for HRK and HRC do not extend to HRP.

4 THEORY

We bound cumulative exploration cost first with supplied geometry (HRK), then with independently calibrated geometry (HRC). The final subsection translates these bounds into a conditional reward comparison. Full geometry proofs precede the supporting reward and archive analysis in the appendix.

4.1 CONDITIONS

We use the observation model of Section 2, writing $x(u, a)$ for the feature map $\psi(u, a)$ in the assumptions below. Mean rewards lie in $[0, 1]$, noise is conditionally centered and σ -sub-Gaussian, the archive is predictable, and features have norm at most L . Appendix A.2 states these requirements formally. Geometry-specific conditions follow.

4.2 HRK: KNOWN ARCHIVE GEOMETRY

HRK uses the generalized design (7), estimate (6), and score (8) defined in Section 3, with the geometry held fixed at Ω . HRP in Algorithm 1 instead re-estimates the geometry periodically.

Assume $\|\theta_a\|_{\Omega^{-1}} \leq S_\Omega$ and $x(u, a)^\top \Omega x(u, a) \leq L_\Omega^2$. For a candidate design matrix X , the effective dimension is

$$d_{\text{eff}}(\Omega; X) = \text{tr} \left[X^\top X (X^\top X + \lambda \Omega^{-1})^{-1} \right]. \quad (10)$$

Let $\bar{d}_{\text{eff},T}(\Omega)$ be a deterministic upper bound on this quantity over all candidate designs through the horizon. Assumption 5 states the full requirements.

The known-geometry theorem needs only the ellipsoid, feature-radius, and complexity-cap conditions. Interpreting Ω as a covariance requires the additional random-effects assumptions used by the HRC calibration theorem: fresh candidate effects are independent, have mean zero and covariance Ω , and the regression uses the centered response $r - b(u)$.

Proposition 1 (HRK: known-geometry exploration-cost bound). *Assume Assumptions 1–3 and Assumption 5. Suppose $|A_T(U)| \leq K_{\max}$. For the decomposition-compatible result, draw one user U*

before routing and set $u_t = U$ for all t . Define

$$c_T^\Omega = 1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right) \quad (11)$$

and, for $\delta \in (0, 1)$,

$$\alpha_{\Omega,T}(\delta) = \max\left\{1, \sigma\sqrt{c_T^\Omega \bar{d}_{\text{eff},T}(\Omega) + 2\log(K_{\max}/\delta)} + \sqrt{\lambda}S_\Omega\right\}. \quad (12)$$

Consider HRK, the oracle-covariance router that sets $\hat{\Omega}_t = \Omega$ for every round and uses the UCB multiplier $\alpha_{\Omega,T}(\delta)$. Then, with probability at least $1 - \delta$,

$$\sum_{t=1}^T L_t(U) \leq B_T^\Omega(\delta) := 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T c_T^\Omega \bar{d}_{\text{eff},T}(\Omega)}. \quad (13)$$

Consequently,

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq B_T^\Omega(\delta) + \delta T. \quad (14)$$

Proof sketch. Lemma 3 gives simultaneous confidence intervals after a union bound over the predictable birth-order slots. Lemma 4 bounds each slot's log-determinant and width sum by $c_T^\Omega \bar{d}_{\text{eff},T}(\Omega)$. Optimism gives $L_t(U) \leq 2\alpha_{\Omega,T}(\delta)\sqrt{x_t(a_t)^\top (V_{a_t,t}^\Omega)^{-1}x_t(a_t)}$ on the confidence event. Summing the clipped widths and applying Cauchy–Schwarz yields (13). Since $0 \leq L_t(U) \leq 1$, the complement of the confidence event contributes at most δT in expectation. Appendix B gives the full derivation. $\square$

4.3 HRC: INDEPENDENT CALIBRATION

HRC uses fresh candidates that are excluded from the subsequent routing archive. For each calibration candidate, it collects centered linear responses under an exogenous design with a positive definite population Gram matrix, fits ordinary least squares, and averages the fitted outer products with a deterministic ridge floor. It then freezes both covariance and routing archive. The confidence radius requires supplied bounds on noise, feature size, calibration diversity, and the parameter ellipsoid. Theorem 15 states the full finite-sample conditions, including a relative-error bound smaller than one.

Remark 1 (HRC: calibration-split guarantee). *For HRC, Theorem 15 proves that the relative covariance error satisfies $\rho_{\Omega,T} = O(T^{-1/10}\sqrt{\log T})$ with high probability. It uses an explicit frequentist confidence radius. Under the theorem's schedule and fixed problem constants,*

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq B_{T,0}^{\text{cal}} + \text{Rem}_T^{\text{cal}}. \quad (15)$$

Moreover,

$$\begin{aligned} B_{T,0}^{\text{cal}} &= O\left(\sqrt{dK_{\max}\bar{d}_{\text{eff},T}(\Omega)T \log T}\right), \\ \text{Rem}_T^{\text{cal}} &= O\left(T^{2/5}(\log T)^{3/2}\right) = o(\sqrt{T}). \end{aligned} \quad (16)$$

This is a theorem for HRC with a fixed post-calibration archive. It is not a theorem for HRP. The leading term in (15) is not the oracle budget: its confidence radius already depends on the covariance-error bound. The displayed remainder isolates only the additional width inflation, failure probability, and, when charged, calibration pulls.

4.4 CONDITIONAL REWARD CONSEQUENCE

An exploration bound gives a sufficient comparison with fixed deployment when the reward available from personalization and archive growth exceeds the exploration budget. The accounting identity and full comparison are given in Appendix D; they are consequences of the exploration analysis, not assumptions used to prove it.

What the bounds do not imply. A smaller effective dimension alone does not establish a smaller budget: the confidence radius must also be compared. With fixed problem parameters and $\delta = 1/T$, both the displayed HRK and ordinary-ridge certificates are $O(\sqrt{T} \log T)$. Moreover, in the fixed-user model with an unchanged feature vector for each candidate, every candidate design has rank at most one under *both* identity and shared geometry. Comparing an ambient-dimension identity bound with a sharper geometry bound does not establish an intrinsic dimension advantage in that setting. Neither the HRK result nor independent HRC calibration controls HRP’s adaptive periodic estimate or its fixed empirical exploration multiplier.

5 EXPERIMENTS

We first test the effect of known geometry, then compare estimated geometries under matched conditions. We next evaluate the configured growing-archive method and a real-LLM preference task. HRB component results are presented separately at the end.

5.1 SHARED GEOMETRY AND EXPLORATION

Known-geometry mechanism: HRK and HRP versus HRB. We construct a diagnostic linear bandit in which $K = 30$ candidate parameters are drawn from a planted rank-three covariance, contexts are dense random unit vectors, and the horizon is $T = 400$. Table 1 compares HRB (isotropic ridge), HRK (true covariance), and HRP (covariance periodically re-estimated from the candidate models), all with the same ridge weight $\lambda = 0.5$. The true geometry cuts cumulative exploration cost by 46–59%, while the online estimate raises it by 7–19%. This diagnostic uses Gaussian parameters, unbounded linear rewards, and a fixed exploration multiplier, so it does not test the bounded-reward theorem or its rate. It only isolates the effect of supplying the correct geometry.

Table 1: Exploration cost with known and estimated archive geometry. Mean cumulative exploration cost over five seeds on planted rank-three problems ($K = 30$, $T = 400$, $\lambda = 0.5$ for all three routers). HRB: HyperRidge-Base; HRK: HyperRidge-Known; HRP: HyperRidge-Periodic. Lower is better. This mechanism diagnostic does not satisfy the assumptions of Proposition 1.

d	HRB	HRK	HRP (ours)	HRK reduction vs. HRB
20	314.1	170.2	375.0	45.8%
50	447.6	213.1	504.9	52.4%
100	455.4	188.1	487.5	58.7%

Independent calibration under the theorem’s assumptions. We separately test HRC with bounded random effects, independent full-rank calibration, centered least squares, and a fixed routing archive. Across three calibration sizes and five seeds, relative covariance error decreases from 0.0243 to 0.0064, with no observed violation of the tested covariance or routing inequalities. These runs use deterministic confidence radii, not measured errors. The conservative certificate requires many calibration observations and does not establish a practical cost advantage. Appendix C.1 reports the protocol and full calibration charge.

Matched covariance controls. Table 2 compares identity, scalar, diagonal, and full geometry on the same nine environments, with five seeds per method. All four methods use a fixed archive, ridge weight 0.5, and exploration multiplier 0.8. Estimated geometries are refreshed every 25 rounds with ridge floor 0.02. The primary endpoint is mean-reward exploration cost, not the observed-reward residual.

General, Workspace, and Support are synthetic preference environments. The GEPA and Hermes settings also use synthetic rewards and do not execute those agent systems. BESPOKE uses an imputed preference proxy; PAHF shopping and PrefEval use constructed proxies; PAHF embodied uses a label-backed proxy. Each has a shared 25-dimensional user-candidate representation. These experiments test the routing estimator on fixed reward models, not task success in the original benchmarks.

Table 2: Matched geometry controls. Mean per-round exploration cost $T^{-1} \sum_t [m_t - \mu(U, a_t)]$, with standard deviation across five seeds; lower is better. All archives are fixed, all ridge weights are 0.5, and the exploration multiplier is 0.8. HRB-NF uses identity geometry; HRS, HRD, and HRP use scalar, diagonal, and full periodically estimated geometry. These are simulated or proxy rewards, not live benchmark executions.

Environment	HRB-NF	HRS	HRD	HRP
General	0.0314 ± 0.0004	0.0164 ± 0.0007	0.0164 ± 0.0007	0.0165 ± 0.0006
Workspace	0.0473 ± 0.0010	0.0149 ± 0.0015	0.0149 ± 0.0015	0.0149 ± 0.0015
Support	0.0400 ± 0.0007	0.0156 ± 0.0008	0.0156 ± 0.0008	0.0156 ± 0.0008
GEPA archive	0.0682 ± 0.0012	0.0338 ± 0.0089	0.0330 ± 0.0092	0.0319 ± 0.0096
Hermes agent	0.0784 ± 0.0007	0.0519 ± 0.0053	0.0521 ± 0.0059	0.0531 ± 0.0064
BESPOKE	0.0497 ± 0.0007	0.0427 ± 0.0042	0.0419 ± 0.0050	0.0410 ± 0.0059
PAHF shopping	0.1259 ± 0.0010	0.0516 ± 0.0047	0.0519 ± 0.0057	0.0630 ± 0.0062
PAHF embodied	0.1256 ± 0.0004	0.1036 ± 0.0010	0.1040 ± 0.0010	0.1043 ± 0.0006
PrefEval	0.0476 ± 0.0004	0.0137 ± 0.0007	0.0137 ± 0.0009	0.0128 ± 0.0008

Averaged equally over the nine environments, HRP reduces per-round exploration cost by 0.0290 relative to identity geometry. Its cost is 0.0010 higher than scalar geometry and 0.0011 higher than diagonal geometry. Thus adaptive regularization helps under this protocol, but the comparison does not establish a distinct benefit from off-diagonal covariance. In these fixed-user experiments, each arm’s repeated feature vector also limits its design rank; a large ambient representation does not by itself create a high-dimensional exploration problem.

5.2 FULL-METHOD COMPARISON

Table 3 compares Static, HRB, HRD, and HRP on the same nine reward environments and five seeds. General, Workspace, and Support use $T = 300$; the others use $T = 250$. The learned routers may add up to eight candidates using the common oracle-assisted gate and reserve-candidate proposer. The gate threshold is 0.15 and its cooldown is 20 rounds. HRB uses ridge weight 1.0, whereas HRD and HRP use 0.5; all use exploration multiplier 0.8.

Table 3: Configured full-method comparison. Selected mean reward $T^{-1} \sum_t \mu(U, a_t)$ with standard deviation across five seeds; higher is better. HRB, HRD, and HRP allow gated archive growth. HRB uses ridge weight 1.0; HRD and HRP use 0.5. All use exploration multiplier 0.8. Unlike Table 2, this comparison does not isolate covariance from regularization strength or archive growth.

Environment	Static	HRB	HRD	HRP
General	0.5784 ± 0.0000	0.8058 ± 0.0009	0.8388 ± 0.0046	0.8386 ± 0.0043
Workspace	0.5848 ± 0.0000	0.6600 ± 0.0010	0.6872 ± 0.0040	0.6877 ± 0.0038
Support	0.5675 ± 0.0000	0.7580 ± 0.0015	0.7975 ± 0.0039	0.7975 ± 0.0039
GEPA archive	0.6487 ± 0.0000	0.6516 ± 0.0011	0.6801 ± 0.0055	0.6816 ± 0.0042
Hermes agent	0.8209 ± 0.0000	0.7870 ± 0.0014	0.8460 ± 0.0043	0.8449 ± 0.0050
BESPOKE	0.6604 ± 0.0000	0.6712 ± 0.0012	0.6956 ± 0.0070	0.6965 ± 0.0078
PAHF shopping	0.5617 ± 0.0000	0.7391 ± 0.0011	0.7444 ± 0.0038	0.7333 ± 0.0073
PAHF embodied	0.8017 ± 0.0000	0.7991 ± 0.0003	0.8960 ± 0.0010	0.8957 ± 0.0006
PrefEval	0.6363 ± 0.0000	0.6845 ± 0.0005	0.7131 ± 0.0033	0.7139 ± 0.0032

HRP improves selected mean reward over HRB in eight of nine environments, with an equally weighted mean difference of 0.0370. This is a comparison of the configured methods, not an isolated covariance effect: regularization strength differs, and action-dependent archive growth may differ as well. The matched experiment in Table 2 addresses the covariance comparison.

Additional fitted-covariance and holdout-covariance diagnostics are described in Appendix K. A covariance fitted from the complete reward matrix is a privileged reference, not HRK’s supplied generative covariance. The holdout experiment also differs from HRC’s independent, full-rank calibration and fixed post-calibration archive. It does not validate Theorem 15; the assumption-matched validation is reported separately in Appendix C.1.

5.3 REAL LLM PREFERENCE EVALUATION

We evaluate generated answers on a held-out PrefEval subset (Zhao et al., 2025). The test set has 600 cases from 200 preference-question groups, covering 20 topics, three preference forms, and two context lengths. Opus 4.8 generates one answer per case and candidate; a blinded GPT-5.6 judge evaluates preference accuracy. Training uses 480 cases and validation uses 240, separate from testing. Five seeds vary the training order, not LLM generation.

The initial archive contains direct answering, a preference reminder, constraint checking, and TF-IDF retrieval. Two proposed prompts pass a validation gate during development and are added halfway through each training trajectory. Each learner receives only its selected candidate’s feedback; the complete answer collection supports paired evaluation. Both archive and primary routing policies are frozen before testing. Validation-best static selects one prompt using all validation rewards, so it has more validation information than the online learners.

Table 4: Real-LLM preference evaluation on 600 held-out cases. Primary frozen-policy accuracy under the blinded judge, averaged over five routing-order seeds. Pointwise 95% intervals use a topic/distance-stratified preference-question-group bootstrap over 200 groups. They condition on development data and one generated answer per case/candidate; the seeds are not independent LLM generations. Validation-best static uses full validation rewards, whereas the learned routers receive selected training feedback.

Method	Accuracy (%)	95% interval
Direct	33.00	[29.17, 36.67]
Reminder	70.33	[66.17, 74.33]
Constraint check	69.67	[65.33, 73.83]
TF-IDF retrieval	56.83	[52.33, 61.33]
Validation-best static	81.33	[78.00, 84.67]
Uniform	64.67	[61.30, 67.83]
Noncontextual UCB	81.33	[78.00, 84.67]
HRB-NF (initial archive)	66.87	[63.00, 70.70]
HRB	77.47	[74.27, 80.57]
HRB (ungated)	77.47	[74.27, 80.57]
HRS	67.50	[63.63, 71.14]
HRD	67.50	[63.63, 71.14]
HRP	67.47	[63.60, 71.13]

HRP reaches 67.47% accuracy, compared with 77.47% for HRB and 81.33% for both validation-best static and noncontextual UCB. The paired HRP-minus-HRB difference is -10.00 percentage points, with a pointwise 95% interval of $[-12.30, -8.00]$. Scalar and diagonal geometry each reach 67.50%. Across the five training orders, none of the three estimated-geometry routers selects either newly admitted prompt. This identifies a new-candidate exploration failure in this implementation; it does not establish that shared geometry must fail in general.

An independent Opus assessment of 720 answers from 40 test groups has 85.97% binary accuracy agreement with the primary judge. Agreement does not establish human validity. This study evaluates LLM-judged conversation continuations with development-time prompt evolution, not on-line human-feedback deployment. Additional protocol and secondary results are described in Appendix K.

5.4 HRB COMPONENT ANALYSIS

In the shared-feature suite, removing routing lowers selected mean reward by 0.1497 on average across nine environments. Removing reflection lowers it by only 0.0072, and reflection helps in five of nine environments. The routing component therefore has more consistent support than archive growth. Appendix L retains the separate category-based component study, including its accounting table and Figures 1–2. Those results concern ordinary-ridge HRB and do not evaluate learned covariance.

6 RELATED WORK

Routing queries across a model or prompt pool to trade off cost and quality is now standard: RouteLLM (Ong et al., 2024) learns a strong/weak router from preference data and RouterBench (Hu et al., 2024) benchmarks such routers, while LLM Bandit (Li, 2025) and NeuralUCB routing (Tsai & Tran, 2026) cast routing as an online contextual bandit.

These methods route at the *model* level over a fixed pool. HyperRidge studies candidate-specific reward models with a common regularization geometry, including archives that may grow online.

Benchmarks such as LaMP (Salemi et al., 2023), PersonalLLM (Zollo et al., 2024), and PrefEval (Zhao et al., 2025) supply user-feedback signals, while PersonaAgent (Zhang et al., 2026) and persistent-memory frameworks (Westh  u  er et al., 2025) maintain per-user personas and evolving profiles.

The closest personalized-bandit antecedent, PAB-F (Jin et al., 2026), models per-user reward as a factorized function of context and a fixed memory/tool/style action set, and studies regret against an in-archive oracle.

Prompt and agent evolution methods, including Promptbreeder (Fernando et al., 2023), EvoPrompt (Guo et al., 2024), Symbolic Learning (Zhou et al., 2024), and GEPA’s reflective Pareto search (Agrawal et al., 2025), improve prompts or archives through offline search rather than online user-conditioned routing.

Hermes self-evolution (Team, 2025b;a) adds constraint-gated growth, which motivates our archive-growth gate.

Classical contextual-bandit methods such as OFUL and LinUCB (Abbasi-Yadkori et al., 2011; Chu et al., 2011; Li et al., 2010) provide standard finite-time tools for controlling exploration in linear bandits, and related structural-vs-online decompositions appear in contextual pricing, offline RL, and multi-task transfer.

7 CONCLUSION AND LIMITATIONS

HyperRidge uses a shared covariance to regularize candidate-specific reward models and their uncertainty. The HRK analysis bounds exploration with the complete geometry-dependent confidence budget. HRC extends the analysis to an independently calibrated and frozen covariance under explicit diversity conditions. The reward identity provides a conditional consequence of these bounds, rather than the basis of the method.

The experiments show a benefit from supplied geometry in planted problems and from adaptive regularization relative to identity controls in the nine-environment study. They do not establish a distinct advantage for full covariance. HRP improves over HRB in most configured proxy comparisons but underperforms it in the real-LLM evaluation, where it fails to select newly admitted candidates during training.

Several limitations remain. HRP’s adaptive covariance estimate is not covered by the HRK or HRC guarantees. The independent HRC validation meets the finite-sample assumptions but requires extensive calibration; it does not test the asymptotic schedule or establish a practical cost advantage. The fixed-user feature model also requires equally sharp bounds for identity and shared geometry before a dimension advantage can be claimed. The proxy studies use oracle-assisted archive growth; the real-LLM study instead uses a development-built archive and model-judged feedback, with one generated answer per case and candidate. Improving periodic estimation requires investigating centering, covariance uncertainty, and new-candidate exploration on development data, followed by an untouched test cohort.

STATEMENT ON THE USE OF AI

AI assistants were used to develop mathematical arguments, design and implement experiments, check code and numerical reporting, and draft and revise the manuscript. Language models also generated responses and judged preferences in the evaluation described in Section 5.3; their roles and evaluation protocol are reported there and in Appendix K. The authors are responsible for the claims, results, and final content.

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A HYPERRIDGE EXPLORATION-COST ANALYSIS

We first establish generalized-ridge confidence and potential bounds. Appendix B derives the HRK exploration budget and its conditional perturbation analysis. Appendix C then proves the HRC calibration result. Reward identities and archive-growth analysis follow in Appendix D and subsequent sections.

A.1 PREDICTABLE CANDIDATE BIRTHS

Lemma 2 (Predictable birth-order archive). *Assume at most $K_{\max}$ candidates appear by round T , and index them by predictable birth order. Candidate slot k is created at a stopping time τ_k . Its identity and feature rule are $\mathcal{O}_{\tau_k-1}$ -measurable, while its latent parameter is fixed at birth and $\mathcal{F}_{\tau_k-1}$ -measurable. Before τ_k , pad that slot's feature and selection indicator by zero. Each padded process is predictable in the analysis filtration, so self-normalized concentration applies conditionally from its birth time. A union bound over the deterministic slots $k \leq K_{\max}$ is valid. The router remains adapted to $\mathcal{O}_t$ and does not observe the latent parameter.*

A.2 BASIC ROUTING CONDITIONS

Assumption 1 (Bounded mean reward). *The mean reward is bounded:*

$$\mu(u, a) \in [0, 1] \quad \text{for all } (u, a). \quad (17)$$

Assumption 2 (Conditionally sub-Gaussian noise). *There exists $\sigma > 0$ with*

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \leq \exp(\lambda^2 \sigma^2 / 2), \quad \lambda \in \mathbb{R}. \quad (18)$$

Assumption 3 (Predictable archive). *The user-specific archive is predictable:*

$$\sigma(A_t(U)) \subseteq \mathcal{O}_{t-1}. \quad (19)$$

Assumption 4 (Bounded features). *The feature map satisfies*

$$\|\psi(u, a)\| \leq L < \infty \quad \text{for all } (u, a). \quad (20)$$

A.3 GENERALIZED-RIDGE OBJECTS

Assumption 5 (Archive geometry and complexity cap). *There is a known candidate-independent baseline $b(u)$ such that*

$$\mu(u, a) = b(u) + x(u, a)^\top \theta_a \in [0, 1]. \quad (21)$$

The feature vector satisfies $x(u, a) \in \mathbb{R}^d$. There is a positive definite matrix Ω and finite constants S_Ω, L_Ω such that

$$\Omega \succ 0, \quad \|\theta_a\|_{\Omega^{-1}} \leq S_\Omega, \quad x(u, a)^\top \Omega x(u, a) \leq L_\Omega^2 \quad (22)$$

for every candidate that can appear through round T . Candidate births, identities, parameters, and features satisfy the predictable birth-order conditions of Lemma 2. For the fixed-user trajectory, let

$$\mathcal{A}_T(U) = \bigcup_{t=1}^T A_t(U) \quad (23)$$

be the set of candidates that appear through round T . Effective dimension is defined in (10). There is a deterministic complexity cap $\bar{d}_{\text{eff}, T}(\Omega)$ such that, almost surely,

$$\max_{1 \leq s \leq T+1} \max_{a \in \mathcal{A}_T(U)} d_{\text{eff}}(\Omega; X_{a,s}) \leq \bar{d}_{\text{eff}, T}(\Omega). \quad (24)$$

For results combined with the four-term decomposition, draw one user U before routing and set $U_s = U$ at every round. The concentration lemmas below also allow a predictable sequence U_s . For a current candidate a , let

$$x_s(a) \in \mathbb{R}^d, \quad (25)$$

$$\mu(U_s, a) = b(U_s) + x_s(a)^\top \theta_a \in [0, 1], \quad (26)$$

$$r_s = b(U_s) + x_s(a_s)^\top \theta_{a_s} + \xi_s, \quad (27)$$

$$y_s = r_s - b(U_s) = x_s(a_s)^\top \theta_{a_s} + \xi_s. \quad (28)$$

The known baseline $b(U_s)$ is common to all available candidates. Candidate parameters share the positive-definite hyper-covariance

$$\theta_a \sim (0, \Omega), \quad (29)$$

$$\mathbb{E}[\theta_a] = 0, \quad (30)$$

$$\text{Cov}(\theta_a) = \Omega, \quad (31)$$

$$\Omega \succ 0. \quad (32)$$

For candidate a , collect its pre- t observations as

$$\mathcal{T}_{a,t} = \{s < t : a_s = a\}, \quad (33)$$

$$X_{a,t} = [x_s(a)^\top]_{s \in \mathcal{T}_{a,t}}, \quad (34)$$

$$y_{a,t} = [y_s]_{s \in \mathcal{T}_{a,t}}, \quad (35)$$

$$\hat{\Omega}_t \succ 0. \quad (36)$$

The last line is the hyper-covariance estimate available before round t .

HyperRidge estimates each candidate by

$$\hat{\theta}_{a,t} = \arg \min_{\theta \in \mathbb{R}^d} \left\{ \|y_{a,t} - X_{a,t}\theta\|_2^2 + \lambda \theta^\top \hat{\Omega}_t^{-1} \theta \right\}, \quad (37)$$

$$\hat{\theta}_{a,t} = \left(X_{a,t}^\top X_{a,t} + \lambda \hat{\Omega}_t^{-1} \right)^{-1} X_{a,t}^\top y_{a,t}, \quad (38)$$

$$V_{a,t}^{\text{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \hat{\Omega}_t^{-1}. \quad (39)$$

The HyperRidge score and selected candidate are

$$\text{UCB}_t^{\text{hyper}}(U_t, a) = b(U_t) + x_t(a)^\top \hat{\theta}_{a,t} + \alpha_t \sqrt{x_t(a)^\top \left(V_{a,t}^{\text{hyper}} \right)^{-1} x_t(a)}, \quad (40)$$

$$a_t(U_t) \in \arg \max_{a \in A_t(U_t)} \text{UCB}_t^{\text{hyper}}(U_t, a). \quad (41)$$

For a generic design matrix X , the effective dimension induced by Ω is

$$d_{\text{eff}}(\Omega; X) = \text{tr} \left[X^\top X (X^\top X + \lambda \Omega^{-1})^{-1} \right]. \quad (42)$$

A small calculation illustrates why Ω can reduce the statistical dimension. Take

$$\Omega = \begin{bmatrix} 1 & 0 \\ 0 & 0.01 \end{bmatrix}, \quad (43)$$

$$X^\top X = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}. \quad (44)$$

With $\lambda = 1$, the effective dimension is

$$\begin{aligned} d_{\text{eff}}(\Omega; X) &= \text{tr} \left[\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix} \right)^{-1} \right] \\ &= \frac{9}{10} + \frac{1}{101} \\ &\approx 0.910. \end{aligned} \quad (45)$$

By contrast, identity ridge gives

$$\begin{aligned} d_{\text{eff}}(I; X) &= \frac{9}{10} + \frac{1}{2} \\ &= 1.400. \end{aligned} \quad (46)$$

Thus the same data matrix has smaller exploration geometry when the archive covariance says the second coordinate is unlikely to matter.

For the estimator itself, take

$$X = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}, \quad (47)$$

$$y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \quad (48)$$

Using the same Ω and $\lambda = 1$, the generalized design is

$$\begin{aligned} V^{\text{hyper}} &= X^\top X + \Omega^{-1} \\ &= \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 0 \\ 0 & 100 \end{bmatrix}. \end{aligned} \quad (49)$$

$$X^\top y = \begin{bmatrix} 5 \\ 0 \end{bmatrix}. \quad (50)$$

$$\begin{aligned} \hat{\theta} &= \begin{bmatrix} 6 & 0 \\ 0 & 100 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 5/6 \\ 0 \end{bmatrix}. \end{aligned} \quad (51)$$

$$\sqrt{x^\top (V^{\text{hyper}})^{-1} x} = \sqrt{1/6}. \quad (52)$$

A.4 SHARED-STRUCTURE ASSUMPTION

The HyperRidge analysis uses the following shared-structure assumption. The diagnostics in Section 5 do not establish that it holds for the evaluated archives.

Assumption 6 (Shared archive geometry). *There is a positive-definite hyper-covariance matrix such that*

$$\Omega \in \mathbb{R}^{d \times d}, \quad \Omega \succ 0, \quad (53)$$

$$\theta_a \sim (0, \Omega). \quad (54)$$

Here this notation means $\mathbb{E}[\theta_a] = 0$ and $\mathbb{E}[\theta_a \theta_a^\top] = \Omega$. The raw mean reward includes a known candidate-independent baseline, and realized parameters satisfy

$$\mu(U_t, a) = b(U_t) + x_t(a)^\top \theta_a \in [0, 1], \quad (55)$$

$$\|\theta_a\|_{\Omega^{-1}} \leq S_\Omega \quad \text{for every } a \in A_t(U_t) \text{ and } t \leq T. \quad (56)$$

The archive is indexed by predictable birth-order slots as in Lemma 2. In particular, each candidate identity and feature rule is observable at birth, its latent parameter is fixed at birth in the analysis filtration, and the features are observable and predictable. For a predictable user stream,

$$\sigma(x_t(a)) \subseteq \mathcal{O}_{t-1} \quad \text{for every } a \in A_t(U_t), \quad (57)$$

$$\mathcal{A}_T^{\text{route}} = \bigcup_{t=1}^T A_t(U_t). \quad (58)$$

$$y_t := r_t - b(U_t) = x_t(a_t)^\top \theta_{a_t} + \xi_t. \quad (59)$$

The noise, feature radius, and effective dimension satisfy

$$\mathbb{E}[\exp(\eta \xi_t) \mid \mathcal{F}_{t-1}] \leq \exp\left(\frac{\eta^2 \sigma^2}{2}\right) \quad \text{for every } \eta \in \mathbb{R}, \quad (60)$$

$$x_t(a)^\top \Omega x_t(a) \leq L_\Omega^2 \quad \text{for every } a \in A_t(U_t), \quad (61)$$

$$\max_{1 \leq s \leq T+1} \max_{a \in \mathcal{A}_T^{\text{route}}} d_{\text{eff}}(\Omega; X_{a,s}) \leq \bar{d}_{\text{eff},T}(\Omega) < \infty \quad \text{almost surely}. \quad (62)$$

The concentration argument uses the realized ellipsoid bound, predictable births, centered reward model, and feature radius. The random-effects law is needed only to interpret and estimate Ω . The deterministic cap in (62) converts the pathwise potential bound into a deterministic finite-time certificate.

A.5 HRK: KNOWN- Ω CONCENTRATION

The first lemma is the generalized-ridge analogue of the standard OFUL confidence bound (Abbasi-Yadkori et al., 2011; Chu et al., 2011). It is stated for known Ω , which is the clean oracle-covariance case. The estimated-covariance case adds a perturbation term and is handled separately.

Lemma 3 (Self-normalized concentration in the Ω geometry). *Let $\Omega \succ 0$ and $\lambda > 0$. Fix a predictable birth-order slot a whose parameter is fixed at birth and satisfies $\|\theta_a\|_{\Omega^{-1}} \leq S_\Omega$. Before its birth, pad its feature and selection indicator by zero. Assume the padded selected feature is $\mathcal{F}_{s-1}$ -measurable and that each selected centered response obeys*

$$y_s = x_s(a)^\top \theta_a + \xi_s, \quad \mathbb{E}[\exp(\eta \xi_s) \mid \mathcal{F}_{s-1}] \leq \exp(\eta^2 \sigma^2 / 2) \quad \text{for every } \eta \in \mathbb{R}. \quad (63)$$

Let the router use the true covariance, $\hat{\Omega}_t = \Omega$, and define

$$V_{a,t}^\Omega = \lambda \Omega^{-1} + \sum_{s \in \mathcal{T}_{a,t}} x_s(a) x_s(a)^\top, \quad (64)$$

$$\hat{\theta}_{a,t}^\Omega = (V_{a,t}^\Omega)^{-1} \sum_{s \in \mathcal{T}_{a,t}} x_s(a) y_s. \quad (65)$$

Define the confidence radius

$$\beta_{a,t}^\Omega(\delta) = \sigma \sqrt{2 \log \left(\frac{\det(V_{a,t}^\Omega)^{1/2}}{\det(\lambda \Omega^{-1})^{1/2} \delta} \right)} + \sqrt{\lambda} S_\Omega. \quad (66)$$

Then, with probability at least $1 - \delta$, uniformly over t ,

$$\|\hat{\theta}_{a,t}^\Omega - \theta_a\|_{V_{a,t}^\Omega} \leq \beta_{a,t}^\Omega(\delta), \quad (67)$$

$$|x^\top (\hat{\theta}_{a,t}^\Omega - \theta_a)| \leq \beta_{a,t}^\Omega(\delta) \sqrt{x^\top (V_{a,t}^\Omega)^{-1} x}. \quad (68)$$

The second row holds for every predictable test feature $x \in \mathbb{R}^d$.

Proof. For this fixed candidate, define the regularizer and noise-weighted feature martingale by

$$A = \lambda \Omega^{-1}, \quad (69)$$

$$S_{a,t} = \sum_{s \in \mathcal{T}_{a,t}} x_s(a) \xi_s. \quad (70)$$

Using the reward model, the estimator error satisfies

$$\begin{aligned} \hat{\theta}_{a,t}^\Omega - \theta_a &= (V_{a,t}^\Omega)^{-1} \sum_{s \in \mathcal{T}_{a,t}} x_s(a) (x_s(a)^\top \theta_a + \xi_s) - \theta_a \\ &= (V_{a,t}^\Omega)^{-1} \left[\sum_{s \in \mathcal{T}_{a,t}} x_s(a) x_s(a)^\top \theta_a + S_{a,t} \right] - \theta_a \\ &= (V_{a,t}^\Omega)^{-1} [(V_{a,t}^\Omega - A) \theta_a + S_{a,t}] - \theta_a \\ &= (V_{a,t}^\Omega)^{-1} S_{a,t} - (V_{a,t}^\Omega)^{-1} A \theta_a. \end{aligned} \quad (71)$$

Taking the $V_{a,t}^\Omega$ -norm, applying the triangle inequality, and using $V_{a,t}^\Omega \succeq A$ give

$$\|\hat{\theta}_{a,t}^\Omega - \theta_a\|_{V_{a,t}^\Omega} \leq \|S_{a,t}\|_{(V_{a,t}^\Omega)^{-1}} + \|A \theta_a\|_{(V_{a,t}^\Omega)^{-1}}. \quad (72)$$

$$\begin{aligned} \|A\theta_a\|_{(V_{a,t}^\Omega)^{-1}}^2 &= \theta_a^\top A (V_{a,t}^\Omega)^{-1} A \theta_a \\ &\leq \theta_a^\top A \theta_a = \lambda \theta_a^\top \Omega^{-1} \theta_a \leq \lambda S_\Omega^2. \end{aligned} \quad (73)$$

It remains to control the martingale term. If $\sigma = 0$, the conditional moment-generating-function assumption implies $\xi_s = 0$ almost surely, so the claim is immediate. Suppose $\sigma > 0$. For fixed $z \in \mathbb{R}^d$, define

$$\begin{aligned} M_{a,t}(z) &= \exp\left(\frac{z^\top S_{a,t}}{\sigma} - \frac{1}{2} z^\top (V_{a,t}^\Omega - A) z\right), \\ \mathbb{E}\left[\exp\left(\frac{z^\top x_s(a)\xi_s}{\sigma} - \frac{(z^\top x_s(a))^2}{2}\right) \middle| \mathcal{F}_{s-1}\right] &\leq 1 \quad \text{at every selected round } s. \end{aligned} \quad (74)$$

At an unselected round the process does not change. Hence $(M_{a,t}(z))_{t \leq T}$ is a nonnegative supermartingale with initial value one.

Let

$$\varphi_A(z) = \frac{\det(A)^{1/2}}{(2\pi)^{d/2}} \exp\left(-\frac{z^\top A z}{2}\right), \quad \mathcal{M}_{a,t} = \int_{\mathbb{R}^d} M_{a,t}(z) \varphi_A(z) dz. \quad (75)$$

The first expression is the Gaussian density with precision A . Tonelli's theorem shows that $(\mathcal{M}_{a,t})_{t \leq T}$ is also a nonnegative supermartingale with initial value one. Completing the square gives

$$\mathcal{M}_{a,t} = \frac{\det(A)^{1/2}}{\det(V_{a,t}^\Omega)^{1/2}} \exp\left(-\frac{\|S_{a,t}\|_{(V_{a,t}^\Omega)^{-1}}^2}{2\sigma^2}\right), \quad (76)$$

$$\Pr\left(\sup_{t \leq T} \mathcal{M}_{a,t} \geq \frac{1}{\delta}\right) \leq \delta. \quad (77)$$

Consequently, with probability at least $1 - \delta$, simultaneously for every $t \leq T$, substituting $A = \lambda \Omega^{-1}$ gives

$$\|S_{a,t}\|_{(V_{a,t}^\Omega)^{-1}} \leq \sigma \sqrt{2 \log\left(\frac{\det(V_{a,t}^\Omega)^{1/2}}{\det(\lambda \Omega^{-1})^{1/2} \delta}\right)}. \quad (78)$$

Combining (72), (73), and (78) proves

$$\|\hat{\theta}_{a,t}^\Omega - \theta_a\|_{V_{a,t}^\Omega} \leq \beta_{a,t}^\Omega(\delta). \quad (79)$$

$$\begin{aligned} |x^\top (\hat{\theta}_{a,t}^\Omega - \theta_a)| &\leq \sqrt{x^\top (V_{a,t}^\Omega)^{-1} x} \|\hat{\theta}_{a,t}^\Omega - \theta_a\|_{V_{a,t}^\Omega} \\ &\leq \beta_{a,t}^\Omega(\delta) \sqrt{x^\top (V_{a,t}^\Omega)^{-1} x}. \end{aligned} \quad (80)$$

The second display applies Cauchy–Schwarz in the $V_{a,t}^\Omega$ -inner product. $\square$

A concrete confidence-bound computation is useful. Suppose

$$\frac{\det(V_{a,t}^\Omega)^{1/2}}{\det(\lambda \Omega^{-1})^{1/2}} = 10, \quad (81)$$

$$\sigma = 0.05, \quad (82)$$

$$\delta = 0.05, \quad (83)$$

$$\sqrt{\lambda} S_\Omega = 0.20. \quad (84)$$

Then the confidence radius is

$$\begin{aligned} \beta_{a,t}^\Omega(\delta) &= 0.05 \sqrt{2 \log(10/0.05)} + 0.20 \\ &\approx 0.05 \sqrt{10.596} + 0.20 \\ &\approx 0.363. \end{aligned} \quad (85)$$

If a candidate has width 0.10, Lemma 3 gives

$$\sqrt{x^\top (V_{a,t}^\Omega)^{-1} x} = 0.10, \quad (86)$$

$$\left| x^\top (\hat{\theta}_{a,t}^\Omega - \theta_a) \right| \leq 0.363 \cdot 0.10 = 0.0363. \quad (87)$$

A.6 ELLIPTICAL POTENTIAL IN THE Ω GEOMETRY

The next lemma is the step that converts per-round confidence widths into a dimension-dependent exploration cost. Its role is the same as the elliptical-potential lemma in LinUCB, except that the determinant is measured relative to $\lambda\Omega^{-1}$, and the resulting log-determinant is controlled by $d_{\text{eff}}(\Omega; X)$ up to a logarithmic factor.

Lemma 4 (Ω -elliptical potential and effective dimension). *Fix one candidate a . Write its selected features in chronological order and define*

$$x_1, \dots, x_n, \quad (88)$$

$$V_0 = \lambda\Omega^{-1}, \quad (89)$$

$$V_s = V_{s-1} + x_s x_s^\top, \quad (90)$$

$$q_s = x_s^\top V_{s-1}^{-1} x_s, \quad (91)$$

$$X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix}. \quad (92)$$

For the covariance-scaled Gram matrix, let

$$M = \lambda^{-1} \Omega^{1/2} X^\top X \Omega^{1/2}. \quad (93)$$

$$R = \lambda_{\max}(M), \quad (94)$$

$$c(R) = \begin{cases} 1, & R = 0, \\ \frac{1+R}{R} \log(1+R), & R > 0. \end{cases} \quad (95)$$

Then

$$\sum_{s=1}^n \min\{1, q_s\} \leq 2 c(R) d_{\text{eff}}(\Omega; X). \quad (96)$$

If additionally $x_s^\top \Omega x_s \leq L_\Omega^2$, then

$$R \leq \frac{nL_\Omega^2}{\lambda}, \quad (97)$$

$$\sum_{s=1}^n \min\{1, q_s\} \leq 2 \left[1 + \log \left(1 + \frac{nL_\Omega^2}{\lambda} \right) \right] d_{\text{eff}}(\Omega; X). \quad (98)$$

Proof. The matrix determinant lemma, its product over s , and the logarithm of that product give

$$\begin{aligned} \det(V_s) &= \det(V_{s-1}) (1 + x_s^\top V_{s-1}^{-1} x_s) = \det(V_{s-1}) (1 + q_s), \\ \frac{\det(V_n)}{\det(V_0)} &= \prod_{s=1}^n (1 + q_s), \\ \log \frac{\det(V_n)}{\det(V_0)} &= \sum_{s=1}^n \log(1 + q_s). \end{aligned} \quad (99)$$

For every $q \geq 0$, $\min\{1, q\} \leq 2 \log(1 + q)$. Hence

$$\begin{aligned} \min\{1, q\} &\leq 2 \log(1 + q), \\ \sum_{s=1}^n \min\{1, q_s\} &\leq 2 \sum_{s=1}^n \log(1 + q_s) = 2 \log \frac{\det(V_n)}{\det(V_0)}. \end{aligned} \quad (100)$$

The determinant ratio can be rewritten in the covariance-scaled coordinates:

$$\begin{aligned} \log \frac{\det(V_n)}{\det(V_0)} &= \log \det \left(I + \lambda^{-1} \Omega^{1/2} X^\top X \Omega^{1/2} \right) \\ &= \log \det(I + M). \end{aligned} \quad (101)$$

Let $\nu_1, \dots, \nu_d$ be the eigenvalues of M . Then

$$\begin{aligned} &\nu_1, \dots, \nu_d, \\ &0 \leq \nu_i \leq R, \\ \log \det(I + M) &= \sum_{i=1}^d \log(1 + \nu_i). \end{aligned} \quad (102)$$

The effective dimension is

$$\begin{aligned} d_{\text{eff}}(\Omega; X) &= \text{tr} \left[X^\top X (X^\top X + \lambda \Omega^{-1})^{-1} \right] \\ &= \text{tr} \left[\Omega^{1/2} X^\top X \Omega^{1/2} \left(\Omega^{1/2} X^\top X \Omega^{1/2} + \lambda I \right)^{-1} \right] \\ &= \sum_{i=1}^d \frac{\nu_i}{1 + \nu_i}. \end{aligned} \quad (103)$$

For $z > 0$, define

$$\begin{aligned} g(z) &= \frac{1+z}{z} \log(1+z). \\ g'(z) &= -\frac{\log(1+z)}{z^2} + \frac{1+1/z}{1+z} = \frac{z - \log(1+z)}{z^2} \geq 0, \\ \log(1+z) &\leq c(R) \frac{z}{1+z} \quad \text{for } 0 \leq z \leq R. \end{aligned} \quad (104)$$

The last row is equivalent to $g(z) \leq g(R)$.

Applying this scalar bound eigenvalue by eigenvalue gives

$$\begin{aligned} \log \det(I + M) &= \sum_{i=1}^d \log(1 + \nu_i) \\ &\leq c(R) \sum_{i=1}^d \frac{\nu_i}{1 + \nu_i} \\ &= c(R) d_{\text{eff}}(\Omega; X). \end{aligned} \quad (105)$$

Combining (100) and (105) proves

$$\sum_{s=1}^n \min\{1, q_s\} \leq 2c(R) d_{\text{eff}}(\Omega; X). \quad (106)$$

It remains to bound R . Since $M \succeq 0$,

$$\begin{aligned} R \leq \text{tr}(M) &= \lambda^{-1} \text{tr} \left(\Omega^{1/2} X^\top X \Omega^{1/2} \right) \\ &= \lambda^{-1} \sum_{s=1}^n x_s^\top \Omega x_s \leq \frac{n L_\Omega^2}{\lambda}. \end{aligned} \quad (107)$$

$$\begin{aligned} c(R) &\leq 1 + \log(1 + R) \\ &\leq 1 + \log \left(1 + \frac{n L_\Omega^2}{\lambda} \right). \end{aligned} \quad (108)$$

The claimed logarithmic bound follows. $\square$

B HRK: KNOWN-GEOMETRY EXPLORATION COST

We now combine Lemma 3 and Lemma 4. The result concerns HRK, which uses the true hyper-covariance Ω . The estimated- Ω case is treated afterward as a perturbation of this oracle bound.

Assumption 7 (Oracle covariance and explicit confidence radius). *The router uses the true positive-definite hyper-covariance. For the fixed-user route, define the realized candidate union and assume*

$$\hat{\Omega}_t = \Omega, \quad (109)$$

$$\Omega \succ 0, \quad (110)$$

$$\mathcal{A}_T(U) := \bigcup_{t=1}^T A_t(U). \quad (111)$$

$$|\mathcal{A}_T(U)| \leq K_{\max}, \quad (112)$$

$$x_t(a)^\top \Omega x_t(a) \leq L_\Omega^2. \quad (113)$$

The archive uses the predictable birth-order slots of Lemma 2. Its realized candidate designs obey the deterministic cap

$$\max_{1 \leq s \leq T+1} \max_{a \in \mathcal{A}_T(U)} d_{\text{eff}}(\Omega; X_{a,s}) \leq \bar{d}_{\text{eff},T}(\Omega) \quad \text{almost surely.} \quad (114)$$

For $\delta \in (0, 1)$, define

$$c_T^\Omega := 1 + \log \left(1 + \frac{TL_\Omega^2}{\lambda} \right) \quad (115)$$

$$\alpha_{\Omega,T}(\delta) := \max \left\{ 1, \sigma \sqrt{c_T^\Omega \bar{d}_{\text{eff},T}(\Omega) + 2 \log(K_{\max}/\delta)} + \sqrt{\lambda} S_\Omega \right\}. \quad (116)$$

The bound uses the largest realized candidate-wise effective dimension. For each a , collect its selected feature rows and define

$$X_{a,T+1} := [x_s(a_s)^\top]_{s \leq T: a_s = a}, \quad (117)$$

$$d_{\text{eff}}(\Omega; X_{a,T+1}) := \text{tr} \left[X_{a,T+1}^\top X_{a,T+1} (X_{a,T+1}^\top X_{a,T+1} + \lambda \Omega^{-1})^{-1} \right], \quad (118)$$

$$d_{\text{eff},T}^{\max}(\Omega) := \max_{a \in \mathcal{A}_T(U)} d_{\text{eff}}(\Omega; X_{a,T+1}). \quad (119)$$

Assumption 7 requires $d_{\text{eff},T}^{\max}(\Omega) \leq \bar{d}_{\text{eff},T}(\Omega)$ almost surely, and imposes the same cap on every earlier design prefix.

Proposition 5 (HRK: known- Ω effective-dimension exploration cost). *Under Assumptions 6 and 7, let HRK use multiplier $\alpha_{\Omega,T}(\delta)$. For the result used in the four-term decomposition, $U_t = U$ for all t . There is an event $\mathcal{E}_\Omega$ with probability at least $1 - \delta$ on which*

$$\sum_{t=1}^T L_t(U) \leq 2\alpha_{\Omega,T}(\delta) \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff},T}(\Omega)}. \quad (120)$$

Consequently, its expected exploration cost satisfies

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq 2\alpha_{\Omega,T}(\delta) \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff},T}(\Omega)} + \delta T. \quad (121)$$

For $T \geq 2$, fixed problem parameters, and $\delta = 1/T$,

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] = O(\sqrt{T} \log T). \quad (122)$$

For fixed δ , the same order holds for the pathwise right-hand side of (120) on its probability- $(1 - \delta)$ event, but the expectation bound (121) retains the linear term δT .

Proof. Index all candidates by the predictable birth-order slots in Lemma 2, using a dummy process for any slot that is never filled. For each of the at most $K_{\max}$ slots, apply Lemma 3 with failure probability $\delta/K_{\max}$. For every realized slot and every $t \leq T$, the determinant calculation in Lemma 4 gives

$$\begin{aligned} \log \frac{\det(V_{a,t}^\Omega)}{\det(\lambda\Omega^{-1})} &\leq c_T^\Omega d_{\text{eff}}(\Omega; X_{a,t}) \\ &\leq c_T^\Omega \bar{d}_{\text{eff},T}(\Omega). \end{aligned} \quad (123)$$

Thus the confidence radius in Lemma 3, with $\delta/K_{\max}$ in place of δ , is at most $\alpha_{\Omega,T}(\delta)$. A union bound over the slots defines an event $\mathcal{E}_\Omega$ with probability at least $1 - \delta$ on which these confidence intervals hold for every candidate after its birth and every $t \leq T$.

On $\mathcal{E}_\Omega$, define the current-archive oracle action, its exploration cost, and the confidence quadratic form by

$$a_t^* \in \arg \max_{a \in A_t(U)} x_t(a)^\top \theta_a, \quad (124)$$

$$L_t(U) = x_t(a_t^*)^\top \theta_{a_t^*} - x_t(a_t)^\top \theta_{a_t}, \quad (125)$$

$$q_t(a) := x_t(a)^\top (V_{a,t}^\Omega)^{-1} x_t(a), \quad q_t := q_t(a_t). \quad (126)$$

The confidence bounds and the router's UCB choice give the single chain

$$\begin{aligned} x_t(a_t^*)^\top \theta_{a_t^*} &\leq x_t(a_t^*)^\top \hat{\theta}_{a_t^*,t}^\Omega + \alpha_{\Omega,T}(\delta) \sqrt{q_t(a_t^*)} \\ &\leq x_t(a_t)^\top \hat{\theta}_{a_t,t}^\Omega + \alpha_{\Omega,T}(\delta) \sqrt{q_t} \\ &\leq x_t(a_t)^\top \theta_{a_t} + 2\alpha_{\Omega,T}(\delta) \sqrt{q_t}. \end{aligned} \quad (127)$$

Subtracting $x_t(a_t)^\top \theta_{a_t}$, then using bounded rewards and $\alpha_{\Omega,T}(\delta) \geq 1$, yields

$$\begin{aligned} L_t(U) &\leq 2\alpha_{\Omega,T}(\delta) \sqrt{q_t}, \\ 0 &\leq L_t(U) \leq 1, \\ L_t(U) &\leq \min\{1, 2\alpha_{\Omega,T}(\delta) \sqrt{q_t}\} \leq 2\alpha_{\Omega,T}(\delta) \min\{1, \sqrt{q_t}\}, \\ \sum_{t=1}^T L_t(U) &\leq 2\alpha_{\Omega,T}(\delta) \sum_{t=1}^T \min\{1, \sqrt{q_t}\}. \end{aligned} \quad (128)$$

Partition the selected rounds by candidate. Let n_a be the number of selections of a , and let $q_{a,s}$ be its uncertainty at its s -th selection:

$$n_a := \sum_{t=1}^T \mathbf{1}\{a_t = a\}. \quad (129)$$

$$q_{a,s} := x_{a,s}^\top (V_{a,s-1}^\Omega)^{-1} x_{a,s}. \quad (130)$$

Then Cauchy–Schwarz gives

$$\begin{aligned} \sum_{t=1}^T \min\{1, \sqrt{q_t}\} &= \sum_{a \in \mathcal{A}_T(U)} \sum_{s=1}^{n_a} \min\{1, \sqrt{q_{a,s}}\} \\ &\leq \sqrt{\left(\sum_{a \in \mathcal{A}_T(U)} n_a \right) \left(\sum_{a \in \mathcal{A}_T(U)} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\} \right)} \\ &= \sqrt{T \sum_{a \in \mathcal{A}_T(U)} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\}}. \end{aligned} \quad (131)$$

Lemma 4 applied separately to each candidate yields

$$\begin{aligned} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\} &\leq 2 \left[1 + \log \left(1 + \frac{n_a L_\Omega^2}{\lambda} \right) \right] d_{\text{eff}}(\Omega; X_{a,T+1}) \\ &\leq 2c_T^\Omega \bar{d}_{\text{eff},T}(\Omega), \end{aligned} \quad (132)$$

$$\sum_{a \in \mathcal{A}_T(U)} \sum_{s=1}^{n_a} \min\{1, q_{a,s}\} \leq 2K_{\max} c_T^\Omega \bar{d}_{\text{eff},T}(\Omega). \quad (133)$$

Substituting (133) into (131) and then into (128) proves

$$\sum_{t=1}^T L_t(U) \leq 2\alpha_{\Omega,T}(\delta) \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff},T}(\Omega)}. \quad (134)$$

This is (120). Taking expectations and splitting on $\mathcal{E}_\Omega$ gives

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}[L_t(U)] &= \mathbb{E} \left[\mathbf{1}_{\mathcal{E}_\Omega} \sum_{t=1}^T L_t(U) \right] + \mathbb{E} \left[\mathbf{1}_{\mathcal{E}_\Omega^c} \sum_{t=1}^T L_t(U) \right] \\ &\leq 2\alpha_{\Omega,T}(\delta) \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff},T}(\Omega)} + \delta T. \end{aligned} \quad (135)$$

This is (121). For fixed problem parameters, $c_T^\Omega = O(\log T)$ and $\alpha_{\Omega,T}(\delta) = O(\sqrt{\log T})$ for either fixed δ or $\delta = 1/T$ with $T \geq 2$. Thus the pathwise right-hand side is $O(\sqrt{T} \log T)$ in both cases. For the expectation bound, choosing $\delta = 1/T$ with $T \geq 2$ also makes the complement term $\delta T = 1$, which proves (122). With fixed δ , that complement term is linear and no sublinear expected bound follows from this conversion. $\square$

The geometry changes the potential factor from an ambient-dimension term to $\bar{d}_{\text{eff},T}(\Omega)$. This does not by itself determine which router has the smaller certificate: the full comparison must also include $\alpha_{\Omega,T}(\delta)$, c_T^Ω , and their LinUCB counterparts.

B.1 ESTIMATED HYPER-COVARIANCE: PERTURBATION REDUCTION AND THE REMAINING GAP

HRP uses a periodically updated estimate $\hat{\Omega}_t$. We therefore separate two issues. First, if $\hat{\Omega}_t$ is close to Ω in relative spectral norm, the oracle proof above is stable. Second, proving that an online archive estimator achieves this closeness is a separate statistical problem; we state it as Conjecture 1 rather than claiming it as proven.

Define the pointwise and uniform relative covariance errors, together with the estimated and oracle design matrices, by

$$\rho_{\Omega,t} := \left\| \Omega^{-1/2} \left(\hat{\Omega}_t - \Omega \right) \Omega^{-1/2} \right\|_{\text{op}}, \quad (136)$$

$$\rho_{\Omega,T} := \max_{1 \leq t \leq T} \rho_{\Omega,t}, \quad (137)$$

$$V_{a,t}^{\hat{\Omega}} := X_{a,t}^\top X_{a,t} + \lambda \hat{\Omega}_t^{-1}, \quad (138)$$

$$V_{a,t}^\Omega := X_{a,t}^\top X_{a,t} + \lambda \Omega^{-1}. \quad (139)$$

For a multiplier $\tilde{\alpha}_T \geq 1$, define the excess width caused by using $\hat{\Omega}_t$ instead of Ω :

$$\begin{aligned} \hat{w}_t &:= \min \left\{ 1, \sqrt{x_t(a_t)^\top \left(V_{a_t,t}^{\hat{\Omega}} \right)^{-1} x_t(a_t)} \right\}, \\ w_t^\Omega &:= \min \left\{ 1, \sqrt{x_t(a_t)^\top \left(V_{a_t,t}^\Omega \right)^{-1} x_t(a_t)} \right\}, \\ [z]_+ &:= \max\{z, 0\}, \end{aligned} \quad (140)$$

$$\text{WidthErr}_T(\hat{\Omega}, \Omega) := 2\tilde{\alpha}_T \sum_{t=1}^T [\hat{w}_t - w_t^\Omega]_+. \quad (141)$$

Lemma 6 (Relative spectral error implies Loewner control). *Assume*

$$\rho_{\Omega,T} \leq \varepsilon, \quad (142)$$

$$0 \leq \varepsilon < 1. \quad (143)$$

Then, for every candidate a and round t ,

$$(1 - \varepsilon)\Omega \preceq \hat{\Omega}_t \preceq (1 + \varepsilon)\Omega. \quad (144)$$

$$\frac{1}{1 + \varepsilon}\Omega^{-1} \preceq \hat{\Omega}_t^{-1} \preceq \frac{1}{1 - \varepsilon}\Omega^{-1}. \quad (145)$$

$$V_{a,t}^{\hat{\Omega}} \succeq \frac{1}{1 + \varepsilon}V_{a,t}^{\Omega}. \quad (146)$$

Proof. The definition of $\rho_{\Omega,T}$, followed by adding I , congruence by $\Omega^{1/2}$, and order-reversing inversion, gives

$$\begin{aligned} -\varepsilon I &\preceq \Omega^{-1/2} (\hat{\Omega}_t - \Omega) \Omega^{-1/2} \preceq \varepsilon I. \\ (1 - \varepsilon)I &\preceq \Omega^{-1/2} \hat{\Omega}_t \Omega^{-1/2} \preceq (1 + \varepsilon)I. \\ (1 - \varepsilon)\Omega &\preceq \hat{\Omega}_t \preceq (1 + \varepsilon)\Omega. \\ \frac{1}{1 + \varepsilon}\Omega^{-1} &\preceq \hat{\Omega}_t^{-1} \preceq \frac{1}{1 - \varepsilon}\Omega^{-1}. \end{aligned} \quad (147)$$

Substituting the lower inverse bound into the design matrix gives

$$\begin{aligned} V_{a,t}^{\hat{\Omega}} &= X_{a,t}^\top X_{a,t} + \lambda \hat{\Omega}_t^{-1} \\ &\succeq X_{a,t}^\top X_{a,t} + \frac{\lambda}{1 + \varepsilon} \Omega^{-1} \\ &\succeq \frac{1}{1 + \varepsilon} (X_{a,t}^\top X_{a,t} + \lambda \Omega^{-1}) \\ &= \frac{1}{1 + \varepsilon} V_{a,t}^{\Omega}. \end{aligned} \quad (148)$$

□

Lemma 7 (Width and effective-dimension stability). *Under the assumptions of Lemma 6, every feature vector x and fixed design matrix X satisfy*

$$x^\top \left(V_{a,t}^{\hat{\Omega}} \right)^{-1} x \leq (1 + \varepsilon) x^\top \left(V_{a,t}^{\Omega} \right)^{-1} x. \quad (149)$$

$$d_{\text{eff}}(\hat{\Omega}_t; X) \leq (1 + \varepsilon) d_{\text{eff}}(\Omega; X). \quad (150)$$

Proof. From (146), inversion, quadratic-form monotonicity, and trace monotonicity with $A = X^\top X \succeq 0$ give

$$\begin{aligned} \left(V_{a,t}^{\hat{\Omega}} \right)^{-1} &\preceq (1 + \varepsilon) \left(V_{a,t}^{\Omega} \right)^{-1}. \\ x^\top \left(V_{a,t}^{\hat{\Omega}} \right)^{-1} x &\leq (1 + \varepsilon) x^\top \left(V_{a,t}^{\Omega} \right)^{-1} x. \end{aligned} \quad (151)$$

$$\begin{aligned} d_{\text{eff}}(\hat{\Omega}_t; X) &= \text{tr} \left[A \left(A + \lambda \hat{\Omega}_t^{-1} \right)^{-1} \right] \\ &\leq (1 + \varepsilon) \text{tr} \left[A \left(A + \lambda \Omega^{-1} \right)^{-1} \right] \\ &= (1 + \varepsilon) d_{\text{eff}}(\Omega; X). \end{aligned} \quad (152)$$

□

The spectral comparison controls confidence widths. A regret statement also needs valid confidence intervals in the estimated geometry.

Proposition 8 (Conditional perturbation bound for estimated HyperRidge). *Assume Assumption 6, suppose at most $K_{\max}$ candidates appear through round T , and let $\hat{\Omega}_t \succ 0$ for every t . Let an estimated-covariance router use a multiplier $\tilde{\alpha}_T \geq 1$, and suppose the event $\mathcal{E}_{\text{conf}}(\hat{\Omega})$ gives simultaneous confidence:*

$$\left| x^\top \left(\hat{\theta}_{a,t}^{\hat{\Omega}} - \theta_a \right) \right| \leq \tilde{\alpha}_T \sqrt{x^\top \left(V_{a,t}^{\hat{\Omega}} \right)^{-1} x} \quad (153)$$

for every available candidate a , every $t \leq T$, and every predictable test feature x . Assume also that

$$\rho_{\Omega, T} \leq \varepsilon < 1. \quad (154)$$

On the intersection of these two events,

$$\text{WidthErr}_T(\widehat{\Omega}, \Omega) \leq 2\tilde{\alpha}_T (\sqrt{1+\varepsilon} - 1) \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff}, T}(\Omega)}, \quad (155)$$

$$\sum_{t=1}^T L_t(U) \leq 2\tilde{\alpha}_T \sqrt{1+\varepsilon} \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff}, T}(\Omega)}. \quad (156)$$

If the intersection has probability at least $1 - \eta$, then the right-hand side of (156) plus ηT bounds the expected cumulative exploration cost.

Proof. Lemma 7 and $\min\{1, cz\} \leq c \min\{1, z\}$ for $c \geq 1$ give

$$\begin{aligned} \hat{w}_t &\leq \sqrt{1+\varepsilon} w_t^\Omega, \\ [\hat{w}_t - w_t^\Omega]_+ &\leq (\sqrt{1+\varepsilon} - 1) w_t^\Omega. \end{aligned} \quad (157)$$

The definition of WidthErr_T and the oracle width-sum bound from the proof of Proposition 5 now give

$$\begin{aligned} \text{WidthErr}_T(\widehat{\Omega}, \Omega) &\leq 2\tilde{\alpha}_T (\sqrt{1+\varepsilon} - 1) \sum_{t=1}^T w_t^\Omega \\ &\leq 2\tilde{\alpha}_T (\sqrt{1+\varepsilon} - 1) \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff}, T}(\Omega)}. \end{aligned} \quad (158)$$

On $\mathcal{E}_{\text{conf}}(\widehat{\Omega})$, the same UCB argument and the first row above yield

$$\begin{aligned} L_t(U) &\leq 2\tilde{\alpha}_T \hat{w}_t \leq 2\tilde{\alpha}_T \sqrt{1+\varepsilon} w_t^\Omega, \\ \sum_{t=1}^T L_t(U) &\leq 2\tilde{\alpha}_T \sqrt{1+\varepsilon} \sum_{t=1}^T w_t^\Omega \\ &\leq 2\tilde{\alpha}_T \sqrt{1+\varepsilon} \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff}, T}(\Omega)}. \end{aligned}$$

This is (156). On the complement of an event with probability $1 - \eta$, bounded rewards give $\sum_{t=1}^T L_t(U) \leq T$, which adds ηT in expectation. $\square$

A numerical perturbation example clarifies the scale:

$$\varepsilon = 0.21, \quad (159)$$

$$\sqrt{1+\varepsilon} = 1.10. \quad (160)$$

Thus the estimated-geometry width is at most 1.10 times the oracle-geometry width on the spectral event. Turning this comparison into an exploration bound also requires the confidence event. Neither event has been proved for the periodic estimator.

Conjecture 1 (Sharper estimated-covariance exploration cost). *There is an estimated-covariance routing rule for which*

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq C \sqrt{K_{\max} \bar{d}_{\text{eff}, T}(\Omega) T \log(1+T) + o(\sqrt{T})}, \quad (161)$$

under assumptions comparable to those used here. The explicit frequentist radius in Theorem 15 does not prove this rate.

The conjecture requires a sharper confidence analysis or a different routing rule. The periodic estimator and its diagnostics do not establish it.

Table 5: Theory map for HyperRidge. HRK and HRC have separate guarantees; HRP does not.

Claim	Mathematical object	Status
HRK confidence	$V_{a,t}^\Omega = X_{a,t}^\top X_{a,t} + \lambda \Omega^{-1}$	Lemma 3
Effective-dimension potential	$\sum_s \min\{1, q_s\}$	Lemma 4
HRK exploration cost	explicit radius times effective-dimension potential	Proposition 5
Estimated- Ω width stability	WidthErr $_T$ under spectral error and confidence	Proposition 8
HRC frozen calibration estimate	finite-sample covariance and exploration bounds	Theorem 15
HRP periodic covariance estimate	confidence and covariance rates	open
Conditional reward consequence	$V_{\text{static}} + \Delta_{\text{route}} + \bar{G}_T - \bar{L}_T$	Theorem 17 and Corollary 22
Estimated-geometry experiments	matched controls and real LLM answers	completed; no general full-covariance advantage
HRC empirical validation	independent full-rank calibration and frozen archive	Appendix C.1; finite-sample synthetic check

B.2 MAP OF HYPERRIDGE CLAIMS TO PROOFS AND OPEN STEPS

Table 5 summarizes which parts of the HyperRidge theory are proved in this section and which are conditional. The inherited four-term decomposition remains the base accounting theorem; the new analysis only sharpens the exploration-cost term $\bar{L}_T$.

C HRC: CALIBRATION-SPLIT COVARIANCE CONTROL

This section analyzes HyperRidge-Calibrated (HRC). It estimates Ω from fresh candidates, freezes the estimate, and then routes over a fixed archive. The result does not cover HRP in Algorithm 1.

Assumption 8 (Exogenous uniform-routing calibration diversity). *Let $\Omega \succ 0$. The calibration phase uses N candidates and collects m rewards per candidate under an exogenous uniform-routing policy. For $i \in \{1, \dots, N\}$ and $j \in \{1, \dots, m\}$, define*

$$z_{i,j} = \Omega^{1/2} x_{i,j}, \quad X_i = \begin{bmatrix} x_{i,1}^\top \\ \vdots \\ x_{i,m}^\top \end{bmatrix}. \quad (162)$$

The arrays $X_1, \dots, X_N$ are independent. Within block i , the vectors $z_{i,1}, \dots, z_{i,m}$ are independent and, for deterministic $\kappa_{\text{cal}} > 0$ and $L_{\text{cal}} < \infty$,

$$\mathbb{E}[z_{i,j} z_{i,j}^\top] \succeq \kappa_{\text{cal}} I_d, \quad \|z_{i,j}\|_2^2 \leq L_{\text{cal}}^2 \quad \text{almost surely.} \quad (163)$$

The full feature array is independent of the candidate effects.

The effects $\theta_1, \dots, \theta_N$ are i.i.d. and satisfy

$$\mathbb{E}\theta_i = 0, \quad \mathbb{E}[\theta_i \theta_i^\top] = \Omega, \quad \|\theta_i\|_{\Omega^{-1}} \leq S_\Omega \quad \text{almost surely.} \quad (164)$$

The reward and centered response are

$$r_{i,j} = b(u_{i,j}) + x_{i,j}^\top \theta_i + \xi_{i,j}, \quad y_{i,j} = r_{i,j} - b(u_{i,j}). \quad (165)$$

Conditional on the full feature array, the pairs (θ_i, ξ_i) , with $\xi_i = (\xi_{i,1}, \dots, \xi_{i,m})^\top$, are independent across i . For each block there is a filtration $\mathcal{H}_{i,0} \subseteq \dots \subseteq \mathcal{H}_{i,m}$ such that the full feature array and θ_i are $\mathcal{H}_{i,0}$ -measurable, $\xi_{i,j}$ is $\mathcal{H}_{i,j}$ -measurable, and

$$\mathbb{E}[\exp(\eta \xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \leq \exp\left(\frac{\eta^2 \sigma^2}{2}\right) \quad \text{almost surely for every } \eta \in \mathbb{R}. \quad (166)$$

The filtration is used only for conditioning: the estimator observes (X_i, y_i) , not θ_i .

Lemma 9 (Net reduction for symmetric quadratic forms). *Let $A \in \mathbb{R}^{d \times d}$ be symmetric. Let $\mathcal{N} \subset \mathbb{S}^{d-1}$ be a $1/4$ -net of the Euclidean unit sphere. Then*

$$\|A\|_{\text{op}} \leq 2 \sup_{v \in \mathcal{N}} |v^\top A v|. \quad (167)$$

Moreover, there exists such a $1/4$ -net with cardinality at most 9^d .

Proof. Set $M = \sup_{v \in \mathcal{N}} |v^\top A v|$. For any $u \in \mathbb{S}^{d-1}$, choose $v \in \mathcal{N}$ with $\|u - v\|_2 \leq 1/4$. Symmetry gives

$$\begin{aligned} |u^\top A u - v^\top A v| &= |(u - v)^\top A u + v^\top A(u - v)| \\ &\leq 2\|u - v\|_2 \|A\|_{\text{op}} \leq \frac{1}{2} \|A\|_{\text{op}}. \end{aligned} \quad (168)$$

Thus $|u^\top A u| \leq M + \|A\|_{\text{op}}/2$. Taking the supremum over u and using $\|A\|_{\text{op}} = \sup_{\|u\|_2=1} |u^\top A u|$ gives $\|A\|_{\text{op}} \leq 2M$.

For the covering-number claim, let $\mathcal{N}$ be a maximal subset of $\mathbb{S}^{d-1}$ such that distinct points of $\mathcal{N}$ have Euclidean distance strictly larger than $1/4$. Maximality implies that $\mathcal{N}$ is a $1/4$ -net. The balls

$$B(v, 1/8) = \{w \in \mathbb{R}^d : \|w - v\|_2 \leq 1/8\}, \quad v \in \mathcal{N}, \quad (169)$$

are pairwise disjoint and contained in $B(0, 9/8)$. Volume comparison yields

$$|\mathcal{N}| \left(\frac{1}{8}\right)^d \leq \left(\frac{9}{8}\right)^d. \quad (170)$$

Thus $|\mathcal{N}| \leq 9^d$. $\square$

Lemma 10 (Independent sub-Gaussian sample covariance). *The universal numerical constant $C_{\text{cov}} = 24$ has the following property. Let $Z_1, \dots, Z_N \in \mathbb{R}^d$ be independent mean-zero random vectors such that, for every i , every $u \in \mathbb{S}^{d-1}$, and every $s \in \mathbb{R}$,*

$$\mathbb{E} \exp(s u^\top Z_i) \leq \exp\left(\frac{s^2 K^2}{2}\right). \quad (171)$$

Let $\Sigma_i = \mathbb{E}[Z_i Z_i^\top]$. Then, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left\| \frac{1}{N} \sum_{i=1}^N (Z_i Z_i^\top - \Sigma_i) \right\|_{\text{op}} \leq C_{\text{cov}} K^2 \left[\sqrt{\frac{d + \log(2/\delta)}{N}} + \frac{d + \log(2/\delta)}{N} \right]. \quad (172)$$

The same conclusion holds conditionally on any sigma-field with respect to which the displayed independence and moment-generating-function hypotheses hold with the same deterministic value of K .

Proof. If $K = 0$, every $Z_i = 0$ almost surely and the claim is immediate. Assume $K > 0$. Fix $u \in \mathbb{S}^{d-1}$. Chernoff's method, applied also to $-u$, gives

$$\Pr(|u^\top Z_i| \geq t) \leq 2 \exp\left(-\frac{t^2}{2K^2}\right) \quad t \geq 0. \quad (173)$$

Equivalently, $u^\top Z_i$ is sub-Gaussian with sub-Gaussian norm bounded by a universal multiple of K . Lemma 2.7.7 of Vershynin, *High-Dimensional Probability*, Cambridge University Press, 2018, shows that $Y_i = (u^\top Z_i)^2 - \mathbb{E}[(u^\top Z_i)^2]$ is centered sub-exponential with norm at most a universal multiple of K^2 . Theorem 2.8.1 of the same reference gives a universal $c_1 > 0$ such that

$$\Pr\left(\left| \frac{1}{N} \sum_{i=1}^N [(u^\top Z_i)^2 - \mathbb{E}(u^\top Z_i)^2] \right| > c_1 K^2 \left[\sqrt{\frac{t}{N}} + \frac{t}{N} \right]\right) \leq 2e^{-t}. \quad (174)$$

Let $\mathcal{N}$ be a $1/4$ -net of $\mathbb{S}^{d-1}$ with $|\mathcal{N}| \leq 9^d$, whose existence follows from Lemma 9. Applying the preceding scalar inequality with $t = d \log 9 + \log(2/\delta)$, followed by a union bound, gives

$$\sup_{u \in \mathcal{N}} \left| u^\top \left[\frac{1}{N} \sum_{i=1}^N (Z_i Z_i^\top - \Sigma_i) \right] u \right| \leq c_1 K^2 \left[\sqrt{\frac{d \log 9 + \log(2/\delta)}{N}} + \frac{d \log 9 + \log(2/\delta)}{N} \right]. \quad (175)$$

Applying Lemma 9 to the symmetric matrix inside the brackets yields

$$\left\| \frac{1}{N} \sum_{i=1}^N (Z_i Z_i^\top - \Sigma_i) \right\|_{\text{op}} \leq 2c_1 K^2 \left[\sqrt{\frac{d \log 9 + \log(2/\delta)}{N}} + \frac{d \log 9 + \log(2/\delta)}{N} \right]. \quad (176)$$

For an explicit constant, write $X = u^\top Z_i$ and $W = X^2 - \mathbb{E}X^2$. The sub-Gaussian tail above gives $\mathbb{E}|X|^{2p} \leq 2(2K^2)^p p!$. If X' is an independent copy of X , Jensen's inequality and $|a-b|^p \leq a^p + b^p$ for $a, b \geq 0$ give, for integers $p \geq 2$,

$$\mathbb{E}|W|^p \leq \mathbb{E}|X^2 - (X')^2|^p \leq 2\mathbb{E}|X|^{2p} \leq \frac{p!}{2} (32K^4)(2K^2)^{p-2}. \quad (177)$$

Expanding the moment-generating function, using $\mathbb{E}W = 0$, and applying $\log(1+v) \leq v$ yields $\log \mathbb{E}e^{sW} \leq 16K^4 s^2 / (1 - 2K^2|s|)$ for $|s| < (2K^2)^{-1}$. Independence and Chernoff's inequality therefore give

$$\Pr\left(\left|\frac{1}{N} \sum_{i=1}^N W_i\right| > 8K^2 \sqrt{\frac{t}{N}} + 2K^2 \frac{t}{N}\right) \leq 2e^{-t}. \quad (178)$$

The same net argument bounds the operator norm by $16K^2 \sqrt{t/N} + 4K^2 t/N$, with $t = d \log 9 + \log(2/\delta)$. Writing $q = d + \log(2/\delta)$, we have $t \leq (\log 9)q$. Since $16\sqrt{\log 9} < 24$ and $4 \log 9 < 24$, this is at most $24K^2[\sqrt{q/N} + q/N]$, proving the stated numerical choice.

For the conditional version, fix a regular conditional distribution given the conditioning sigma-field. Under that conditional distribution, the same proof applies with the same deterministic K and constants. Integrating its failure probability completes the conditional claim. $\square$

Lemma 11 (Calibration Gram lower tail under exogenous diversity). *Assume Assumption 8. For each calibration candidate define*

$$G_i = X_i^\top X_i, \quad H_i = \Omega^{1/2} G_i \Omega^{1/2} = \sum_{j=1}^m z_{i,j} z_{i,j}^\top. \quad (179)$$

For every $\delta_{\text{gram}} \in (0, 1)$, if

$$m \geq \frac{8L_{\text{cal}}^2}{\kappa_{\text{cal}}} \log\left(\frac{dN}{\delta_{\text{gram}}}\right), \quad (180)$$

then, with probability at least $1 - \delta_{\text{gram}}$,

$$\begin{aligned} H_i &\succeq \frac{\kappa_{\text{cal}} m}{2} I_d \quad (i = 1, \dots, N), \\ \left\| \Omega^{-1/2} G_i^{-1} \Omega^{-1/2} \right\|_{\text{op}} &\leq \frac{2}{\kappa_{\text{cal}} m} \quad (i = 1, \dots, N). \end{aligned} \quad (181)$$

Every G_i is invertible on this event.

Proof. Fix i and set $A_{i,j} = z_{i,j} z_{i,j}^\top$. By Assumption 8, the matrices $A_{i,1}, \dots, A_{i,m}$ are independent positive-semidefinite random matrices satisfying

$$0 \preceq A_{i,j} \preceq L_{\text{cal}}^2 I_d, \quad \lambda_{\min}\left(\sum_{j=1}^m \mathbb{E} A_{i,j}\right) \geq \kappa_{\text{cal}} m. \quad (182)$$

Tropp's matrix Chernoff lower-tail inequality for independent positive-semidefinite summands bounded above by $R I_d$ states that, with $\mu_{\min} = \lambda_{\min}(\sum_j \mathbb{E} A_{i,j})$, for every $\varepsilon \in [0, 1]$,

$$\Pr\left(\lambda_{\min}\left(\sum_{j=1}^m A_{i,j}\right) \leq (1 - \varepsilon)\mu_{\min}\right) \leq d \exp\left(-\frac{\varepsilon^2 \mu_{\min}}{2R}\right). \quad (183)$$

This is Theorem 5.1.1 of Tropp, *An Introduction to Matrix Concentration Inequalities*, Foundations and Trends in Machine Learning, 2015, applied with $R = L_{\text{cal}}^2$. Taking $\varepsilon = 1/2$ gives

$$\Pr\left(H_i \not\succeq \frac{\kappa_{\text{cal}} m}{2} I_d\right) \leq d \exp\left(-\frac{\kappa_{\text{cal}} m}{8L_{\text{cal}}^2}\right). \quad (184)$$

The sample-size condition makes the right-hand side at most δ_{gram}/N ; a union bound proves the simultaneous Gram event.

On this event,

$$\Omega^{1/2} G_i \Omega^{1/2} \succeq \frac{\kappa_{\text{cal}} m}{2} I_d. \quad (185)$$

Thus G_i is invertible. Inverting the Loewner inequality and identifying the inverse give

$$\begin{aligned} (\Omega^{1/2} G_i \Omega^{1/2})^{-1} &\preceq \frac{2}{\kappa_{\text{cal}} m} I_d, \\ (\Omega^{1/2} G_i \Omega^{1/2})^{-1} &= \Omega^{-1/2} G_i^{-1} \Omega^{-1/2}, \end{aligned} \quad (186)$$

the asserted operator-norm bound follows. $\square$

Lemma 12 (Conditional calibration-block covariance control). *Assume Assumption 8. Write $X_{1:N} = (X_1, \dots, X_N)$. Condition on a realization of the full calibration feature array for which*

$$H_i = \Omega^{1/2} X_i^\top X_i \Omega^{1/2} \succeq \frac{\kappa_{\text{cal}} m}{2} I_d \quad \text{for every } i = 1, \dots, N. \quad (187)$$

For each i , define

$$y_i = \begin{bmatrix} r_{i,1} - b(u_{i,1}) \\ \vdots \\ r_{i,m} - b(u_{i,m}) \end{bmatrix} = X_i \theta_i + \xi_i, \quad (188)$$

and

$$\begin{aligned} \tilde{\theta}_i &= G_i^{-1} X_i^\top y_i, \quad Z_i = \Omega^{-1/2} \tilde{\theta}_i, \\ \tau_m &= \frac{2\sigma^2}{\kappa_{\text{cal}} m}, \quad K_m^2 = S_\Omega^2 + \tau_m. \end{aligned} \quad (189)$$

Then, conditional on the feature array, $Z_1, \dots, Z_N$ are independent mean-zero random vectors. Moreover,

$$\mathbb{E}[Z_i Z_i^\top \mid X_{1:N}] = I_d + R_i, \quad R_i \succeq 0, \quad \|R_i\|_{\text{op}} \leq \tau_m, \quad (190)$$

and, for every $u \in \mathbb{S}^{d-1}$ and every $s \in \mathbb{R}$,

$$\mathbb{E}[\exp(su^\top Z_i) \mid X_{1:N}] \leq \exp\left(\frac{s^2 K_m^2}{2}\right). \quad (191)$$

Proof. The least-squares identity gives

$$\begin{aligned} \tilde{\theta}_i &= G_i^{-1} X_i^\top (X_i \theta_i + \xi_i) \\ &= \theta_i + G_i^{-1} X_i^\top \xi_i. \end{aligned} \quad (192)$$

Set

$$\begin{aligned} U_i &= \Omega^{-1/2} \theta_i, \quad Z_i = U_i + E_i, \\ E_i &= \Omega^{-1/2} G_i^{-1} X_i^\top \xi_i. \end{aligned} \quad (193)$$

The random-effects assumptions imply, conditional on the full calibration feature array,

$$\mathbb{E}[U_i \mid X_{1:N}] = 0, \quad \mathbb{E}[U_i U_i^\top \mid X_{1:N}] = I_d, \quad \|U_i\|_2 \leq S_\Omega \quad \text{almost surely.} \quad (194)$$

The conditional sub-Gaussian noise assumption implies $\mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}] = 0$. Indeed, Jensen's inequality gives, for $h > 0$,

$$\exp(h \mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}]) \leq \mathbb{E}[\exp(h \xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \leq \exp\left(\frac{h^2 \sigma^2}{2}\right), \quad (195)$$

so $\mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}] \leq h\sigma^2/2$. Letting $h \downarrow 0$ gives the upper bound 0. Applying the same argument to $-\xi_{i,j}$ gives the lower bound 0. Therefore the conditional mean is zero. By iterated conditioning,

$$\mathbb{E}[\xi_i \mid X_{1:N}, \theta_i] = 0, \quad \mathbb{E}[E_i \mid X_{1:N}, \theta_i] = 0. \quad (196)$$

Consequently, the cross terms vanish:

$$\mathbb{E}[U_i E_i^\top \mid X_{1:N}] = 0, \quad \mathbb{E}[E_i U_i^\top \mid X_{1:N}] = 0. \quad (197)$$

Thus

$$\mathbb{E}[Z_i Z_i^\top \mid X_{1:N}] = I_d + R_i, \quad R_i = \mathbb{E}[E_i E_i^\top \mid X_{1:N}] \succeq 0. \quad (198)$$

To bound R_i , fix $v \in \mathbb{S}^{d-1}$ and set $b_i = X_i G_i^{-1} \Omega^{-1/2} v$. Conditional on the feature array, b_i is deterministic and

$$v^\top R_i v = \mathbb{E}[(b_i^\top \xi_i)^2 \mid X_{1:N}]. \quad (199)$$

For $j < k$, the random variable $\xi_{i,j}$ is $\mathcal{H}_{i,k-1}$ -measurable. Hence

$$\begin{aligned} \mathbb{E}[\xi_{i,j} \xi_{i,k} \mid X_{1:N}] &= \mathbb{E}[\xi_{i,j} \mathbb{E}[\xi_{i,k} \mid \mathcal{H}_{i,k-1}] \mid X_{1:N}] \\ &= 0. \end{aligned} \quad (200)$$

The conditional sub-Gaussian inequality also implies $\mathbb{E}[\xi_{i,j}^2 \mid \mathcal{H}_{i,j-1}] \leq \sigma^2$. To see this, average the MGF bounds at h and $-h$:

$$\mathbb{E}[\cosh(h \xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \leq \exp\left(\frac{h^2 \sigma^2}{2}\right). \quad (201)$$

Conditional Fatou's lemma and $2(\cosh(hx) - 1)/h^2 \rightarrow x^2$ yield

$$\begin{aligned} \mathbb{E}[\xi_{i,j}^2 \mid \mathcal{H}_{i,j-1}] &\leq \liminf_{h \downarrow 0} \frac{2}{h^2} \left[\exp\left(\frac{h^2 \sigma^2}{2}\right) - 1 \right] \\ &= \sigma^2. \end{aligned} \quad (202)$$

The zero cross-covariances and variance bound now give

$$\begin{aligned} v^\top R_i v &\leq \sigma^2 \|b_i\|_2^2, \\ \|b_i\|_2^2 &= v^\top \Omega^{-1/2} G_i^{-1} X_i^\top X_i G_i^{-1} \Omega^{-1/2} v \\ &= v^\top \Omega^{-1/2} G_i^{-1} \Omega^{-1/2} v. \end{aligned} \quad (203)$$

On the conditioned Gram event,

$$\Omega^{-1/2} G_i^{-1} \Omega^{-1/2} = (\Omega^{1/2} G_i \Omega^{1/2})^{-1} \preceq \frac{2}{\kappa_{\text{cal}} m} I_d. \quad (204)$$

Therefore

$$v^\top R_i v \leq \frac{2\sigma^2}{\kappa_{\text{cal}} m} = \tau_m. \quad (205)$$

Taking the supremum over v gives $\|R_i\|_{\text{op}} \leq \tau_m$.

It remains to prove the one-dimensional moment-generating-function bound. Fix $u \in \mathbb{S}^{d-1}$. Since $\mathbb{E}[u^\top U_i \mid X_{1:N}] = 0$ and $|u^\top U_i| \leq \|U_i\|_2 \leq S_\Omega$ almost surely, Hoeffding's lemma gives

$$\mathbb{E}[\exp(s u^\top U_i) \mid X_{1:N}] \leq \exp\left(\frac{s^2 S_\Omega^2}{2}\right). \quad (206)$$

For the noise part, set $b_i = X_i G_i^{-1} \Omega^{-1/2} u$, which is deterministic conditional on the full feature array. Iterating the conditional sub-Gaussian inequality over $j = m, m-1, \dots, 1$ gives

$$\mathbb{E}[\exp(s b_i^\top \xi_i) \mid X_{1:N}, \theta_i] \leq \exp\left(\frac{s^2 \sigma^2 \|b_i\|_2^2}{2}\right). \quad (207)$$

The Gram-event calculation above gives $\|b_i\|_2^2 \leq 2/(\kappa_{\text{cal}} m)$. Therefore,

$$\mathbb{E}[\exp(s u^\top E_i) \mid X_{1:N}, \theta_i] \leq \exp\left(\frac{s^2 \tau_m}{2}\right). \quad (208)$$

Since U_i is a function of θ_i ,

$$\begin{aligned}\mathbb{E}[\exp(su^\top Z_i) \mid X_{1:N}] &= \mathbb{E}[\exp(su^\top U_i) \mathbb{E}[\exp(su^\top E_i) \mid X_{1:N}, \theta_i] \mid X_{1:N}] \\ &\leq \exp\left(\frac{s^2 \tau_m}{2}\right) \mathbb{E}[\exp(su^\top U_i) \mid X_{1:N}] \\ &\leq \exp\left(\frac{s^2 (S_\Omega^2 + \tau_m)}{2}\right).\end{aligned}\tag{209}$$

Finally, Assumption 8 makes the pairs (θ_i, ξ_i) independent across i , conditional on the feature array. Since Z_i is a function of (X_i, θ_i, ξ_i) , the Z_i are conditionally independent. $\square$

Lemma 13 (Relative covariance error for the calibration estimator). *Assume Assumption 8. Define*

$$\hat{\Omega} = \frac{1}{N} \sum_{i=1}^N \tilde{\theta}_i \tilde{\theta}_i^\top + \gamma I_d, \quad \gamma > 0,\tag{210}$$

where $\tilde{\theta}_i = G_i^{-1} X_i^\top y_i$ on the event that G_i is invertible, and $\tilde{\theta}_i = 0$ otherwise. Let

$$\tau_m = \frac{2\sigma^2}{\kappa_{\text{cal}} m}, \quad K_m^2 = S_\Omega^2 + \tau_m.\tag{211}$$

For $\delta_{\text{gram}}, \delta_{\text{cov}} \in (0, 1)$, suppose

$$m \geq \frac{8L_{\text{cal}}^2}{\kappa_{\text{cal}}} \log\left(\frac{dN}{\delta_{\text{gram}}}\right).\tag{212}$$

Define

$$\begin{aligned}\varepsilon_{\text{cov}}(N, m, \gamma, \delta_{\text{cov}}) &= C_{\text{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\text{cov}})}{N}} + \frac{d + \log(2/\delta_{\text{cov}})}{N} \right] \\ &\quad + \tau_m + \gamma \|\Omega^{-1}\|_{\text{op}}.\end{aligned}\tag{213}$$

Then, with probability at least $1 - \delta_{\text{gram}} - \delta_{\text{cov}}$,

$$\left\| \Omega^{-1/2} (\hat{\Omega} - \Omega) \Omega^{-1/2} \right\|_{\text{op}} \leq \varepsilon_{\text{cov}}(N, m, \gamma, \delta_{\text{cov}}).\tag{214}$$

Proof. Let

$$\mathcal{E}_{\text{gram}} = \left\{ H_i \succeq \frac{\kappa_{\text{cal}} m}{2} I_d \text{ for every } i = 1, \dots, N \right\}.\tag{215}$$

Lemma 11 gives $\Pr(\mathcal{E}_{\text{gram}}) \geq 1 - \delta_{\text{gram}}$. Condition on a realization of the full calibration feature array in $\mathcal{E}_{\text{gram}}$. By Lemma 12, the vectors $Z_i = \Omega^{-1/2} \tilde{\theta}_i$ are conditionally independent and satisfy

$$\mathbb{E}[Z_i Z_i^\top \mid X_{1:N}] = I_d + R_i, \quad R_i \succeq 0, \quad \|R_i\|_{\text{op}} \leq \tau_m,\tag{216}$$

and, for every $u \in \mathbb{S}^{d-1}$ and $s \in \mathbb{R}$,

$$\mathbb{E}[\exp(su^\top Z_i) \mid X_{1:N}] \leq \exp\left(\frac{s^2 K_m^2}{2}\right).\tag{217}$$

The conditional form of Lemma 10 therefore gives an event of conditional probability at least $1 - \delta_{\text{cov}}$ on which

$$\begin{aligned}\left\| \frac{1}{N} \sum_{i=1}^N [Z_i Z_i^\top - \mathbb{E}(Z_i Z_i^\top \mid X_{1:N})] \right\|_{\text{op}} &\leq C_{\text{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\text{cov}})}{N}} \right. \\ &\quad \left. + \frac{d + \log(2/\delta_{\text{cov}})}{N} \right].\end{aligned}\tag{218}$$

On the intersection of this event and $\mathcal{E}_{\text{gram}}$,

$$\begin{aligned} \left\| \frac{1}{N} \sum_{i=1}^N Z_i Z_i^\top - I_d \right\|_{\text{op}} &\leq \left\| \frac{1}{N} \sum_{i=1}^N [Z_i Z_i^\top - \mathbb{E}(Z_i Z_i^\top \mid X_{1:N})] \right\|_{\text{op}} + \left\| \frac{1}{N} \sum_{i=1}^N R_i \right\|_{\text{op}} \\ &\leq C_{\text{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\text{cov}})}{N}} + \frac{d + \log(2/\delta_{\text{cov}})}{N} \right] + \tau_m. \end{aligned} \quad (219)$$

Here

$$\left\| \frac{1}{N} \sum_{i=1}^N R_i \right\|_{\text{op}} \leq \frac{1}{N} \sum_{i=1}^N \|R_i\|_{\text{op}} \leq \tau_m. \quad (220)$$

Finally,

$$\begin{aligned} \Omega^{-1/2} \hat{\Omega} \Omega^{-1/2} &= \frac{1}{N} \sum_{i=1}^N Z_i Z_i^\top + \gamma \Omega^{-1}, \\ \Omega^{-1/2} (\hat{\Omega} - \Omega) \Omega^{-1/2} &= \left(\frac{1}{N} \sum_{i=1}^N Z_i Z_i^\top - I_d \right) + \gamma \Omega^{-1}. \end{aligned} \quad (221)$$

The triangle inequality gives

$$\left\| \Omega^{-1/2} (\hat{\Omega} - \Omega) \Omega^{-1/2} \right\|_{\text{op}} \leq \varepsilon_{\text{cov}}(N, m, \gamma, \delta_{\text{cov}}). \quad (222)$$

The unconditional probability of the intersection of the Gram event and the conditional covariance event is at least $1 - \delta_{\text{gram}} - \delta_{\text{cov}}$ by the law of total probability and a union bound. $\square$

Lemma 14 (Confidence under a frozen estimated covariance). *Assume the post-calibration routing model of Assumption 6. Assume the post-calibration routing archive is fixed before routing begins and contains at most K_{max} candidates. Let $\mathcal{F}_0^{\text{route}}$ contain all calibration data, the fixed routing archive, and the latent parameter of every routing candidate. Let $\hat{\Omega} \succ 0$ be $\mathcal{F}_0^{\text{route}}$ -measurable and frozen during routing. Assume the post-routing conditional sub-Gaussian bound (60) holds for a routing filtration that contains $\mathcal{F}_0^{\text{route}}$. Suppose*

$$\left\| \Omega^{-1/2} (\hat{\Omega} - \Omega) \Omega^{-1/2} \right\|_{\text{op}} \leq \varepsilon, \quad 0 \leq \varepsilon < 1. \quad (223)$$

For candidate a , define

$$V_{a,t}^{\hat{\Omega}} = X_{a,t}^\top X_{a,t} + \lambda \hat{\Omega}^{-1}, \quad \hat{\theta}_{a,t}^{\hat{\Omega}} = (V_{a,t}^{\hat{\Omega}})^{-1} X_{a,t}^\top y_{a,t}. \quad (224)$$

Define

$$\alpha_{\hat{\Omega},T}(\varepsilon, \delta_{\text{route}}) = \max \left\{ 1, \sigma \sqrt{d \log \left(1 + \frac{(1+\varepsilon) T L_\Omega^2}{\lambda} \right)} + 2 \log \left(\frac{K_{\text{max}}}{\delta_{\text{route}}} \right) + \sqrt{\frac{\lambda}{1-\varepsilon}} S_\Omega \right\}. \quad (225)$$

Then, conditional on $\mathcal{F}_0^{\text{route}}$, with probability at least $1 - \delta_{\text{route}}$, uniformly over all routing candidates a , all $t \leq T$, and all predictable test features x ,

$$\left| x^\top (\hat{\theta}_{a,t}^{\hat{\Omega}} - \theta_a) \right| \leq \alpha_{\hat{\Omega},T}(\varepsilon, \delta_{\text{route}}) \sqrt{x^\top (V_{a,t}^{\hat{\Omega}})^{-1} x}. \quad (226)$$

Proof. Fix a routing candidate a . Conditional on $\mathcal{F}_0^{\text{route}}$, the matrix $A = \lambda \hat{\Omega}^{-1}$ is deterministic and positive definite. Set

$$S_{a,t} = \sum_{s \in \mathcal{T}_{a,t}} x_s(a) \xi_s, \quad V_{a,t}^{\hat{\Omega}} = A + \sum_{s \in \mathcal{T}_{a,t}} x_s(a) x_s(a)^\top. \quad (227)$$

The reward model and the triangle inequality give

$$\begin{aligned} \hat{\theta}_{a,t}^{\hat{\Omega}} - \theta_a &= (V_{a,t}^{\hat{\Omega}})^{-1} S_{a,t} - (V_{a,t}^{\hat{\Omega}})^{-1} A \theta_a. \\ \|\hat{\theta}_{a,t}^{\hat{\Omega}} - \theta_a\|_{V_{a,t}^{\hat{\Omega}}} &\leq \|S_{a,t}\|_{(V_{a,t}^{\hat{\Omega}})^{-1}} + \|A \theta_a\|_{(V_{a,t}^{\hat{\Omega}})^{-1}}. \end{aligned} \quad (228)$$

Because $V_{a,t}^{\widehat{\Omega}} \succeq A$,

$$\begin{aligned} \|A\theta_a\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}}^2 &\leq \theta_a^\top A \theta_a = \lambda \theta_a^\top \widehat{\Omega}^{-1} \theta_a. \\ (1-\varepsilon)\Omega &\preceq \widehat{\Omega} \preceq (1+\varepsilon)\Omega, \quad \widehat{\Omega}^{-1} \preceq \frac{1}{1-\varepsilon}\Omega^{-1}. \end{aligned} \quad (229)$$

$$\|A\theta_a\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}} \leq \sqrt{\frac{\lambda}{1-\varepsilon}} S_\Omega.$$

It remains to control the martingale term. If $\sigma = 0$, conditional sub-Gaussianity implies $S_{a,t} = 0$ almost surely and the claim follows from the bias bound. Suppose $\sigma > 0$. For fixed $q \in \mathbb{R}^d$, define

$$M_t(q) = \exp\left(\frac{q^\top S_{a,t}}{\sigma} - \frac{1}{2} q^\top (V_{a,t}^{\widehat{\Omega}} - A) q\right). \quad (230)$$

$$\mathbb{E}\left[\exp\left(\frac{q^\top x_t(a)\xi_t}{\sigma} - \frac{(q^\top x_t(a))^2}{2}\right) \middle| \mathcal{F}_{t-1}\right] \leq 1 \quad \text{when round } t \text{ selects } a.$$

If round t does not select a , then $S_{a,t}$ and $V_{a,t}^{\widehat{\Omega}}$ do not change. Therefore $(M_t(q))_{t \leq T}$ is a nonnegative supermartingale with initial value one.

Let Q be Gaussian with precision A . Its density and the corresponding mixture process are

$$\varphi_A(q) = \frac{\det(A)^{1/2}}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2} q^\top A q\right), \quad \mathcal{M}_t = \int_{\mathbb{R}^d} M_t(q) \varphi_A(q) dq. \quad (231)$$

Tonelli's theorem and the supermartingale property of $M_t(q)$ imply that $(\mathcal{M}_t)_{t \leq T}$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$. Completing the square and then applying Ville's inequality give

$$\mathcal{M}_t = \frac{\det(A)^{1/2}}{\det(V_{a,t}^{\widehat{\Omega}})^{1/2}} \exp\left(-\frac{\|S_{a,t}\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}}^2}{2\sigma^2}\right).$$

$$\Pr\left(\sup_{t \leq T} \mathcal{M}_t \geq \frac{1}{\delta_a}\right) \leq \delta_a, \quad (232)$$

$$\|S_{a,t}\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}} \leq \sigma \sqrt{2 \log\left(\frac{\det(V_{a,t}^{\widehat{\Omega}})^{1/2}}{\det(A)^{1/2} \delta_a}\right)} \quad \text{uniformly over } t \leq T.$$

The last row holds with probability at least $1 - \delta_a$. To bound its determinant ratio, use $A = \lambda \widehat{\Omega}^{-1}$ and $\widehat{\Omega} \preceq (1+\varepsilon)\Omega$:

$$\begin{aligned} \frac{\det(V_{a,t}^{\widehat{\Omega}})}{\det(A)} &= \det\left(I + \lambda^{-1} \widehat{\Omega}^{1/2} X_{a,t}^\top X_{a,t} \widehat{\Omega}^{1/2}\right). \\ \text{tr}\left(\lambda^{-1} \widehat{\Omega}^{1/2} X_{a,t}^\top X_{a,t} \widehat{\Omega}^{1/2}\right) &= \lambda^{-1} \text{tr}(X_{a,t} \widehat{\Omega} X_{a,t}^\top) \\ &\leq \frac{1+\varepsilon}{\lambda} \sum_{s \in \mathcal{T}_{a,t}} x_s(a)^\top \Omega x_s(a) \leq \frac{(1+\varepsilon) T L_\Omega^2}{\lambda}. \end{aligned} \quad (233)$$

For any positive-semidefinite $d \times d$ matrix M with eigenvalues $\nu_1, \dots, \nu_d$,

$$\begin{aligned} \det(I + M) &= \prod_{k=1}^d (1 + \nu_k) \leq \left(1 + \sum_{k=1}^d \nu_k\right)^d = (1 + \text{tr } M)^d. \\ 2 \log\left(\frac{\det(V_{a,t}^{\widehat{\Omega}})^{1/2}}{\det(A)^{1/2} \delta_a}\right) &\leq d \log\left(1 + \frac{(1+\varepsilon) T L_\Omega^2}{\lambda}\right) + 2 \log(1/\delta_a). \end{aligned} \quad (234)$$

Taking $\delta_a = \delta_{\text{route}}/K_{\max}$ and applying a union bound over the fixed routing archive gives, with probability at least $1 - \delta_{\text{route}}$, the simultaneous bound

$$\begin{aligned} \|\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a\|_{V_{a,t}^{\widehat{\Omega}}} &\leq \alpha_{\widehat{\Omega},T}(\varepsilon, \delta_{\text{route}}) \\ \left|x^\top (\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a)\right| &\leq \|\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a\|_{V_{a,t}^{\widehat{\Omega}}} \sqrt{x^\top (V_{a,t}^{\widehat{\Omega}})^{-1} x}. \end{aligned} \quad (235)$$

The two rows hold for every candidate a , every $t \leq T$, and every predictable test feature x . The second row is Cauchy–Schwarz in the $V_{a,t}^\Omega$ -inner product. Substituting the preceding simultaneous parameter bound proves the claim. $\square$

Theorem 15 (HRC: calibration-split frozen-covariance guarantee). *Fix a post-calibration routing horizon T . Consider the following HRC procedure.*

For the version combined with the four-term decomposition, draw one routing user U before the post-calibration phase and hold that user fixed for all T routing rounds.

Before routing begins, the algorithm runs a calibration phase on $N = N_T$ calibration candidates that are not used in the subsequent routing archive. For each calibration candidate i , the algorithm collects $m = m_T$ calibration rewards under the exogenous uniform-routing calibration policy:

$$y_{i,j} := r_{i,j} - b(u_{i,j}) = x_{i,j}^\top \theta_i + \xi_{i,j}, \quad j = 1, \dots, m. \quad (236)$$

Let

$$\begin{aligned} X_i &= \begin{bmatrix} x_{i,1}^\top \\ \vdots \\ x_{i,m}^\top \end{bmatrix}, & G_i &= X_i^\top X_i, \\ \tilde{\theta}_i &= \begin{cases} G_i^{-1} X_i^\top y_i, & G_i \text{ invertible,} \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (237)$$

For a deterministic ridge floor $\gamma = \gamma_T > 0$, define

$$\begin{aligned} \hat{\Omega} &= \frac{1}{N} \sum_{i=1}^N \tilde{\theta}_i \tilde{\theta}_i^\top + \gamma I_d. \\ \hat{\Omega}_t &= \hat{\Omega}, \quad t = 1, \dots, T. \end{aligned} \quad (238)$$

The post-calibration routing archive is fixed before routing begins and has cardinality at most $K_{\max}$. The routing UCB rule uses

$$\alpha_T = \alpha_{\hat{\Omega},T}(\varepsilon_*, \delta_{\text{route}}), \quad (239)$$

where $\alpha_{\hat{\Omega},T}$ is defined in Lemma 14, and where ε_ is the deterministic quantity defined below.*

Assume Assumption 8 and the post-calibration routing assumptions of Assumption 6. Let

$$\begin{aligned} \tau_m &= \frac{2\sigma^2}{\kappa_{\text{cal}} m}, & K_m^2 &= S_\Omega^2 + \tau_m, \\ m &\geq \frac{8L_{\text{cal}}^2}{\kappa_{\text{cal}}} \log\left(\frac{dN}{\delta_{\text{gram}}}\right). \end{aligned} \quad (240)$$

Here $\delta_{\text{gram}}, \delta_{\text{cov}}, \delta_{\text{route}} \in (0, 1)$. Define

$$\begin{aligned} \varepsilon_* &= C_{\text{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\text{cov}})}{N}} + \frac{d + \log(2/\delta_{\text{cov}})}{N} \right] \\ &\quad + \tau_m + \gamma \|\Omega^{-1}\|_{\text{op}}. \end{aligned} \quad (241)$$

Computing this radius requires supplied deterministic bounds for $S_\Omega, \sigma, \kappa_{\text{cal}}, L_{\text{cal}}, L_\Omega$, and $\|\Omega^{-1}\|_{\text{op}}$. Without them, the certificate is oracle-tuned. Let $\mathcal{F}_0^{\text{route}}$ contain all calibration data, the fixed post-calibration archive, and the latent parameter of every routing candidate. Assume the post-routing conditional sub-Gaussian bound holds after this sigma-field is added to the routing filtration, and assume $\varepsilon_ < 1$. Define*

$$c_T^\Omega = 1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right), \quad B_{T,0}^{\text{cal}} = 2\alpha_T \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff},T}(\Omega)}. \quad (242)$$

Then, with probability at least $1 - \delta_{\text{gram}} - \delta_{\text{cov}}$,

$$\rho_{\Omega,T} := \max_{1 \leq t \leq T} \left\| \Omega^{-1/2} (\hat{\Omega}_t - \Omega) \Omega^{-1/2} \right\|_{\text{op}} \leq \varepsilon_*. \quad (243)$$

In particular, taking $\delta_{\text{gram}} = \delta_{\text{cov}} = \delta/2$ gives the covariance-error control with probability at least $1 - \delta$.

With probability at least $1 - \delta_{\text{gram}} - \delta_{\text{cov}} - \delta_{\text{route}}$,

$$\sum_{t=1}^T L_t(U) \leq \sqrt{1 + \varepsilon_*} B_{T,0}^{\text{cal}}. \quad (244)$$

Define the uncharged and charged remainder terms and record the resulting expectation bounds:

$$\begin{aligned} \text{Rem}_T^{\text{cal}} &= (\sqrt{1 + \varepsilon_*} - 1) B_{T,0}^{\text{cal}} + (\delta_{\text{gram}} + \delta_{\text{cov}} + \delta_{\text{route}})T. \\ \sum_{t=1}^T \mathbb{E}[L_t(U)] &\leq B_{T,0}^{\text{cal}} + \text{Rem}_T^{\text{cal}}. \\ \text{Rem}_T^{\text{charged}} &= Nm + \text{Rem}_T^{\text{cal}}. \\ \sum_{t=1}^T \mathbb{E}[L_t(U)] + Nm &\leq B_{T,0}^{\text{cal}} + \text{Rem}_T^{\text{charged}}. \end{aligned} \quad (245)$$

The last two rows assign each of the Nm calibration pulls a worst-case unit opportunity-cost charge. For fixed $d, K_{\text{max}}, S_{\Omega}, \sigma, \kappa_{\text{cal}}, L_{\text{cal}}, L_{\Omega}, \lambda$, and $\|\Omega^{-1}\|_{\text{op}}$, and uniformly bounded $\bar{d}_{\text{eff},T}(\Omega)$, take

$$\begin{aligned} N &= \lceil T^{1/5} \rceil, & m &= \lceil T^{1/5} \rceil, & \gamma &= T^{-1/5}, \\ \delta_{\text{gram}} &= \delta_{\text{cov}} = \delta_{\text{route}} = T^{-2}. \end{aligned} \quad (246)$$

For all sufficiently large T , the resulting rates are

$$\begin{aligned} \varepsilon_* &< 1, & \varepsilon_* &= O\left(T^{-1/10} \sqrt{\log T}\right). \\ \alpha_T &= O(\sqrt{d \log T}), & B_{T,0}^{\text{cal}} &= O\left(\sqrt{d K_{\text{max}} \bar{d}_{\text{eff},T}(\Omega) T \log T}\right), \\ \text{Rem}_T^{\text{cal}} &= O\left(T^{2/5} (\log T)^{3/2}\right) = o(\sqrt{T}). \end{aligned} \quad (247)$$

The same rate holds for $\text{Rem}_T^{\text{charged}}$.

Proof. Because $\hat{\Omega}_t = \hat{\Omega}$ for every post-calibration routing round, Lemma 13 gives the event

$$\mathcal{E}_{\text{cov}} := \left\{ \left\| \Omega^{-1/2} (\hat{\Omega} - \Omega) \Omega^{-1/2} \right\|_{\text{op}} \leq \varepsilon_* \right\}, \quad \Pr(\mathcal{E}_{\text{cov}}) \geq 1 - \delta_{\text{gram}} - \delta_{\text{cov}}. \quad (248)$$

On $\mathcal{E}_{\text{cov}}, \rho_{\Omega,T} \leq \varepsilon_*$, and

$$(1 - \varepsilon_*)\Omega \preceq \hat{\Omega} \preceq (1 + \varepsilon_*)\Omega. \quad (249)$$

By Lemma 14, conditional on $\mathcal{F}_0^{\text{route}}$, with probability at least $1 - \delta_{\text{route}}$, the simultaneous routing confidence event

$$\mathcal{E}_{\text{route}} := \left\{ \left| x^\top (\hat{\theta}_{a,t}^{\hat{\Omega}} - \theta_a) \right| \leq \alpha_T \sqrt{x^\top (V_{a,t}^{\hat{\Omega}})^{-1} x} \text{ for all } a, t, x \right\} \quad (250)$$

holds, where x ranges over predictable test features. Its conditional probability bound integrates to an unconditional one. Hence $\mathcal{E} = \mathcal{E}_{\text{cov}} \cap \mathcal{E}_{\text{route}}$ satisfies $\Pr(\mathcal{E}) \geq 1 - \delta_{\text{gram}} - \delta_{\text{cov}} - \delta_{\text{route}}$.

On $\mathcal{E}$, fix t , let $a_t^* \in \arg \max_{a \in A_t(U)} x_t(a)^\top \theta_a$, and define the local confidence width

$$w_{a,t}^{\hat{\Omega}} := \sqrt{x_t(a)^\top (V_{a,t}^{\hat{\Omega}})^{-1} x_t(a)}. \quad (251)$$

The confidence event and the UCB choice of a_t yield

$$\begin{aligned} x_t(a_t^*)^\top \theta_{a_t^*} &\leq x_t(a_t^*)^\top \hat{\theta}_{a_t^*,t}^{\hat{\Omega}} + \alpha_T w_{a_t^*,t}^{\hat{\Omega}} \\ &\leq x_t(a_t)^\top \hat{\theta}_{a_t,t}^{\hat{\Omega}} + \alpha_T w_{a_t,t}^{\hat{\Omega}} \\ &\leq x_t(a_t)^\top \theta_{a_t} + 2\alpha_T w_{a_t,t}^{\hat{\Omega}}. \end{aligned} \quad (252)$$

Thus, because $0 \leq L_t(U) \leq 1$ and $\alpha_T \geq 1$,

$$L_t(U) \leq 2\alpha_T \min\{1, w_{a,t}^{\hat{\Omega}}\}. \quad (253)$$

The covariance event gives $\hat{\Omega}^{-1} \succeq (1 + \varepsilon_*)^{-1} \Omega^{-1}$. Therefore,

$$\begin{aligned} V_{a,t}^{\hat{\Omega}} &= X_{a,t}^\top X_{a,t} + \lambda \hat{\Omega}^{-1} \\ &\succeq X_{a,t}^\top X_{a,t} + \frac{\lambda}{1 + \varepsilon_*} \Omega^{-1} \\ &\succeq \frac{1}{1 + \varepsilon_*} (X_{a,t}^\top X_{a,t} + \lambda \Omega^{-1}) = \frac{1}{1 + \varepsilon_*} V_{a,t}^\Omega. \\ (V_{a,t}^{\hat{\Omega}})^{-1} &\preceq (1 + \varepsilon_*) (V_{a,t}^\Omega)^{-1}. \\ w_{a,t}^{\hat{\Omega}} &\leq \sqrt{1 + \varepsilon_*} w_{a,t}^\Omega, \quad w_{a,t}^\Omega := \sqrt{x_t(a)^\top (V_{a,t}^\Omega)^{-1} x_t(a)}. \end{aligned} \quad (254)$$

Since $\min\{1, cz\} \leq c \min\{1, z\}$ for $c \geq 1$ and $z \geq 0$,

$$L_t(U) \leq 2\alpha_T \sqrt{1 + \varepsilon_*} \min\{1, w_{a,t}^\Omega\}. \quad (255)$$

Summing over t , partitioning selected rounds by candidate, applying Cauchy–Schwarz, and then applying the already-proved Ω -elliptical-potential lemma candidate by candidate yields

$$\sum_{t=1}^T \min\{1, w_{a,t}^\Omega\} \leq \sqrt{2K_{\max} T c_T^\Omega \bar{d}_{\text{eff},T}(\Omega)}. \quad (256)$$

Thus, on $\mathcal{E}$,

$$\begin{aligned} \sum_{t=1}^T L_t(U) &\leq \sqrt{1 + \varepsilon_*} B_{T,0}^{\text{cal}} \\ &= B_{T,0}^{\text{cal}} + (\sqrt{1 + \varepsilon_*} - 1) B_{T,0}^{\text{cal}}. \end{aligned} \quad (257)$$

On $\mathcal{E}^c$, $0 \leq L_t(U) \leq 1$ gives $\sum_{t=1}^T L_t(U) \leq T$. Taking expectations therefore yields

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}[L_t(U)] &\leq B_{T,0}^{\text{cal}} + (\sqrt{1 + \varepsilon_*} - 1) B_{T,0}^{\text{cal}} \\ &\quad + (\delta_{\text{gram}} + \delta_{\text{cov}} + \delta_{\text{route}}) T. \end{aligned} \quad (258)$$

This is the stated bound with $\text{Rem}_T^{\text{cal}}$. Under the declared unit opportunity-cost convention, the Nm calibration pulls add at most Nm , giving $\text{Rem}_T^{\text{charged}}$.

It remains to verify the stated asymptotic schedule. With fixed problem constants,

$$\begin{aligned} \sqrt{\frac{d + \log(2/\delta_{\text{cov}})}{N}} &= O\left(T^{-1/10} \sqrt{\log T}\right), \quad \frac{d + \log(2/\delta_{\text{cov}})}{N} = O(T^{-1/5} \log T), \\ \tau_m &= O(T^{-1/5}), \quad \gamma \|\Omega^{-1}\|_{\text{op}} = O(T^{-1/5}), \\ \varepsilon_* &= O\left(T^{-1/10} \sqrt{\log T}\right). \end{aligned} \quad (259)$$

Since $T^{-1/10} \sqrt{\log T} \rightarrow 0$, there exists $T_0 < \infty$ such that $\varepsilon_* < 1$ for all $T \geq T_0$. The Gram lower-tail condition also holds for all sufficiently large T , because its right-hand side is $O(\log T)$ whereas $m = \lceil T^{1/5} \rceil$.

The explicit radius in Lemma 14, the bound $c_T^\Omega = O(\log T)$, and the uniformly bounded effective-dimension cap give

$$\begin{aligned}
\alpha_T &= O(\sqrt{d \log T}), \\
B_{T,0}^{\text{cal}} &= O\left(\sqrt{d K_{\max} \bar{d}_{\text{eff},T}(\Omega) T \log T}\right). \\
\sqrt{1 + \varepsilon_*} - 1 &= \frac{\varepsilon_*}{\sqrt{1 + \varepsilon_*} + 1} \leq \varepsilon_*. \\
(\sqrt{1 + \varepsilon_*} - 1) B_{T,0}^{\text{cal}} &= O\left(T^{-1/10} \sqrt{\log T}\right) O(\sqrt{T} \log T) = O\left(T^{2/5} (\log T)^{3/2}\right). \\
Nm &= O(T^{2/5}), \\
(\delta_{\text{gram}} + \delta_{\text{cov}} + \delta_{\text{route}})T &= 3T^{-1}. \\
\text{Rem}_T^{\text{charged}} &= O(T^{2/5}) + O\left(T^{2/5} (\log T)^{3/2}\right) + O(T^{-1}) = o(\sqrt{T}),
\end{aligned} \tag{260}$$

because $T^{-1/10} (\log T)^{3/2} \rightarrow 0$. The uncharged remainder is no larger. $\square$

C.1 FINITE-SAMPLE HRC VALIDATION

This experiment implements the finite-sample procedure in Theorem 15, not the heuristic hold-out diagnostic in Appendix K. Set $d = 2$ and let Q rotate the plane by $\pi/6$. Fresh candidate effects are $\theta_i = Q \text{diag}(1, 1/2)v_i$, where v_i has independent Rademacher entries. Thus $\Omega = Q \text{diag}(1, 1/4)Q^\top$, $\mathbb{E}\theta_i = 0$, and $\|\theta_i\|_{\Omega^{-1}} = \sqrt{2}$. Calibration features are $x_{i,j} = s_{i,j}/8$, with independent Rademacher vectors $s_{i,j}$ independent of all effects. Responses use the known baseline $b = 1/2$ and independent Gaussian noise with standard deviation 0.05, without clipping. Since $|x^\top \theta| \leq \sqrt{5/2}/8 < 1/2$, the mean rewards are bounded.

We use $m = 1280$ samples for each of $N \in \{32768, 131072, 524288\}$ calibration candidates, centered ordinary least squares, and ridge floor $\gamma = N^{-1/2}$. The supplied deterministic bounds are $S_\Omega = \sqrt{2}$, $\kappa_{\text{cal}} = 1/256$, $L_{\text{cal}}^2 = L_\Omega^2 = 1/32$, and $\|\Omega^{-1}\|_{\text{op}} = 4$. All three configurations meet the theorem’s Gram sample-size condition. The covariance radius uses $C_{\text{cov}} = 24$, as established in Lemma 10. We set $\delta_{\text{gram}} = \delta_{\text{cov}} = \delta_{\text{route}} = 0.01$ and $\lambda = 0.5$. The resulting ε_* is below one before any observations are drawn, and HRC uses exactly $\alpha_{\hat{\Omega},T}(\varepsilon_*, \delta_{\text{route}})$.

Gaussian calibration is sampled through its exact sufficient statistics. If $C_i \sim \text{Binomial}(m, 1/2)$, the prescribed feature design has

$$G_i = \frac{1}{64} \begin{bmatrix} m & 2C_i - m \\ 2C_i - m & m \end{bmatrix}, \quad X_i^\top y_i \mid G_i, \theta_i \sim \mathcal{N}(G_i \theta_i, \sigma^2 G_i). \tag{261}$$

This generates the same OLS estimator as explicitly drawing every feature-response pair. Singular blocks use the theorem’s zero-estimate rule and are not resampled. The sample count Nm is charged in full; it is a modeled calibration cost, not a count of actual LLM calls.

After calibration, twelve fresh candidates form a fixed routing archive. One independently sampled user feature is held fixed for $T = 2000$ rounds. Candidate effects and routing noise are independent of calibration. Only the chosen candidate’s centered response updates the router; latent means and effects are used solely for evaluation. The estimated covariance is never refreshed. The fixed-user design has effective dimension at most one for each candidate, including under identity regularization.

Across the fifteen calibration runs, no singular block or violation of the tested Gram, covariance, simultaneous parameter-confidence, or routing inequality was observed. Five seeds do not empirically establish the nominal high-probability guarantee; the proof supplies that guarantee. The finite-sample radius decreases as N increases, reducing HRC’s exploration bonus and observed exploration cost. The calibration charge, however, greatly exceeds the routing horizon. At the smallest N , even the uncharged routing bound exceeds T . These results validate the specified statistical procedure and illustrate its conservative constants; they do not establish efficient calibration, a high-dimensional advantage, or the theorem’s asymptotic schedule.

Table 6: Finite-sample HRC validation with independent calibration and a fixed routing archive. Means and standard deviations use five seeds at each calibration size; $d = 2$, $m = 1280$, $K = 12$, and $T = 2000$. The deterministic bound is $\sqrt{1 + \varepsilon_*} B_{T,0}^{\text{cal}}$; Nm is the modeled calibration-observation count and unit opportunity-cost charge, not an LLM-call count. Gaussian calibration is simulated through exact sufficient statistics. The conservative calibration cost precludes a cost-advantage claim.

N	$Nm/10^6$	ρ_Ω	ε_*	$\sum_t L_t$	Bound
32768	41.94	0.0243 ± 0.0009	0.7505	73.6 ± 24.5	3153.8
131072	167.77	0.0116 ± 0.0001	0.3731	39.0 ± 13.0	1873.5
524288	671.09	0.0064 ± 0.0004	0.1864	32.0 ± 11.2	1561.8

D SUPPORTING REWARD AND ARCHIVE RESULTS

This section collects the reward identity and auxiliary base-router results. They explain consequences of exploration bounds and properties of archive growth; they are not the basis of the shared-covariance estimator. References here to observed-reward accounting concern the HRB component study in Section L.

D.1 REWARD DEFINITIONS

An exploration bound gives a reward comparison through an accounting identity. Let the routed value be $R_T = T^{-1} \sum_t \mathbb{E}[\mu(U, a_t(U))]$, $V_{\text{static}}(A_1) = \max_a \mathbb{E}_U[\mu(U, a)]$ the best fixed initial candidate’s value, $\Delta_{\text{route}}(A_1) = \mathbb{E}[m_1(U)] - V_{\text{static}}(A_1)$ the initial archive’s routing gain, and $\bar{G}_T = T^{-1} \sum_t \mathbb{E}[m_t(U) - m_1(U)]$ the gain from archive growth. Then

$$\text{routed value} = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \quad (262)$$

Thus a bound $\sum_t \mathbb{E}[L_t(U)] \leq B_T$ is sufficient for $R_T > V_{\text{static}}(A_1)$ when $\Delta_{\text{route}}(A_1) + \bar{G}_T > B_T/T$. This is a consequence of the exploration guarantee, not an additional assumption used to prove it.

We define the four terms of (262).

The *default reward* is $V_{\text{static}}(A) = \max_{a \in A} \mathbb{E}_U[\mu(U, a)]$: average each candidate’s reward over users, then select the single best candidate for everyone. It measures the performance of a fixed policy without personalization.

The *oracle routing value* is $V_{\text{route}}(A) = \mathbb{E}_U[\max_{a \in A} \mu(U, a)]$: select the best candidate for each user, then average over users. It is the highest population-mean reward available from the fixed archive when user preferences are known.

Their difference is the *routing gain* $\Delta_{\text{route}}(A) = V_{\text{route}}(A) - V_{\text{static}}(A) \geq 0$: the reward available from matching candidates to users. The four-term decomposition separates this structural gain from archive growth and exploration. If users are nearly homogeneous, or one candidate is best for everyone, then $\Delta_{\text{route}}(A) \approx 0$. If some users prefer concise replies and others prefer detailed explanations, $\Delta_{\text{route}}(A)$ can be large. Estimating this gain requires rewards for multiple candidates on held-out users.

These quantities say *why* personalization matters but not *when* it pays off; the dominance condition of Corollary 18 answers the latter by weighing Δ_{route} against the exploration cost.

The *evolution gain* at round t is $G_t(U) = m_t(U) - m_1(U)$, the improvement in the best candidate available to user U since round 1. For example, if the initial best template has mean satisfaction $m_1(U) = 0.70$ and a new template raises it to $m_t(U) = 0.78$, then $G_t(U) = 0.08$. If no better candidate appears, $G_t(U) = 0$. The simulator knows μ and computes $m_t(U)$ exactly; a deployed system would need to estimate it on held-out interactions.

The *exploration cost* is $L_t(U) = m_t(U) - \mu(U, a_t(U))$, the reward lost by selecting a candidate other than the current best. If the best template has mean satisfaction 0.78 and the selected template

has mean 0.60, the cost is 0.18. The simulator knows both means, but the HRB component accounting reports $m_t - r_t$, an observed-reward residual. A deployed system would need estimates of both means and their uncertainty.

The horizon-averaged terms are $\bar{G}_T = \frac{1}{T} \sum_{t=1}^T \mathbb{E}[G_t(U)]$ and $\bar{L}_T = \frac{1}{T} \sum_{t=1}^T \mathbb{E}[L_t(U)]$.

D.2 REWARD ACCOUNTING

The average mean reward of Algorithm 2 over horizon T satisfies (262). Computing all four terms requires the current-archive optimum and the selected candidate's mean reward; observed rewards alone are not enough.

Example. Consider two candidates and two equally likely user types with means $\mu(u_1, a_1) = 0.9$, $\mu(u_2, a_1) = 0.4$, $\mu(u_1, a_2) = 0.4$, $\mu(u_2, a_2) = 0.9$.

Then $V_{\text{static}} = 0.65$ (the best single candidate averages 0.65), while $V_{\text{route}} = 0.9$ (matching each user to their best gives 0.9), so $\Delta_{\text{route}} = 0.25$.

With no archive growth ($\bar{G}_T = 0$), the decomposition gives: routed value = $0.65 + 0.25 - \bar{L}_T$.

Routing is profitable as long as the exploration cost $\bar{L}_T < 0.25$.

When a held-out reward model supplies $\mu(u, a)$, V_{static} and Δ_{route} can be estimated from a held-out user sample: $V_{\text{static}}(A_1) = \max_a N^{-1} \sum_u \mu(u, a)$ and $\Delta_{\text{route}}(A_1) = N^{-1} \sum_u \max_a \mu(u, a) - V_{\text{static}}(A_1)$.

Computing $\bar{G}_T$ and $\bar{L}_T$ also requires the archive means $m_t(u)$ and the selected mean $\mu(u, a_t(u))$. The component accounting uses $m_t(u) - r_t$ instead, which is a different observed-reward residual.

The first result establishes when routing gain is strictly positive:

Lemma 16 (Static-vs-routed value). *For every finite archive A ,*

$$\Delta_{\text{route}}(A) \geq 0, \quad (263)$$

with equality if and only if there exists $a^ \in A$ such that*

$$\mu(U, a^*) = \max_{a \in A} \mu(U, a) \quad \text{almost surely.} \quad (264)$$

Thus $\Delta_{\text{route}}(A) > 0$ exactly when no single candidate is optimal for almost every user.

The proof is in Appendix E.

Reward identity.

Theorem 17 (Exact decomposition on the common initial archive). *Let A_1 be deterministic, finite, and shared by all users. For every round t and every horizon $T \geq 1$,*

$$\mathbb{E}[\mu(U, a_t(U))] = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \quad (265)$$

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mu(U, a_t(U))] = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \quad (266)$$

Proof sketch (full proof in Appendix E). By definition,

$$\begin{aligned} \mu(U, a_t(U)) &= m_t(U) - L_t(U) \\ &= m_1(U) + G_t(U) - L_t(U). \end{aligned}$$

Taking expectations over all model and online randomness and using Lemma 16 gives (265). Averaging over t gives (266). $\square$

Theorem 17 says that the reward achieved by the personalized routing policy decomposes into a structural piece on the initial archive ($V_{\text{static}} + \Delta_{\text{route}}$), an online discovery piece ($\bar{G}_T$), and a negative online learning piece ($-\bar{L}_T$).

In a simulator that logs mean rewards, $\bar{L}_T$ is computed from $m_t(u) - \mu(u, a_t(u))$. The HRB component results instead report $m_t(u) - r_t$, so they do not directly measure $\bar{L}_T$.

Dominance condition.

Corollary 18 (When personalized routing beats the best static initial candidate). *For any horizon T ,*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mu(U, a_t(U))] > V_{\text{static}}(A_1) \iff \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \quad (267)$$

On a fixed archive (no growth, $\bar{G}_T = 0$), this collapses to

$$\Delta_{\text{route}}(A_1) > \bar{L}_T. \quad (268)$$

In other words, personalized routing is beneficial when the routing gain and evolution gain together exceed the exploration cost.

The condition can be estimated when the evaluation protocol records mean rewards for every current candidate. A held-out user sample estimates $\Delta_{\text{route}}(A_1)$; the archive and selected-candidate means estimate $\bar{G}_T$ and $\bar{L}_T$. The reported experiments use the observed-reward residual instead, so they do not verify this condition.

Lemma 16 applies $\mathbb{E}[\max_a X_a] \geq \max_a \mathbb{E}[X_a]$ and characterizes its equality case. The static comparator $V_{\text{static}}(A_1) = \max_a \mathbb{E}_U[\mu(U, a)]$ selects one candidate by population-mean reward. A heuristic choice such as $\arg \max_a q(a)$ uses intrinsic quality instead and need not attain this comparator. The dominance criterion (267) uses the population-mean definition.

This accounting follows one user drawn from a population. It converts a valid exploration bound into a reward comparison for that trajectory model.

Supporting lemmas. Lemma 2, stated with the geometry proofs, supplies the birth-order conditioning used by the bandit bounds. Lemma 19 below controls how often the failure rule can trigger a mutation.

Lemma 19 (Failure-prioritization). *Under the trigger (524), the expected number of reflection mutations across T rounds is bounded by*

$$\mathbb{E}[\text{\#mutations}] \leq \sum_{t=1}^T \Pr(L_t(U) + |\xi_t| > \tau) \leq T \sup_{t \leq T} \Pr(L_t(U) + |\xi_t| > \tau), \quad (269)$$

which controls archive bloat without changing the regret analysis of the router itself.

The failure-prioritization gate is the algorithmic counterpart of the constraint-gated self-evolution line reviewed in Section 6, but here it is embedded inside a user-conditioned routing rule.

Positioning of the decomposition. The comparison to personalized bandits can now be stated in the paper’s notation.

A fixed in-archive oracle comparison folds the structural routing gain into the oracle value, whereas (266) factors it out as $\Delta_{\text{route}}(A_1)$.

A fixed action set also removes the archive-improvement term, whereas HyperRidge keeps $\bar{G}_T$ explicit and uses Lemma 2 to ensure that online archive growth does not depend on future randomness.

Offline prompt-evolution methods can improve the archive, but without user-conditioned routing they do not separate the routing gain $\Delta_{\text{route}}(A_1)$ from the evolution gain $\bar{G}_T$.

Finite-time bound on the exploration cost. Theorem 17 is an exact equality but does not by itself bound the size of the exploration cost $\bar{L}_T$.

To obtain a finite-time lower bound on the router’s average reward, we need a controlled estimate of $\bar{L}_T$.

Rather than re-deriving such a bound from scratch for our particular router, we state it as an external assumption (Assumption 9 below) and substitute it into the decomposition.

Assumption 9 (External exploration-cost bound). *There exists a deterministic sequence B_T such that*

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq B_T \quad \text{for every } T \geq 1. \quad (270)$$

Corollary 20 (Plug-in bound). *Under Assumptions 1–4 and 9,*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mu(U, a_t(U))] \geq V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \quad (271)$$

For the theoretical LinUCB instantiation in Proposition 29, set the UCB multiplier to $\alpha = \beta_T(\delta)$, where

$$\beta_T(\delta) = \sigma \sqrt{2 \log(K_{\max}/\delta) + d \log\left(1 + \frac{TL_x^2}{\lambda d}\right)} + \sqrt{\lambda} S. \quad (272)$$

Then one may take

$$B_T = 2\beta_T(\delta) \sqrt{2TK_{\max}d \log\left(1 + \frac{TL_x^2}{\lambda d}\right)} + T\delta. \quad (273)$$

For $T \geq 2$, with $\delta = 1/T$ and fixed problem parameters, including $K_{\max}$ and d , $B_T/T = O(\log T/\sqrt{T})$. The reported experiments use a fixed heuristic α and are not covered by this certificate.

Discovery-time bound. Discovery is the most fragile term in (262) because it depends on the proposer $\mathcal{F}$.

We bound the number of attempts until a useful candidate appears under an explicit per-attempt repair probability.

Assumption 10 (Repair probability). *There exists $q_\epsilon \in (0, 1]$ such that for each cluster g with current gap $\Delta_g > \epsilon$, each reflection attempt produces an ϵ -improving child with probability at least q_ϵ , independent of past attempts conditional on the trigger firing.*

Theorem 21 (First-useful-specialist time). *Under Assumption 10, after m reflection attempts on cluster g ,*

$$\Pr(\text{some attempt improves cluster } g \text{ by } \epsilon) \geq 1 - (1 - q_\epsilon)^m. \quad (274)$$

The experimental proposer uses privileged information, but the experiments do not estimate q_ϵ for a fixed ϵ . This bound is therefore not empirically tested.

The non-oracle case is left as future work.

Section 5 reports the first three terms in (266) and the separate observed-reward residual $\bar{L}_T^{\text{obs}}$.

Proposition 1 supplies the known-geometry budget used in the comparison below. Both this budget and the ordinary-ridge certificate contain a confidence radius and an elliptical-potential term; a smaller potential alone is insufficient.

D.3 GEOMETRY AND EXPLORATION

The dominance condition is

$$\Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \quad (275)$$

Therefore, any certified reduction in $\bar{L}_T$ lowers the margin needed for personalization to pay. The qualification “certified” matters: the theorem does not claim that an arbitrary covariance estimate, or the periodic estimate in Algorithm 1, reduces exploration.

Write $R_T^{\text{hyper}} = T^{-1} \sum_{t=1}^T \mathbb{E}[\mu(U, a_t(U))]$. Equation (266) applies directly. This section replaces the ordinary dimension-dependent control of $\bar{L}_T = T^{-1} \sum_{t=1}^T \mathbb{E}[L_t(U)]$ with an archive-geometry-dependent control.

As a concrete scale calculation, take $K_{\max} = 16$, $d = 100$, and $\bar{d}_{\text{eff},T}(\Omega) = 9$. If the two confidence radii and logarithmic potential factors are equal, the width-sum part changes by the ratio

$$\frac{\sqrt{K_{\max} \bar{d}_{\text{eff},T}(\Omega) T}}{\sqrt{K_{\max} d T}} = \sqrt{\frac{9}{100}} = 0.3. \quad (276)$$

This is a geometric illustration, not the final budget ratio. The final comparison must also include the two confidence radii.

Corollary 22 (Dominance in more regimes). *Define the structural-and-evolution margin by*

$$M_T = \Delta_{\text{route}}(A_1) + \bar{G}_T. \quad (277)$$

Define the HyperRidge exploration budget by

$$B_T^{\text{hyper}} = B_T^{\Omega}(\delta) + \delta T. \quad (278)$$

If

$$M_T > \frac{B_T^{\text{hyper}}}{T}, \quad (279)$$

then HyperRidge satisfies the dominance condition

$$R_T^{\text{hyper}} > V_{\text{static}}(A_1). \quad (280)$$

Moreover, compared with an ordinary LinUCB budget

$$B_T^{\text{LinUCB}} = B_T^{\text{lin}}(\delta) + \delta T, \quad (281)$$

HyperRidge certifies all margins in the additional interval

$$\frac{B_T^{\text{hyper}}}{T} < M_T \leq \frac{B_T^{\text{LinUCB}}}{T}, \quad (282)$$

whenever

$$B_T^{\text{hyper}} < B_T^{\text{LinUCB}}. \quad (283)$$

The proof is in Appendix D.4.

This corollary states the comparison without assuming its outcome. HRK certifies more margins only when its complete finite-time budget is smaller. The HRC budget may be substituted separately from Theorem 15; no claim is made for HRP.

D.4 PROOF OF THE HYPERRIDGE DOMINANCE COROLLARY

Proof of Corollary 22. Substituting Proposition 1 into the inherited decomposition gives

$$\begin{aligned} R_T^{\text{hyper}} &= V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T \\ &\geq V_{\text{static}}(A_1) + M_T - \frac{B_T^{\text{hyper}}}{T}. \end{aligned} \quad (284)$$

Condition (279) implies (280). The interval claim follows by comparing (278) and (281). $\square$

D.5 EQUIVALENT SUMMARY EXPRESSIONS

Writing the identity directly in terms of average reward gives

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \quad (285)$$

The corresponding strict comparison with fixed deployment is

$$\Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T, \quad (286)$$

The complete HRK high-probability budget used in this comparison is

$$B_T^{\Omega}(\delta) = 2\alpha_{\Omega,T}(\delta) \sqrt{2K_{\max} T c_T^{\Omega} \bar{d}_{\text{eff},T}(\Omega)}, \quad (287)$$

E PRELIMINARIES AND THE EXACT DECOMPOSITION

This appendix defines the probability model and proves the exact four-term decomposition.

The proof is an accounting identity; it does not use a contextual-bandit regret theorem and does not assume that reflection is realistic.

E.1 PROBABILITY SPACE AND ONLINE OBJECTS

All random variables are defined on a common probability space $(\Xi, \mathcal{F}, \mathbb{P})$, and expectations average over both the population draw and the online trajectory. The evaluation user is $U \sim \Pi$ with values in $\mathcal{U}$, and candidates belong to $\mathcal{A}$. The common initial archive is $A_1 \subseteq \mathcal{A}$; before round t , the current archive is $A_t(U) \subseteq \mathcal{A}$, and the router selects $a_t(U) \in A_t(U)$. Mean and observed rewards satisfy

$$\mu : \mathcal{U} \times \mathcal{A} \rightarrow [0, 1], \quad r_t = \mu(U, a_t(U)) + \xi_t. \quad (288)$$

For the later plug-in bounds, let $\mathcal{F}_0 \subseteq \dots \subseteq \mathcal{F}_T$ be the online filtration and $x_t(a) \in \mathbb{R}^d$ the feature vector for candidate a . The exact decomposition uses only bounded rewards and the archive conditions; the noise and feature assumptions are recorded for the later results.

Assumption 11 (A1: bounded mean reward). *For every user u and candidate a , $0 \leq \mu(u, a) \leq 1$.*

Assumption 12 (A2: finite common initial archive). *The initial archive is deterministic, nonempty, and finite: $1 \leq |A_1| < \infty$.*

Assumption 13 (A3: finite append-only online archives). *For every round, $1 \leq |A_t(U)| < \infty$ almost surely, and the archive is append-only: $A_t(U) \subseteq A_{t+1}(U)$.*

Assumption 14 (A4: predictable archive). *Before the router chooses at round t , the archive is measurable with respect to the past: $\sigma(A_t(U)) \subseteq \mathcal{F}_{t-1}$.*

Assumption 15 (A5: centered conditionally sub-Gaussian noise). *For every $\lambda \in \mathbb{R}$,*

$$\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0, \quad \mathbb{E}[e^{\lambda \xi_t} \mid \mathcal{F}_{t-1}] \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right). \quad (289)$$

Assumption 16 (A6: bounded features). *Every available candidate satisfies $\|x_t(a)\|_2 \leq L$.*

E.2 VALUES AND ACCOUNTING TERMS

For a deterministic finite archive A , define its static value, routed value, and heterogeneity gain by

$$\begin{aligned} V_{\text{static}}(A) &= \max_{a \in A} \mathbb{E}[\mu(U, a)], & V_{\text{route}}(A) &= \mathbb{E}\left[\max_{a \in A} \mu(U, a)\right], \\ \Delta_{\text{route}}(A) &= V_{\text{route}}(A) - V_{\text{static}}(A). \end{aligned} \quad (290)$$

For the online archive, define

$$\begin{aligned} m_t(U) &= \max_{a \in A_t(U)} \mu(U, a), & G_t(U) &= m_t(U) - m_1(U), & L_t(U) &= m_t(U) - \mu(U, a_t(U)), \\ \bar{G}_T &= \frac{1}{T} \sum_{t=1}^T \mathbb{E}[G_t(U)], & \bar{L}_T &= \frac{1}{T} \sum_{t=1}^T \mathbb{E}[L_t(U)]. \end{aligned} \quad (291)$$

Proposition 23 (Integrability and signs of the accounting terms). *Under Assumptions 11–13, the current-archive optimum satisfies*

$$m_t(U), G_t(U), L_t(U) \in [0, 1], \quad \mathbb{E}|m_t(U)| + \mathbb{E}|G_t(U)| + \mathbb{E}|L_t(U)| \leq 3. \quad (292)$$

Proof. Bounded rewards give $0 \leq m_t(U) \leq 1$. Since $a_t(U) \in A_t(U)$, and since append-only growth gives $A_1 \subseteq A_t(U)$,

$$\begin{aligned} 0 &\leq m_t(U) - \mu(U, a_t(U)) = L_t(U) \leq 1, \\ 0 &\leq m_t(U) - m_1(U) = G_t(U) \leq 1. \end{aligned} \quad (293)$$

The integrability bound follows by summing these three pointwise bounds. $\square$

E.3 SUPPORTING LEMMAS FOR THE FOUR-TERM DECOMPOSITION

Lemma 24 (Static-versus-routed value). *For every deterministic, finite, nonempty archive A , $\Delta_{\text{route}}(A) \geq 0$. Equality holds if and only if some $a^* \in A$ is pointwise optimal almost surely:*

$$\mu(U, a^*) = \max_{a \in A} \mu(U, a) \quad \text{almost surely.} \quad (294)$$

In particular, the inequality is strict if every candidate is suboptimal with positive probability:

$$\forall a \in A, \quad \mathbb{P}\left(\mu(U, a) < \max_{b \in A} \mu(U, b)\right) > 0. \quad (295)$$

Proof. Because A is finite, choose $a^* \in \arg \max_{a \in A} \mathbb{E}[\mu(U, a)]$. Pointwise maximality gives

$$\Delta_{\text{route}}(A) = \mathbb{E}\left[\max_{a \in A} \mu(U, a) - \mu(U, a^*)\right] \geq 0. \quad (296)$$

The integrand is nonnegative, so its expectation is zero exactly when it is zero almost surely. This proves the equality condition. Conversely, an almost-surely pointwise-optimal candidate attains both $V_{\text{static}}(A)$ and $V_{\text{route}}(A)$. Under the stated strictness condition, the integrand is positive with positive probability for every possible static maximizer, so its expectation is positive. $\square$

Lemma 25 (Pointwise routed-value accounting). *For every round t , the preceding definitions imply*

$$\mu(U, a_t(U)) = m_1(U) + G_t(U) - L_t(U). \quad (297)$$

Proof. Rearrange $L_t(U) = m_t(U) - \mu(U, a_t(U))$ and substitute $m_t(U) = m_1(U) + G_t(U)$. $\square$

E.4 EXACT REWARD IDENTITY

Theorem 26 (Exact decomposition over the common initial archive). *This restates Theorem 17. Let $U \sim \Pi$, let $1 \leq |A_1| < \infty$, and suppose $a_t(U) \in A_t(U)$. Then, for every round t ,*

$$\mathbb{E}[\mu(U, a_t(U))] = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \quad (298)$$

Define

$$R_T = \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mu(U, a_t(U))], \quad \bar{G}_T = \frac{1}{T} \sum_{t=1}^T \mathbb{E}[G_t(U)], \quad \bar{L}_T = \frac{1}{T} \sum_{t=1}^T \mathbb{E}[L_t(U)]. \quad (299)$$

Then

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \quad (300)$$

Proof. Proposition 23 ensures integrability. Taking expectations in Lemma 25 gives

$$\begin{aligned} \mathbb{E}[\mu(U, a_t(U))] &= \mathbb{E}[m_1(U)] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)] \\ &= V_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)] \\ &= V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \end{aligned} \quad (301)$$

Summing over t and dividing by T proves (300). $\square$

The theorem concerns mean rewards. Under Assumption 15, the same expectation governs unclipped observed rewards because

$$\begin{aligned} \mathbb{E}[r_t] &= \mathbb{E}[\mu(U, a_t(U))] + \mathbb{E}[\xi_t] \\ &= \mathbb{E}[\mu(U, a_t(U))]. \end{aligned} \quad (302)$$

This link does not apply to the experimental simulator’s clipped observations in (521) when μ denotes the pre-clipping mean. The following General-environment row instead checks the separate observed-reward accounting identity. Its measured terms are

$$V_{\text{static}}(A_1) = 0.570, \quad \Delta_{\text{route}}(A_1) = 0.280, \quad \bar{G}_T = 0.051, \quad \bar{L}_T^{\text{obs}} = 0.036. \quad (303)$$

Therefore,

$$\begin{aligned} V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} &= 0.570 + 0.280 + 0.051 - 0.036 \\ &= 0.865, \end{aligned} \quad (304)$$

which matches the reported observed routed value 0.864 up to table rounding. Because $\bar{L}_T^{\text{obs}}$ is defined from the same routed rewards, this is an algebraic consistency check rather than empirical validation of Theorem 26.

F EXPLORATION-COST BOUNDS

This section proves the supporting plug-in statement for the exploration cost.

The proof does not assert a new regret rate; it shows how any valid bound on the cumulative exploration cost plugs into the four-term decomposition of Theorem 26.

Assumption 17 (External expected exploration-cost bound). *For $T \in \mathbb{N}$, there is a deterministic $B_T \geq 0$ such that*

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq B_T. \quad (305)$$

The next lemma records the conditioning fact that makes growing archives compatible with standard martingale arguments, provided children are appended only after the current reward is observed.

Lemma 27 (Predictable archive conditioning). *Let the pre-action sigma-field be*

$$\mathcal{H}_{t-1} = \sigma(U, A_1, \{a_s(U), r_s, A_{s+1}(U) : 1 \leq s < t\}). \quad (306)$$

Assume $A_t(U)$ is $\mathcal{H}_{t-1}$ -measurable and at most $K_{\max}$ candidates appear by time T . Index them by deterministic birth-order slots $k \leq K_{\max}$, with stopping-time birth τ_k and $\tau_k = \infty$ for an unfilled slot. On $\{\tau_k \leq T\}$, the identity $a^{(k)}$ and its feature rule are $\mathcal{H}_{\tau_k-1}$ -measurable. Define the padded feature and selection indicator

$$\tilde{x}_{t,k} = \mathbf{1}\{\tau_k \leq t\} \psi(U, a^{(k)}), \quad I_{t,k} = \mathbf{1}\{\tau_k \leq t, a_t(U) = a^{(k)}\}. \quad (307)$$

Assume both are $\mathcal{H}_{t-1}$ -measurable: they are zero before birth, and the slot identity is in the pre-action history afterward. Also assume

$$\mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] = 0, \quad \mathbb{E}[\exp(s\xi_t) \mid \mathcal{H}_{t-1}] \leq \exp\left(\frac{s^2\sigma^2}{2}\right) \quad \text{for all } s \in \mathbb{R}. \quad (308)$$

Then, for each deterministic slot k ,

$$Z_{t,k} = I_{t,k} \tilde{x}_{t,k} \xi_t, \quad M_{n,k} = \sum_{t=1}^n Z_{t,k} \quad (309)$$

defines a vector-valued martingale with respect to the post-round filtration.

Proof. Fix k . Since $I_{t,k} \tilde{x}_{t,k}$ is $\mathcal{H}_{t-1}$ -measurable,

$$\mathbb{E}[Z_{t,k} \mid \mathcal{H}_{t-1}] = I_{t,k} \tilde{x}_{t,k} \mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] = 0. \quad (310)$$

Moreover, for any direction v ,

$$\mathbb{E}[\exp(s v^\top Z_{t,k}) \mid \mathcal{H}_{t-1}] \leq \exp\left(\frac{s^2\sigma^2}{2} I_{t,k} (v^\top \tilde{x}_{t,k})^2\right). \quad (311)$$

Thus the increments are conditionally mean-zero (and sub-Gaussian in every predictable direction), so their partial sums form the claimed martingale. $\square$

The following elementary conversion is useful because many router analyses produce high-probability regret bounds, whereas Theorem 26 is stated in expectation.

Lemma 28 (From high-probability tax to expected tax). *Suppose $0 \leq \mu(U, a) \leq 1$, let $S_T^L = \sum_{t=1}^T L_t(U)$, and let $\tilde{B}_T \geq 0$ and $\delta \in (0, 1)$ satisfy*

$$\mathbb{P}(S_T^L \leq \tilde{B}_T) \geq 1 - \delta. \quad (312)$$

Then

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq \tilde{B}_T + T\delta. \quad (313)$$

Proof. Because $a_t(U) \in A_t(U)$ and rewards lie in $[0, 1]$, $0 \leq L_t(U) \leq 1$, hence $0 \leq S_T^L \leq T$. For $\mathcal{E}_T^L = \{S_T^L \leq \tilde{B}_T\}$,

$$\begin{aligned} \mathbb{E}[S_T^L] &= \mathbb{E}[S_T^L \mathbf{1}\{\mathcal{E}_T^L\}] + \mathbb{E}[S_T^L \mathbf{1}\{(\mathcal{E}_T^L)^c\}] \\ &\leq \tilde{B}_T + T \mathbb{P}((\mathcal{E}_T^L)^c) \\ &\leq \tilde{B}_T + T\delta. \end{aligned} \quad (314)$$

Linearity of expectation completes the proof. $\square$

We next give one sufficient route to Assumption 17.

This proposition is not a new contribution of the paper; it is a standard disjoint-linear UCB certificate written only to clarify what must be true for the plug-in theorem to inherit a familiar rate.

Proposition 29 (Disjoint-linear certificate for the external tax bound). *Let $d \in \mathbb{N}$ and $\mathcal{A}_T(U) = \bigcup_{t=1}^T A_t(U)$, with $|\mathcal{A}_T(U)| \leq K_{\max}$. Use the deterministic birth-order slots from Lemma 27. Let $(\mathcal{G}_t)$ be an analysis filtration with $\mathcal{H}_t \subseteq \mathcal{G}_t$. Each τ_k is an $\mathcal{H}_t$ -stopping time; the candidate identity and feature rule are $\mathcal{H}_{\tau_k-1}$ -measurable, while its latent parameter is fixed at birth and $\mathcal{G}_{\tau_k-1}$ -measurable. The zero-padded process is $\mathcal{G}_t$ -predictable. The router remains $\mathcal{H}_t$ -adapted and does not observe the latent parameter.*

For every realized candidate a , suppose

$$x_t(a) = \psi(U, a) \in \mathbb{R}^d, \quad \|x_t(a)\|_2 \leq L_x, \quad \|\theta_a^*\|_2 \leq S, \quad \mu(U, a) = x_t(a)^\top \theta_a^*. \quad (315)$$

Let $r_t = \mu(U, a_t(U)) + \xi_t$, where the centered sub-Gaussian conditions of Lemma 27 also hold conditionally on $\mathcal{G}_{t-1}$. For $\lambda \geq L_x^2 > 0$, define

$$\begin{aligned} V_{t,a} &= \lambda I_d + \sum_{s=1}^{t-1} \mathbf{1}\{a_s(U) = a\} x_s(a) x_s(a)^\top, \\ \hat{\theta}_{t,a} &= V_{t,a}^{-1} \sum_{s=1}^{t-1} \mathbf{1}\{a_s(U) = a\} x_s(a) r_s. \end{aligned} \quad (316)$$

For $\delta \in (0, 1)$, let

$$\beta_T(\delta) = \sigma \sqrt{2 \log\left(\frac{K_{\max}}{\delta}\right) + d \log\left(1 + \frac{TL_x^2}{\lambda d}\right)} + \sqrt{\lambda} S. \quad (317)$$

If the router uses the disjoint UCB rule

$$a_t(U) \in \arg \max_{a \in A_t(U)} \left\{ x_t(a)^\top \hat{\theta}_{t,a} + \beta_T(\delta) \sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)} \right\} \quad (318)$$

then, with probability at least $1 - \delta$,

$$\sum_{t=1}^T L_t(U) \leq B_T^{\text{lin}}(\delta), \quad (319)$$

where

$$B_T^{\text{lin}}(\delta) = 2\beta_T(\delta) \sqrt{2TK_{\max}d \log\left(1 + \frac{TL_x^2}{\lambda d}\right)}. \quad (320)$$

Consequently,

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq B_T^{\text{lin}}(\delta) + T\delta. \quad (321)$$

Proof. Write $a_t = a_t(U)$. For each deterministic slot k , let

$$M_{t,k} = \sum_{s=1}^{t-1} \mathbf{1}\{a_s = a^{(k)}\} x_s(a^{(k)}) \xi_s, \quad (322)$$

with zero summands before τ_k . Conditional on the slot's birth history, the noise and measurability assumptions make this a vector-valued martingale in the analysis filtration. The self-normalized martingale inequality gives

$$\mathbb{P}\left(\exists t \leq T : \|M_{t,k}\|_{V_{t,a^{(k)}}^{-1}} > \sigma \sqrt{2 \log\left(\frac{K_{\max}}{\delta}\right) + \log\left(\frac{\det V_{t,a^{(k)}}}{\det(\lambda I_d)}\right)}\right) \leq \frac{\delta}{K_{\max}}. \quad (323)$$

The trace bound gives, uniformly in $t \leq T$,

$$\begin{aligned} \log\left(\frac{\det V_{t,a^{(k)}}}{\det(\lambda I_d)}\right) &\leq d \log\left(1 + \frac{1}{\lambda d} \sum_{s=1}^{t-1} \mathbf{1}\{a_s = a^{(k)}\} \|x_s(a^{(k)})\|_2^2\right) \\ &\leq d \log\left(1 + \frac{TL_x^2}{\lambda d}\right). \end{aligned} \quad (324)$$

Union-bounding over the deterministic slots yields, with probability at least $1 - \delta$,

$$\forall t \leq T, \forall a \in A_t(U) : \left| x_t(a)^\top (\hat{\theta}_{t,a} - \theta_a^*) \right| \leq \beta_T(\delta) \sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)}. \quad (325)$$

On this event, write $\hat{\mu}_t(a) = x_t(a)^\top \hat{\theta}_{t,a}$, $z_t(a) = \sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)}$, and choose

$$a_t^* \in \arg \max_{a \in A_t(U)} \mu(U, a). \quad (326)$$

Confidence, the UCB choice, and confidence again give

$$\begin{aligned} \mu(U, a_t^*) &\leq \hat{\mu}_t(a_t^*) + \beta_T(\delta) z_t(a_t^*) \\ &\leq \hat{\mu}_t(a_t) + \beta_T(\delta) z_t(a_t) \\ &\leq \mu(U, a_t) + 2\beta_T(\delta) z_t(a_t). \end{aligned} \quad (327)$$

Thus $L_t(U) \leq 2\beta_T(\delta) z_t(a_t)$. Also $z_t(a_t)^2 \leq \|x_t(a_t)\|_2^2 / \lambda \leq 1$. The determinant recursion for each arm and a sum over at most $K_{\max}$ arms give

$$\begin{aligned} \sum_{t=1}^T \mathbf{1}\{a_t = a\} z_t(a)^2 &\leq 2 \log\left(\frac{\det V_{T+1,a}}{\det(\lambda I_d)}\right) \leq 2d \log\left(1 + \frac{TL_x^2}{\lambda d}\right), \\ \sum_{t=1}^T z_t(a_t)^2 &\leq 2K_{\max} d \log\left(1 + \frac{TL_x^2}{\lambda d}\right). \end{aligned} \quad (328)$$

Combining this with Cauchy–Schwarz yields

$$\begin{aligned} \sum_{t=1}^T L_t(U) &\leq 2\beta_T(\delta) \sum_{t=1}^T z_t(a_t) \\ &\leq 2\beta_T(\delta) \sqrt{2TK_{\max} d \log\left(1 + \frac{TL_x^2}{\lambda d}\right)} \\ &= B_T^{\text{lin}}(\delta). \end{aligned} \quad (329)$$

Lemma 28 gives the expected bound. $\square$

Theorem 30 (Conditional plug-in regret for HyperRidge-Base). *Use R_T , $\bar{G}_T$, and $\bar{L}_T$ from Theorem 26. If*

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq B_T \quad (330)$$

then

$$R_T \geq V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \quad (331)$$

Proof. The assumption gives $\bar{L}_T \leq B_T/T$. Substitute this into the exact decomposition in Theorem 26. $\square$

Corollary 31 (Plug-in dominance condition). *Under Theorem 30, personalized routing beats the best static initial candidate whenever*

$$\Delta_{\text{route}}(A_1) + \bar{G}_T > \frac{B_T}{T}. \quad (332)$$

In that case, $R_T > V_{\text{static}}(A_1)$. On a fixed archive, where $\bar{G}_T = 0$, the condition becomes

$$\Delta_{\text{route}}(A_1) > \frac{B_T}{T}. \quad (333)$$

Proof. Subtract $V_{\text{static}}(A_1)$ from the bound in Theorem 30. The stated condition makes the remaining right-hand side positive. $\square$

Corollary 32 (HRB plug-in instance: disjoint LinUCB). *If the conditions of Proposition 29 hold, then*

$$R_T \geq V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T^{\text{lin}}(\delta) + T\delta}{T}. \quad (334)$$

For $\delta = 1/T$, and with $K_{\max}$, d , L_x , λ , S , and σ fixed, the tax term satisfies

$$\frac{B_T^{\text{lin}}(1/T) + 1}{T} = O\left(\frac{\log T}{\sqrt{T}}\right) \quad (335)$$

Proof. Apply Theorem 30 with $B_T = B_T^{\text{lin}}(\delta) + T\delta$. For $T \geq 2$ and $\delta = 1/T$, the definition of B_T^{lin} gives

$$\begin{aligned} \frac{B_T^{\text{lin}}(1/T) + 1}{T} &= \frac{2\beta_T(1/T) \sqrt{2TK_{\max}d \log\left(1 + \frac{TL_x^2}{\lambda d}\right)} + 1}{T} \\ &= O\left(\frac{\log T}{\sqrt{T}}\right), \end{aligned} \quad (336)$$

under fixed problem parameters. $\square$

As a scale check, suppose a valid certificate on the workspace environment gave $B_T/T = 0.050$. Table 10 reports $V_{\text{static}}(A_1) = 0.577$, $\Delta_{\text{route}}(A_1) = 0.124$, and $\bar{G}_T = 0.041$. The plug-in bound would then be

$$\begin{aligned} V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T} &= 0.577 + 0.124 + 0.041 - 0.050 \\ &= 0.692. \end{aligned} \quad (337)$$

This exceeds the static comparator by 0.115. The empirical exploration cost in Section 5 is diagnostic, not a certificate; using it only as a scale check gives $0.577 + 0.124 + 0.041 - 0.043 = 0.699$, which matches the reported routed value up to rounding. Proposition 29 states the additional conditions needed to replace that diagnostic curve by a certified B_T .

G DISCOVERY TIME AND ARCHIVE GROWTH

This appendix proves the discovery-time result in Section 4.

The theorem is not a contraction theorem for language-model reflection; it is a conditional stopping-time statement for the first useful child under an explicit repair-probability assumption.

Assumption 18 (Conditional repair probability for an active cluster). *Let $\mathcal{G} = \{1, \dots, K_{\text{cl}}\}$, let $g(U) \in \mathcal{G}$ denote the user’s cluster, and assume every cluster considered below has positive mass*

$$\pi_g := \mathbb{P}(g(U) = g) > 0. \quad (338)$$

For fixed g , draw $U_g \sim \Pi(\cdot \mid g(U) = g)$. Fix $\epsilon > 0$ and $q_\epsilon \in (0, 1]$. Let $\tau_{g,j}$ be the time of the j -th active repair attempt, with $\tau_{g,j} = \infty$ if it never occurs. The objects below are defined on $\{\tau_{g,j} < \infty\}$; any result invoking the first m attempts assumes they occur almost surely. Let $\mathcal{H}_{g,j}$ denote the information available immediately before drawing the child, and set

$$A_{g,j}^- = A_{\tau_{g,j}}(U_g), \quad A_{g,j}^+ = A_{\tau_{g,j}+1}(U_g). \quad (339)$$

Define the success indicator

$$I_{g,j}^{(\epsilon)} = \begin{cases} \mathbf{1}\left\{\max_{a \in A_{g,j}^+} \mu(U_g, a) - \max_{a \in A_{g,j}^-} \mu(U_g, a) \geq \epsilon\right\}, & \tau_{g,j} < \infty, \\ 0, & \tau_{g,j} = \infty. \end{cases} \quad (340)$$

The event that the first j attempts all fail is

$$F_{g,j}^{(\epsilon)} = \bigcap_{\ell=1}^j \{I_{g,\ell}^{(\epsilon)} = 0\}. \quad (341)$$

Assume $F_{g,j-1}^{(\epsilon)} \in \mathcal{H}_{g,j}$ and, before the first success,

$$\mathbb{P}\left(I_{g,j}^{(\epsilon)} = 1 \mid \mathcal{H}_{g,j}\right) \geq q_\epsilon \quad \text{on } F_{g,j-1}^{(\epsilon)}. \quad (342)$$

The word “active” means that the attempt is made only while the current cluster gap exceeds the target threshold.

The proof uses only (342); it does not use a contraction factor.

Theorem 33 (First-useful-specialist time). *Fix $g \in \mathcal{G}$, $\epsilon > 0$, and $m \in \mathbb{N}$, and define*

$$S_{g,m}^{(\epsilon)} = \left\{ \sum_{j=1}^m I_{g,j}^{(\epsilon)} \geq 1 \right\}. \quad (343)$$

If Assumption 18 holds for the first m active attempts, then

$$\mathbb{P}\left(F_{g,m}^{(\epsilon)}\right) \leq (1 - q_\epsilon)^m, \quad \mathbb{P}\left(S_{g,m}^{(\epsilon)}\right) \geq 1 - (1 - q_\epsilon)^m. \quad (344)$$

Proof. Set $F_{g,0}^{(\epsilon)} = \Xi$. Since $F_{g,j-1}^{(\epsilon)} \in \mathcal{H}_{g,j}$, the tower property and (342) give

$$\begin{aligned} \mathbb{P}\left(F_{g,j}^{(\epsilon)}\right) &= \mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\} \mathbf{1}\{I_{g,j}^{(\epsilon)} = 0\}\right] \\ &= \mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\} \mathbb{P}\left(I_{g,j}^{(\epsilon)} = 0 \mid \mathcal{H}_{g,j}\right)\right] \\ &\leq (1 - q_\epsilon) \mathbb{P}\left(F_{g,j-1}^{(\epsilon)}\right). \end{aligned} \quad (345)$$

Iteration yields

$$\mathbb{P}\left(F_{g,m}^{(\epsilon)}\right) \leq (1 - q_\epsilon)^m. \quad (346)$$

Taking complements proves the success bound. $\square$

The theorem is intentionally conditional. The experimental simulator uses privileged peak-category information, but its child qualities remain random and the runs do not estimate q_ϵ for a fixed ϵ . Thus Theorem 33 states the quantity that a reflection study must estimate; the reported experiments do not test its assumption.

Corollary 34 (Attempt complexity for one useful specialist). *For $\delta \in (0, 1)$, the exact number of active attempts sufficient for success probability at least $1 - \delta$ is*

$$m_\delta^{\text{exact}} = \begin{cases} 1, & q_\epsilon = 1, \\ \left\lceil \frac{\log \delta}{\log(1 - q_\epsilon)} \right\rceil, & 0 < q_\epsilon < 1. \end{cases} \quad (347)$$

A simpler sufficient choice is

$$m_\delta = \left\lceil \frac{1}{q_\epsilon} \log\left(\frac{1}{\delta}\right) \right\rceil. \quad (348)$$

Either choice satisfies $\mathbb{P}(S_{g,m}^{(\epsilon)}) \geq 1 - \delta$.

Proof. If $q_\epsilon = 1$, one attempt succeeds almost surely. If $0 < q_\epsilon < 1$, the exact choice satisfies

$$m_\delta^{\text{exact}} \log(1 - q_\epsilon) \leq \log \delta, \quad (1 - q_\epsilon)^{m_\delta^{\text{exact}}} \leq \delta. \quad (349)$$

For the simpler choice, $1 - q_\epsilon \leq e^{-q_\epsilon}$, so

$$(1 - q_\epsilon)^{m_\delta} \leq e^{-q_\epsilon m_\delta} \leq \delta. \quad (350)$$

Apply Theorem 33. $\square$

For example, if a non-oracle proposer had $q_\epsilon = 0.20$ and $\delta = 0.05$, then

$$m_\delta = \left\lceil \frac{1}{0.20} \log(20) \right\rceil = 15, \quad 1 - (1 - 0.20)^{15} = 1 - 0.8^{15} \approx 0.965. \quad (351)$$

Corollary 35 (Cluster coverage by active attempts). *Let $\mathcal{G}_{\text{act}} \subseteq \mathcal{G}$ contain K_{act} active clusters. Suppose cluster g has repair probability $q_{\epsilon,g} \in (0, 1]$ and receives $m_g \in \mathbb{N}$ active attempts. Define full coverage by*

$$C_{\text{all}}^{(\epsilon)} = \bigcap_{g \in \mathcal{G}_{\text{act}}} S_{g,m_g}^{(\epsilon)}. \quad (352)$$

Then

$$\mathbb{P}(C_{\text{all}}^{(\epsilon)}) \geq 1 - \sum_{g \in \mathcal{G}_{\text{act}}} (1 - q_{\epsilon,g})^{m_g}. \quad (353)$$

If $q_{\min} = \min_{g \in \mathcal{G}_{\text{act}}} q_{\epsilon,g}$ and every cluster receives m attempts, then

$$\mathbb{P}(C_{\text{all}}^{(\epsilon)}) \geq 1 - K_{\text{act}}(1 - q_{\min})^m. \quad (354)$$

A sufficient budget for coverage probability at least $1 - \delta$ is

$$m = \left\lceil \frac{1}{q_{\min}} \log\left(\frac{K_{\text{act}}}{\delta}\right) \right\rceil. \quad (355)$$

Proof. Theorem 33 and a union bound give

$$\begin{aligned} \mathbb{P}\left(\left(C_{\text{all}}^{(\epsilon)}\right)^c\right) &\leq \sum_{g \in \mathcal{G}_{\text{act}}} \mathbb{P}\left(\left(S_{g,m_g}^{(\epsilon)}\right)^c\right) \\ &\leq \sum_{g \in \mathcal{G}_{\text{act}}} (1 - q_{\epsilon,g})^{m_g}. \end{aligned} \quad (356)$$

Taking complements proves the first claim. For uniform m ,

$$\begin{aligned} \sum_{g \in \mathcal{G}_{\text{act}}} (1 - q_{\epsilon,g})^m &\leq K_{\text{act}}(1 - q_{\min})^m, \\ K_{\text{act}}(1 - q_{\min})^m &\leq K_{\text{act}}e^{-q_{\min}m} \leq \delta. \end{aligned} \quad (357)$$

$\square$

For $K_{\text{act}} = 3$, $q_{\min} = 0.20$, and $\delta = 0.05$, the sufficient uniform budget is

$$m = \lceil 5 \log(60) \rceil = 21. \quad (358)$$

Proposition 36 (Rollout and archive-size complexity under the implemented gate). *Let $T, K_{\max} \in \mathbb{N}$, $K_1 = |A_1|$, and $\kappa \in \mathbb{N}_0$. For user u , let $J_t(u) \in \{0, 1\}$ indicate a reflection attempt after round t , and set $M_T(u) = \sum_{t=1}^T J_t(u)$. Assume the archive changes only when the gate inserts one child, obeys the cap, and enforces the cooldown:*

$$\begin{aligned} |A_{t+1}(u)| &= |A_t(u)| + J_t(u), & |A_t(u)| &\leq K_{\max}, \\ J_s(u) &= J_t(u) = 1, \quad s < t &\implies t - s &\geq \kappa + 1. \end{aligned} \quad (359)$$

Then

$$M_T(u) \leq \min \left\{ K_{\max} - K_1, 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \quad (360)$$

Moreover, $|A_{T+1}(u)| \leq K_{\max}$ and $\sum_{t=1}^T |A_t(u)| \leq TK_{\max}$. If each attempt costs $c_{\text{ref}} \geq 0$ additional child evaluations, then

$$T + c_{\text{ref}} M_T(u) \leq T + c_{\text{ref}} \min \left\{ K_{\max} - K_1, 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \quad (361)$$

Proof. Telescoping the archive updates gives

$$|A_{T+1}(u)| = K_1 + M_T(u) \leq K_{\max}, \quad (362)$$

so $M_T(u) \leq K_{\max} - K_1$. If $M_T(u) \geq 1$, order the reflection times as $1 \leq s_1 < \dots < s_{M_T(u)} \leq T$. The cooldown gives

$$s_j \geq 1 + (j-1)(\kappa+1), \quad T \geq 1 + (M_T(u)-1)(\kappa+1). \quad (363)$$

Rearranging yields the cooldown bound on $M_T(u)$; combining it with the cap proves the stated minimum. Finally,

$$\sum_{t=1}^T |A_t(u)| \leq TK_{\max}, \quad \text{Rollouts}_T(u) = T + c_{\text{ref}} M_T(u), \quad (364)$$

and substitution proves the rollout bound. $\square$

With the experimental values $T = 300$, $K_1 = 12$, $K_{\max} = 24$, and $\kappa = 20$,

$$M_T(u) \leq \min \left\{ 24 - 12, 1 + \left\lfloor \frac{299}{21} \right\rfloor \right\} = 12, \quad (365)$$

$$\sum_{t=1}^{300} |A_t(u)| \leq 300 \cdot 24 = 7200,$$

$$T + M_T(u) \leq 300 + 12 = 312.$$

Proposition 37 (Discovery reserve induced by first useful specialists). *Let R_T be the routed value from Theorem 26, and define the static-comparator reserve*

$$\rho_T = R_T - V_{\text{static}}(A_1). \quad (366)$$

Let $\pi_g = \mathbb{P}(g(U) = g) > 0$ for $g \in \mathcal{G}_{\text{act}}$, and fix $s \in \{0, \dots, T\}$. With $\tau_{g,j} = \infty$ for an attempt that never occurs, define

$$N_g(s) = \sum_{j \geq 1} \mathbf{1}\{\tau_{g,j} \leq s\}, \quad (367)$$

$$p_g(s) = \mathbb{P}(N_g(s) \geq m_g \mid g(U) = g).$$

The joint probability that the required attempts occur by round s and one of them succeeds is

$$h_g(s) = \mathbb{P}\left(S_{g,m_g}^{(\epsilon_g)} \cap \{N_g(s) \geq m_g\} \mid g(U) = g\right). \quad (368)$$

Assume a successful ϵ_g -useful child inserted by round s persists and contributes at least ϵ_g to every later discovery gain in that cluster:

$$G_t(U) \geq \epsilon_g \quad \text{for all } t > s \text{ on } \{g(U) = g\} \cap S_{g,m_g}^{(\epsilon_g)} \cap \{N_g(s) \geq m_g\}. \quad (369)$$

If

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq B_T \quad (370)$$

then

$$\rho_T \geq \Delta_{\text{route}}(A_1) - \frac{B_T}{T} + \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g \epsilon_g h_g(s). \quad (371)$$

Consequently, personalized routing beats the best static initial candidate if

$$\Delta_{\text{route}}(A_1) - \frac{B_T}{T} + \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g \epsilon_g h_g(s) > 0. \quad (372)$$

The regret relative to that static candidate,

$$\mathcal{R}_T^{\text{static}} = TV_{\text{static}}(A_1) - \sum_{t=1}^T \mathbb{E}[\mu(U, a_t(U))], \quad (373)$$

satisfies

$$\frac{1}{T} \mathcal{R}_T^{\text{static}} \leq -\Delta_{\text{route}}(A_1) + \frac{B_T}{T} - \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g \epsilon_g h_g(s). \quad (374)$$

Proof. Theorem 26 and the exploration certificate give

$$\rho_T = \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \quad \bar{L}_T \leq \frac{B_T}{T}. \quad (375)$$

For every $t > s$, persistence gives

$$\mathbb{E}[G_t(U)] \geq \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g \epsilon_g h_g(s). \quad (376)$$

Since $G_t(U) \geq 0$, averaging over the final $T-s$ rounds yields

$$\bar{G}_T \geq \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g \epsilon_g h_g(s). \quad (377)$$

Substitution proves the reserve bound, and its strict positivity is $\rho_T > 0$. Finally,

$$\begin{aligned} \frac{1}{T} \mathcal{R}_T^{\text{static}} &= V_{\text{static}}(A_1) - R_T \\ &= -\rho_T, \end{aligned} \quad (378)$$

which proves the regret bound. $\square$

In general, $h_g(s)$ cannot be replaced by $p_g(s)[1 - (1 - q_{\epsilon_g,g})^{m_g}]$: the event that m_g attempts are scheduled may depend on earlier successes or failures. That product is valid only under an additional exogenous attempt-scheduling condition that preserves the per-attempt success bound after conditioning on $\{N_g(s) \geq m_g\}$.

This reserve links Theorem 33 to the decomposition, but does not certify the oracle-assisted mechanism in the reported experiments. Those experiments report structural terms and an observed-reward residual. For the General-environment row in Table 10,

$$\begin{aligned} \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} &= 0.280 + 0.051 - 0.036 \\ &= 0.295, \\ V_{\text{static}}(A_1) + 0.295 &= 0.570 + 0.295 \\ &= 0.865, \end{aligned} \quad (379)$$

matching the reported $R_T^{\text{obs}} \approx 0.864$ up to rounding. This checks only the observed-reward identity; it is not a test of the mean-reward decomposition or a non-oracle proof of discovery.

H FURTHER ACCOUNTING RESULTS AND COUNTEREXAMPLES

This appendix gives algebraic consequences and counterexamples that delimit the interpretation of the decomposition.

H.1 DOMINANCE CONDITION

Corollary 38 (Dominance condition). *The routed policy beats the best static initial candidate if and only if*

$$R_T > V_{\text{static}}(A_1) \iff \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \quad (380)$$

If $A_t(U) = A_1$ for every t , this reduces to

$$R_T > V_{\text{static}}(A_1) \iff \Delta_{\text{route}}(A_1) > \bar{L}_T. \quad (381)$$

Proof. Subtract $V_{\text{static}}(A_1)$ from Theorem 26:

$$R_T - V_{\text{static}}(A_1) = \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \quad (382)$$

Strict positivity is equivalent to the first claim. A fixed archive has $G_t(U) = 0$, proving the second. $\square$

For example, $\Delta_{\text{route}}(A_1) = 0.12$, $\bar{G}_T = 0.04$, and $\bar{L}_T = 0.03$ give reserve $0.12 + 0.04 - 0.03 = 0.13 > 0$.

H.2 SIGN FACTS AND USER-SPECIFIC ARCHIVES

Assumption 19 (Append-only archive for sign statements). *Almost surely, $A_1 \subseteq A_t(U)$ and $a_t(U) \in A_t(U)$.*

Lemma 39 (Range of discovery gain and exploration cost). *If $\mu(u, a) \in [0, 1]$, then under Assumption 19,*

$$L_t(U), G_t(U) \in [0, 1] \text{ almost surely, } \bar{L}_T, \bar{G}_T \in [0, 1]. \quad (383)$$

Proof. Feasibility and append-only growth imply

$$m_t(U) \geq \mu(U, a_t(U)), \quad m_t(U) \geq m_1(U). \quad (384)$$

The reward range bounds both differences above by one. Averaging preserves the interval. $\square$

Proposition 40 (Per-user archive copies preserve the decomposition). *For each user u , let A_t^u be a user-specific archive path with $A_1^u = A_1$, $a_t^u \in A_t^u$, and $m_t^u = \max_{a \in A_t^u} \mu(u, a)$. If the population draw selects the corresponding trajectory,*

$$A_t(U) = A_t^U, \quad a_t(U) = a_t^U. \quad (385)$$

then Theorem 26 remains valid:

$$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \quad (386)$$

Proof. For every fixed user and trajectory,

$$\mu(u, a_t^u) = m_1^u + G_t^u - L_t^u. \quad (387)$$

Taking the trajectory expectation conditional on u , then integrating over U , gives

$$\mathbb{E}[\mu(U, a_t(U))] = \mathbb{E}[m_1^U] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \quad (388)$$

The common initial archive implies

$$\begin{aligned} \mathbb{E}[m_1^U] &= \mathbb{E}\left[\max_{a \in A_1} \mu(U, a)\right] \\ &= V_{\text{route}}(A_1) \\ &= V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1). \end{aligned} \quad (389)$$

Averaging over t completes the proof. $\square$

H.3 FINITE-SAMPLE IDENTITY AND COMPARATOR CHOICE

Proposition 41 (Finite-sample decomposition estimator). *For N sampled users $u_1, \dots, u_N$, let $a_{i,t} \in A_{i,t}$ and $m_{i,t} = \max_{a \in A_{i,t}} \mu(u_i, a)$. Define*

$$\begin{aligned}\hat{R}_T &= \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \mu(u_i, a_{i,t}), \\ \hat{V}_{\text{route}}(A_1) &= \frac{1}{N} \sum_{i=1}^N \max_{a \in A_1} \mu(u_i, a), \\ \hat{V}_{\text{static}}(A_1) &= \max_{a \in A_1} \frac{1}{N} \sum_{i=1}^N \mu(u_i, a), \\ \hat{\Delta}_{\text{route}}(A_1) &= \hat{V}_{\text{route}}(A_1) - \hat{V}_{\text{static}}(A_1), \\ \hat{G}_T &= \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T (m_{i,t} - m_{i,1}), \\ \hat{L}_T &= \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T (m_{i,t} - \mu(u_i, a_{i,t})).\end{aligned}\tag{390}$$

Then

$$\hat{R}_T = \hat{V}_{\text{static}}(A_1) + \hat{\Delta}_{\text{route}}(A_1) + \hat{G}_T - \hat{L}_T.\tag{391}$$

Proof. For each (i, t) ,

$$\mu(u_i, a_{i,t}) = m_{i,1} + (m_{i,t} - m_{i,1}) - (m_{i,t} - \mu(u_i, a_{i,t})).\tag{392}$$

Summing and dividing by NT gives

$$\begin{aligned}\hat{R}_T &= \frac{1}{N} \sum_{i=1}^N m_{i,1} + \hat{G}_T - \hat{L}_T \\ &= \hat{V}_{\text{route}}(A_1) + \hat{G}_T - \hat{L}_T.\end{aligned}\tag{393}$$

Substitute the definition of $\hat{\Delta}_{\text{route}}(A_1)$. $\square$

Proposition 41 applies when the selected-arm means are used. The rounded residuals in Table 10 instead use observed rewards and the matching observed residual, so they check a parallel algebraic identity but do not test the proposition.

For the three completed synthetic environments,

$$\begin{aligned}\varepsilon_{\text{general}} &= 0.570 + 0.280 + 0.051 - 0.036 - 0.864 \\ &= 0.001.\end{aligned}\tag{394}$$

$$\begin{aligned}\varepsilon_{\text{workspace}} &= 0.577 + 0.124 + 0.041 - 0.043 - 0.699 \\ &= 0.000.\end{aligned}\tag{395}$$

$$\begin{aligned}\varepsilon_{\text{support}} &= 0.579 + 0.215 + 0.061 - 0.039 - 0.816 \\ &= 0.000.\end{aligned}\tag{396}$$

The nonzero residual in (394) is a rounding residual, not an additional term in the decomposition.

Proposition 42 (Bias from a heuristic static comparator). *Let $h(A_1) \in A_1$ be a heuristic choice, and define*

$$V_h(A_1) = \mathbb{E}[\mu(U, h(A_1))].\tag{397}$$

For $\Delta_h(A_1) = V_{\text{route}}(A_1) - V_h(A_1)$,

$$\Delta_h(A_1) - \Delta_{\text{route}}(A_1) = V_{\text{static}}(A_1) - V_h(A_1) \geq 0.\tag{398}$$

Equality holds exactly when $h(A_1)$ is population-static optimal.

Proof.

$$\begin{aligned}\Delta_h(A_1) - \Delta_{\text{route}}(A_1) &= [V_{\text{route}}(A_1) - V_h(A_1)] - [V_{\text{route}}(A_1) - V_{\text{static}}(A_1)] \\ &= V_{\text{static}}(A_1) - V_h(A_1) \geq 0,\end{aligned}\quad (399)$$

because $V_{\text{static}}(A_1)$ maximizes over all static candidates. $\square$

H.4 COUNTEREXAMPLES DELIMITING THE CLAIMS

Proposition 43 (Counterexample: homogeneous users have no structural routing gain). *Let $A_t(U) = A_1 = \{a_1, a_2\}$, with $\mu(U, a_1) = 0.8$ and $\mu(U, a_2) = 0.6$ almost surely. If the router selects a_2 on an expected $\eta > 0$ fraction of rounds, then $R_T < V_{\text{static}}(A_1)$.*

Proof.

$$V_{\text{static}}(A_1) = 0.8, \quad V_{\text{route}}(A_1) = 0.8, \quad \Delta_{\text{route}}(A_1) = 0, \quad \bar{G}_T = 0. \quad (400)$$

Selecting a_2 costs 0.2, so $\bar{L}_T = 0.2\eta$. Hence

$$R_T = 0.8 - 0.2\eta < 0.8 = V_{\text{static}}(A_1). \quad (401)$$

$\square$

Proposition 44 (Counterexample: a heuristic static baseline can overstate heterogeneity). *Let $\mathbb{P}(U = u_1) = \mathbb{P}(U = u_2) = 1/2$, $A_1 = \{a_1, a_2\}$, and*

$$\begin{array}{c|cc} & a_1 & a_2 \\ \hline u_1 & 0.9 & 0.8 \\ u_2 & 0.1 & 0.7 \end{array}. \quad (402)$$

If heuristic scores satisfy $q(a_1) > q(a_2)$, so that $h(A_1) = a_1$, then $\Delta_{\text{route}}(A_1) = 0.05$ but $\Delta_h(A_1) = 0.30$.

Proof.

$$V_h(A_1) = \frac{1}{2}(0.9 + 0.1) = 0.50, \quad \mathbb{E}[\mu(U, a_1)] = 0.50, \quad \mathbb{E}[\mu(U, a_2)] = \frac{1}{2}(0.8 + 0.7) = 0.75. \quad (403)$$

The routed oracle value is

$$\begin{aligned}V_{\text{route}}(A_1) &= \frac{1}{2}[\max\{0.9, 0.8\} + \max\{0.1, 0.7\}] = 0.80, \\ \Delta_{\text{route}}(A_1) &= 0.80 - 0.75 = 0.05, \quad \Delta_h(A_1) = 0.80 - 0.50 = 0.30.\end{aligned}\quad (404)$$

$\square$

Proposition 45 (Counterexample: oracle-assisted reflection makes the discovery bound vacuous). *For cluster g , define the repair probability in Theorem 33 by*

$$q_{\epsilon, g} = \inf_j \mathbb{P}(\text{attempt } j \text{ is } \epsilon\text{-improving} \mid \mathcal{F}_{\tau_{g, j-1}}). \quad (405)$$

If the proposer reads the hidden peak category and deterministically creates an ϵ -improving child, then $q_{\epsilon, g} = 1$ and the lower bound is one for every $m \geq 1$. If $q_{\epsilon, g} = 0$, the bound is zero for every finite m .

Proof.

$$1 - (1 - 1)^m = 1, \quad 1 - (1 - 0)^m = 0. \quad (406)$$

Thus the result only reflects the assumed repair probability. Validating a non-oracle proposer requires a separate nontrivial lower bound on $q_{\epsilon, g}$. $\square$

Proposition 46 (Counterexample: without append-only archives, discovery need not be nonnegative). *Let $\mathbb{P}(U = u) = 1$, $A_1 = \{a^*\}$, and $A_2 = \{b\}$, with $\mu(u, a^*) = 1$ and $\mu(u, b) = 0$. Then $G_2(U) = -1$. The decomposition remains algebraically true, but the term G_t can no longer be interpreted as a gain.*

Proof.

$$m_1(U) = 1, \quad m_2(U) = 0, \quad G_2(U) = -1, \quad L_2(U) = 0. \quad (407)$$

Yet

$$\mu(u, b) = m_1(U) + G_2(U) - L_2(U) = 1 - 1 - 0 = 0. \quad (408)$$

Thus append-only growth is needed for the sign interpretation, not the identity. $\square$

H.5 MAP FROM THEORY CLAIMS TO PROOFS

Table 7: Proof map for the supporting reward-accounting and HRB claims.

Claim	Formal content	Proof or qualification
Lemma 24	Static-vs-routed value satisfies $\mathbb{E}[\max_{a \in A} \mu(U, a)] \geq \max_{a \in A} \mathbb{E}[\mu(U, a)]$, with strictness only under positive-mass ranking disagreement.	Proved above in Appendix E; Proposition 43 shows the homogeneous boundary case.
Lemma 25	Adding children after a round preserves predictability of the next action set.	Proved above in Appendix E; Assumption 19 records the extra sign condition used only for $G_t \geq 0$.
Lemma 27	High-gap failure triggering controls the number of mutation attempts without changing the router’s filtration argument.	Proved above in Appendix E; Proposition 45 clarifies that trigger evidence is not evidence for non-oracle repair quality.
Theorem 26	Exact four-term decomposition: $R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T$.	Proved above in Appendix E; Proposition 40 covers user-specific archive copies and Proposition 41 gives the finite-sample estimator.
Theorem 30	Conditional plug-in bound obtained from an externally supplied exploration-cost certificate B_T .	Proved above in Appendix E; no new regret rate is claimed for the HRB implementation.
Theorem 33	First-useful-specialist probability: $\mathbb{P}(S_{g,m}^{(\epsilon)}) \geq 1 - (1 - q_{\epsilon,g})^m$.	Proved above in Appendix E; Proposition 45 explains why oracle-assisted reflection makes the bound vacuous as empirical validation.
Corollary 18	Personalized routing beats the best static initial candidate exactly when $\Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T$.	Restated and proved in Corollary 38.
Proposition 37	A first-useful-specialist reserve can be inserted into the decomposition under persistence and exploration-cost certificates.	Proved above in Appendix E; it is a conditional reserve statement, not a guarantee for the oracle-assisted simulator.
Comparator correction	The theorem comparator is $V_{\text{static}}(A_1)$, not a heuristic intrinsic-quality choice.	Proved in Proposition 42; the numerical failure mode is Proposition 44.

Together, these results close the appendix proof story: HyperRidge’s supported claim is the four-term decomposition over the common initial archive, with conditional learning and discovery terms made explicit, and with the counterexamples identifying when stronger dominance or reflection claims would be unsound.

I AUXILIARY BOUNDS

This block makes explicit the measurability, boundedness, trigger-count, finite-cluster, and contraction constants used by Theorems 26, 30, and 33. It does not add a new mechanism.

Lemma 47 (Measurability and bounded ranges). *Fix t . Suppose the user-specific archive has the finite predictable representation*

$$A_t(U) = \{a_{t,1}(U), \dots, a_{t,N_t(U)}(U)\}. \quad (409)$$

Assume $N_t(U) < \infty$ almost surely, each listed candidate is $\mathcal{F}_{t-1}$ -measurable, and μ is measurable with values in $[0, 1]$. Then $m_t(U) = \max_{a \in A_t(U)} \mu(U, a)$ is measurable, $L_t(U) \in [0, 1]$, and $G_t(U) \in [-1, 1]$. If $A_1 \subseteq A_t(U)$, then $G_t(U) \in [0, 1]$.

Proof. For each n and $z \in \mathbb{R}$,

$$\{m_t(U) \leq z\} \cap \{N_t(U) = n\} = \bigcap_{j=1}^n \{\mu(U, a_{t,j}(U)) \leq z\} \cap \{N_t(U) = n\}. \quad (410)$$

This finite intersection is measurable; taking the countable union over n proves measurability of $m_t(U)$. The reward range gives $m_t(U) \in [0, 1]$. Feasibility gives $\mu(U, a_t(U)) \leq m_t(U)$, hence $L_t(U) \in [0, 1]$. Since G_t is the difference of two values in $[0, 1]$, it lies in $[-1, 1]$; append-only growth gives $m_t(U) \geq m_1(U)$. $\square$

For example, $m_1(U) = 0.60$, $m_t(U) = 0.75$, and $\mu(U, a_t(U)) = 0.70$ give $G_t(U) = 0.15$ and $L_t(U) = 0.05$.

Lemma 48 (From a high-probability tax certificate to an expected tax bound). *Let $S_T = \sum_{t=1}^T L_t(U)$, with $L_t(U) \in [0, 1]$. If $\mathbb{P}(S_T \leq b_T) \geq 1 - \delta$, then*

$$\mathbb{E}[S_T] \leq b_T + \delta T. \quad (411)$$

Thus Theorem 30 applies with $B_T = b_T + \delta T$.

Proof. For $\mathcal{E}_T = \{S_T \leq b_T\}$, use $S_T \leq T$:

$$\begin{aligned} \mathbb{E}[S_T] &= \mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T\}] + \mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T^c\}] \\ &\leq b_T + T \mathbb{P}(\mathcal{E}_T^c) \leq b_T + \delta T. \end{aligned} \quad (412)$$

$\square$

For $b_T = 12$, $T = 300$, and $\delta = 0.01$, this gives $\mathbb{E}[S_T] \leq 15$, or $\mathbb{E}[S_T]/T \leq 0.05$.

Proposition 49 (Explicit constants for high-gap triggering). *For $\tau, \sigma > 0$, define the trigger, trigger count, and observation model by*

$$I_t^{(\tau)} = \mathbf{1}\{m_t(U) - r_t > \tau\}, \quad (413)$$

$$N_T^{(\tau)} = \sum_{t=1}^T I_t^{(\tau)}, \quad (414)$$

$$r_t = \mu(U, a_t(U)) + \xi_t. \quad (415)$$

Suppose that, for every $\lambda \in \mathbb{R}$,

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right). \quad (416)$$

Then

$$\mathbb{E}[N_T^{(\tau)}] \leq \frac{2}{\tau} \sum_{t=1}^T \mathbb{E}[L_t(U)] + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right). \quad (417)$$

In particular, if

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq B_T, \quad (418)$$

then

$$\mathbb{E}[N_T^{(\tau)}] \leq \frac{2B_T}{\tau} + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right). \quad (419)$$

Proof. Since $m_t(U) - r_t = L_t(U) - \xi_t$,

$$\begin{aligned} I_t^{(\tau)} &\leq \mathbf{1}\{L_t(U) + |\xi_t| > \tau\} \\ &\leq \mathbf{1}\{L_t(U) > \tau/2\} + \mathbf{1}\{|\xi_t| > \tau/2\}. \end{aligned} \quad (420)$$

Markov's inequality and the sub-Gaussian tail bound give

$$\mathbb{P}\{L_t(U) > \tau/2\} \leq \frac{2}{\tau} \mathbb{E}[L_t(U)], \quad \mathbb{P}\{|\xi_t| > \tau/2\} \leq 2 \exp\left(-\frac{\tau^2}{8\sigma^2}\right). \quad (421)$$

Taking expectations and summing over t gives

$$\mathbb{E}[N_T^{(\tau)}] \leq \frac{2}{\tau} \sum_{t=1}^T \mathbb{E}[L_t(U)] + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right). \quad (422)$$

This proves (417); substituting (418) gives (419). $\square$

For $T = 100$, $B_T = 4$, $\tau = 0.20$, and $\sigma = 0.05$, the bound is

$$\begin{aligned} \mathbb{E}[N_T^{(\tau)}] &\leq \frac{2 \cdot 4}{0.20} + 2 \cdot 100 \exp\left(-\frac{0.20^2}{8 \cdot 0.05^2}\right) \\ &= 40 + 200e^{-2} \approx 67.07. \end{aligned} \quad (423)$$

This constant is conservative; it is used only to justify the direction of the trigger-count control, not as an empirical estimate of reflection quality.

Proposition 50 (Finite-cluster margin lower bound for heterogeneity). *Let A be a finite archive, and partition the user population into $K_{\text{cl}} < \infty$ clusters. Define the cluster index, mass, and conditional candidate value by*

$$g(U) \in \{1, \dots, K_{\text{cl}}\}, \quad (424)$$

$$\pi_g = \mathbb{P}(g(U) = g), \quad (425)$$

$$\nu_{g,a} = \mathbb{E}[\mu(U, a) \mid g(U) = g] \quad (\pi_g > 0). \quad (426)$$

Set $\nu_{g,a} = 0$ when $\pi_g = 0$. For each static candidate, define its cluster-wise deficit and the clusters with deficit at least $\eta > 0$:

$$d_g(a) = \max_{b \in A} \nu_{g,b} - \nu_{g,a}, \quad (427)$$

$$H_a(\eta) = \{g : d_g(a) \geq \eta\}. \quad (428)$$

If every static candidate has such a deficit on cluster mass at least ρ ,

$$\inf_{a \in A} \sum_{g \in H_a(\eta)} \pi_g \geq \rho. \quad (429)$$

Then the fixed-archive heterogeneity gap satisfies

$$\Delta_{\text{route}}(A) \geq \rho\eta. \quad (430)$$

Proof. For each cluster, $\mathbb{E}[\max_{b \in A} \mu(U, b) \mid g(U) = g] \geq \max_{b \in A} \nu_{g,b}$. Hence

$$V_{\text{route}}(A) \geq \sum_{g=1}^{K_{\text{cl}}} \pi_g \max_{b \in A} \nu_{g,b}, \quad (431)$$

$$V_{\text{static}}(A) = \max_{a \in A} \sum_{g=1}^{K_{\text{cl}}} \pi_g \nu_{g,a}. \quad (432)$$

Because A is finite, choose

$$a^* \in \arg \max_{a \in A} \sum_{g=1}^{K_{\text{cl}}} \pi_g \nu_{g,a}. \quad (433)$$

Then

$$\begin{aligned}
\Delta_{\text{route}}(A) &\geq \sum_{g=1}^{K_{\text{cl}}} \pi_g d_g(a^*). \\
&\geq \sum_{g \in H_{a^*}(\eta)} \pi_g d_g(a^*) \\
&\geq \eta \sum_{g \in H_{a^*}(\eta)} \pi_g \geq \rho\eta.
\end{aligned} \tag{434}$$

□

The bound is tight. For equal cluster masses and value matrix

$$\begin{array}{c|cc}
& a_1 & a_2 \\
\hline
g=1 & 0.80 & 0.50 \\
g=2 & 0.50 & 0.80
\end{array} \cdot \tag{435}$$

we have $\pi_1 = \pi_2 = 1/2$, $\rho = 1/2$, and $\eta = 0.30$. Therefore

$$\Delta_{\text{route}}(A) \geq \rho\eta = 0.15, \quad \Delta_{\text{route}}(A) = 0.80 - \frac{1}{2}(0.80 + 0.50) = 0.15. \tag{436}$$

Proposition 51 (Contraction repair condition implies an explicit discovery-time bound). *Fix a cluster g , an initial gap $\Delta_{g,0} > 0$, and a target $0 < \epsilon < \Delta_{g,0}$. Let $Z_j \in \{0, 1\}$ be independent success indicators satisfying $\mathbb{P}(Z_j = 1) \geq q$ for $q \in (0, 1]$. For $0 < \gamma < 1$, assume*

$$\Delta_{g,j} \leq (1 - \gamma Z_j) \Delta_{g,j-1}. \tag{437}$$

Define the required and realized success counts by

$$s_\epsilon = \left\lceil \frac{\log(\Delta_{g,0}/\epsilon)}{-\log(1 - \gamma)} \right\rceil, \tag{438}$$

$$S_m = \sum_{j=1}^m Z_j \quad (m \in \mathbb{N}). \tag{439}$$

If $mq \geq 2s_\epsilon$, then

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \leq \exp\left(-\frac{mq}{8}\right). \tag{440}$$

Consequently, for $\delta \in (0, 1)$, the sufficient condition

$$m \geq \frac{2s_\epsilon + 8 \log(1/\delta)}{q} \tag{441}$$

to obtain

$$\mathbb{P}(\Delta_{g,m} \leq \epsilon) \geq 1 - \delta. \tag{442}$$

Proof. Iteration and the binary nature of Z_j give

$$\Delta_{g,m} \leq \Delta_{g,0} \prod_{j=1}^m (1 - \gamma Z_j) = \Delta_{g,0} (1 - \gamma)^{S_m}. \tag{443}$$

Thus $S_m \geq s_\epsilon$ implies

$$\Delta_{g,m} \leq \Delta_{g,0} (1 - \gamma)^{s_\epsilon} \leq \epsilon. \tag{444}$$

Let

$$\lambda = \log 2. \tag{445}$$

For any $s \leq mq/2$, Markov's inequality applied to $\exp(-\lambda S_m)$ gives

$$\begin{aligned}
\mathbb{P}(S_m \leq s) &= \mathbb{P}(e^{-\lambda S_m} \geq e^{-\lambda s}) \\
&\leq e^{\lambda s} \mathbb{E}[e^{-\lambda S_m}].
\end{aligned} \tag{446}$$

Using independence and $\mathbb{P}(Z_j = 1) \geq q$,

$$\begin{aligned}\mathbb{E}[e^{-\lambda S_m}] &= \prod_{j=1}^m \mathbb{E}[e^{-\lambda Z_j}] \\ &= \prod_{j=1}^m (1 - \mathbb{P}(Z_j = 1) + \mathbb{P}(Z_j = 1)e^{-\lambda}) \\ &\leq (1 - q + qe^{-\lambda})^m \\ &\leq \exp[-mq(1 - e^{-\lambda})].\end{aligned}\tag{447}$$

With $s = s_\epsilon$ and $s_\epsilon \leq mq/2$,

$$\mathbb{P}(S_m < s_\epsilon) \leq \exp\left[\lambda \frac{mq}{2} - mq(1 - e^{-\lambda})\right].\tag{448}$$

Since $\lambda = \log 2$ and $e^{-\lambda} = 1/2$,

$$\begin{aligned}\lambda \frac{mq}{2} - mq(1 - e^{-\lambda}) &= mq \left(\frac{\log 2}{2} - \frac{1}{2} \right) \\ &= -\frac{mq}{2}(1 - \log 2) \\ &\leq -\frac{mq}{8}.\end{aligned}\tag{449}$$

This proves

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \leq \exp\left(-\frac{mq}{8}\right).\tag{450}$$

If (441) holds, then

$$\begin{aligned}mq &\geq 2s_\epsilon, \\ mq &\geq 8 \log(1/\delta).\end{aligned}\tag{451}$$

Hence

$$\begin{aligned}\mathbb{P}(\Delta_{g,m} > \epsilon) &\leq \exp[-\log(1/\delta)] \\ &= \delta.\end{aligned}\tag{452}$$

□

For $\Delta_{g,0} = 0.40$, $\epsilon = 0.05$, $\gamma = 0.50$, $q = 0.25$, and $\delta = 0.05$,

$$s_\epsilon = \left\lceil \frac{\log 8}{\log 2} \right\rceil = 3,\tag{453}$$

$$m \geq \frac{2 \cdot 3 + 8 \log(20)}{0.25} \approx 119.86.\tag{454}$$

Thus $m = 120$ suffices for the stated 95% guarantee under this stronger contraction model. The oracle-assisted experiments do not estimate q or γ ; Section 5 therefore reports discovery as a measured decomposition term, not as validation of Proposition 51.

J FINITE-SAMPLE ACCOUNTING AND ESTIMATION

This appendix derives finite-sample bounds for the terms of the decomposition.

Lemma 52 (Append-only boundedness of the decomposition terms). *Assume bounded rewards, append-only archives, and feasible actions:*

$$0 \leq \mu(u, a) \leq 1,\tag{455}$$

$$A_1(U) \subseteq A_2(U) \subseteq \dots \subseteq A_T(U),\tag{456}$$

$$a_t(U) \in A_t(U).\tag{457}$$

Define

$$m_t(U) = \max_{a \in A_t(U)} \mu(U, a), \quad (458)$$

$$G_t(U) = m_t(U) - m_1(U), \quad (459)$$

$$L_t(U) = m_t(U) - \mu(U, a_t(U)). \quad (460)$$

Then $0 \leq G_t(U), L_t(U) \leq 1$ for every t , and G_t is nondecreasing:

$$0 \leq G_t(U) \leq 1, \quad 0 \leq L_t(U) \leq 1, \quad G_{t+1}(U) \geq G_t(U). \quad (461)$$

Proof.

$$\begin{aligned} 0 &\leq m_1(U) \leq m_t(U) \leq m_{t+1}(U) \leq 1, \\ 0 &\leq m_t(U) - \mu(U, a_t(U)) \leq 1. \end{aligned} \quad (462)$$

The first line follows from archive inclusion and the reward range; the second follows from feasibility. Substitution into the definitions of G_t and L_t proves all three claims. $\square$

Lemma 53 (From high-probability current-archive regret to expected exploration cost). *Let*

$$S_T^L = \sum_{t=1}^T L_t(U), \quad 0 \leq L_t(U) \leq 1. \quad (463)$$

If

$$\mathbb{P}(S_T^L \leq C_T(\delta)) \geq 1 - \delta. \quad (464)$$

then

$$\mathbb{E}[S_T^L] \leq C_T(\delta) + \delta T. \quad (465)$$

Consequently, the external exploration-cost premise (270) holds with

$$B_T = C_T(\delta) + \delta T. \quad (466)$$

In particular, if

$$C_T(\delta) = c_0 \sqrt{T(\zeta_T + \log(1/\delta))}, \quad (467)$$

$$\delta = \frac{1}{T}, \quad (468)$$

then

$$B_T \leq c_0 \sqrt{T(\zeta_T + \log T)} + 1. \quad (469)$$

Proof. Apply Lemma 48 with $S_T = S_T^L$ and $b_T = C_T(\delta)$. Substituting the displayed form of C_T and $\delta = 1/T$ gives the last claim. $\square$

For example, $C_{300}(1/300) = 12$ gives

$$B_{300} \leq 12 + \frac{300}{300} = 13, \quad \frac{B_{300}}{300} \leq 0.0433. \quad (470)$$

Lemma 54 (Explicit constants for high-gap trigger counts). *For $\tau, \sigma > 0$, define*

$$r_t = \mu(U, a_t(U)) + \xi_t, \quad (471)$$

$$I_t(\tau) = \mathbf{1}\{m_t(U) - r_t > \tau\}, \quad (472)$$

$$N_T(\tau) = \sum_{t=1}^T I_t(\tau). \quad (473)$$

Assume that

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right) \quad (\lambda \in \mathbb{R}). \quad (474)$$

Then

$$\mathbb{E}[N_T(\tau)] \leq T \wedge \left[\frac{2}{\tau} \sum_{t=1}^T \mathbb{E}[L_t(U)] + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right) \right]. \quad (475)$$

Under the external exploration-cost bound

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \leq B_T, \quad (476)$$

the trigger count obeys

$$\mathbb{E}[N_T(\tau)] \leq T \wedge \left[\frac{2B_T}{\tau} + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right) \right]. \quad (477)$$

Proof. Proposition 49, under the same notation, gives the second term inside the minimum. The deterministic inequality $N_T(\tau) \leq T$ gives the first. Substituting (476) proves the final claim. $\square$

For $T = 300$, $\tau = 0.15$, $\sigma = 0.05$, and $B_T = 12$,

$$\begin{aligned} \frac{2B_T}{\tau} + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right) &= \frac{24}{0.15} + 600 \exp(-1.125) \\ &\approx 354.8, \quad \mathbb{E}[N_T(0.15)] \leq 300. \end{aligned} \quad (478)$$

Thus the deterministic cap dominates. This loose bound does not imply the empirical archive-growth claim in Section 5.

Proposition 55 (Cooldown and archive cap imply a deterministic archive-size bound). *Let $K_1 = |A_1| \leq K_{\max}$, and let M_T be the number of unique children appended through round T . When $M_T \geq 1$, write the append times as $1 \leq s_1 < \dots < s_{M_T} \leq T$. Assume the archive is append-only, exactly one child is inserted at each s_j , the cap $|A_t(U)| \leq K_{\max}$ is enforced, and, for $\kappa \in \mathbb{N}$,*

$$s_{j+1} - s_j \geq \kappa + 1. \quad (479)$$

Then the number of appended children satisfies

$$M_T \leq \min\left\{ K_{\max} - K_1, 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \quad (480)$$

Consequently, for every $t \in \{1, \dots, T+1\}$,

$$|A_t(U)| \leq K_1 + \min\left\{ K_{\max} - K_1, 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \quad (481)$$

Proof. The case $M_T = 0$ is immediate. Otherwise, the cooldown gives

$$s_j \geq 1 + (j-1)(\kappa+1), \quad 1 + (M_T-1)(\kappa+1) \leq T. \quad (482)$$

Thus $M_T \leq 1 + \lfloor (T-1)/(\kappa+1) \rfloor$. Unique insertion and the cap also give $K_1 + M_T \leq K_{\max}$. Their minimum proves (480), and $|A_t(U)| \leq K_1 + M_T$ proves the second claim. $\square$

For $K_1 = 12$, $K_{\max} = 24$, $T = 300$, and $\kappa = 20$,

$$\begin{aligned} 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor &= 1 + \left\lfloor \frac{299}{21} \right\rfloor = 15, \\ M_T &\leq \min\{15, 12\} = 12, \quad |A_t(U)| \leq 24. \end{aligned} \quad (483)$$

Proposition 56 (Logged decomposition estimators and finite-sample radii). *Consider N i.i.d. logged one-user trajectories generated by the same routing protocol from a deterministic finite archive A_1 , with $K_1 = |A_1|$. User draws, online noise, and archive paths are independent across trajectories, and each path is append-only. For trajectory n and round t , define*

$$m_{n,t} = \max_{a \in A_{n,t}} \mu(U_n, a), \quad (484)$$

$$G_{n,t} = m_{n,t} - m_{n,1}, \quad (485)$$

$$L_{n,t} = m_{n,t} - \mu(U_n, a_{n,t}). \quad (486)$$

Define the six empirical terms

$$\widehat{\mathcal{V}}_T = \frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T \mu(U_n, a_{n,t}), \quad (487)$$

$$\widehat{V}_{\text{route}}(A_1) = \frac{1}{N} \sum_{n=1}^N m_{n,1}, \quad (488)$$

$$\widehat{V}_{\text{static}}(A_1) = \max_{a \in A_1} \frac{1}{N} \sum_{n=1}^N \mu(U_n, a), \quad (489)$$

$$\widehat{\Delta}_{\text{route}}(A_1) = \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1), \quad (490)$$

$$\widehat{G}_T = \frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T G_{n,t}, \quad (491)$$

$$\widehat{L}_T = \frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T L_{n,t}. \quad (492)$$

These terms satisfy the exact logged decomposition

$$\widehat{\mathcal{V}}_T = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{G}_T - \widehat{L}_T. \quad (493)$$

For $\delta \in (0, 1)$, define

$$\mathcal{V}_T = \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mu(U, a_t(U))], \quad (494)$$

$$\varepsilon_{\text{sc}}(N, \delta) = \sqrt{\frac{\log(10/\delta)}{2N}}, \quad (495)$$

$$\varepsilon_{\text{st}}(N, K_1, \delta) = \sqrt{\frac{\log(10K_1/\delta)}{2N}}. \quad (496)$$

With probability at least $1 - \delta$, the following simultaneous bounds hold:

$$\begin{aligned} \left| \widehat{V}_{\text{static}}(A_1) - V_{\text{static}}(A_1) \right| &\leq \varepsilon_{\text{st}}(N, K_1, \delta), \\ \left| \widehat{V}_{\text{route}}(A_1) - V_{\text{route}}(A_1) \right| &\leq \varepsilon_{\text{sc}}(N, \delta), \\ \left| \widehat{G}_T - \bar{G}_T \right| &\leq \varepsilon_{\text{sc}}(N, \delta), \\ \left| \widehat{L}_T - \bar{L}_T \right| &\leq \varepsilon_{\text{sc}}(N, \delta), \\ \left| \widehat{\mathcal{V}}_T - \mathcal{V}_T \right| &\leq \varepsilon_{\text{sc}}(N, \delta). \end{aligned} \quad (497)$$

Consequently,

$$\left| \widehat{\Delta}_{\text{route}}(A_1) - \Delta_{\text{route}}(A_1) \right| \leq \varepsilon_{\text{st}}(N, K_1, \delta) + \varepsilon_{\text{sc}}(N, \delta). \quad (498)$$

Finally, define the dominance margin and its estimator without reusing the archive-count symbol M_T :

$$\mathcal{M}_T = \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \quad (499)$$

$$\widehat{\mathcal{M}}_T = \widehat{\Delta}_{\text{route}}(A_1) + \widehat{G}_T - \widehat{L}_T. \quad (500)$$

$$|\widehat{\mathcal{M}}_T - \mathcal{M}_T| \leq \varepsilon_{\text{st}}(N, K_1, \delta) + 3\varepsilon_{\text{sc}}(N, \delta). \quad (501)$$

Proof. For the exact logged identity, use the pointwise relation

$$\begin{aligned} \mu(U_n, a_{n,t}) &= m_{n,t} - L_{n,t} \\ &= m_{n,1} + G_{n,t} - L_{n,t}. \end{aligned} \quad (502)$$

Averaging gives

$$\begin{aligned}
\hat{\mathcal{V}}_T &= \frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T (m_{n,1} + G_{n,t} - L_{n,t}) \\
&= \frac{1}{N} \sum_{n=1}^N m_{n,1} + \hat{G}_T - \hat{L}_T \\
&= \hat{V}_{\text{route}}(A_1) + \hat{G}_T - \hat{L}_T \\
&= \hat{V}_{\text{static}}(A_1) + \hat{\Delta}_{\text{route}}(A_1) + \hat{G}_T - \hat{L}_T.
\end{aligned} \tag{503}$$

For each $a \in A_1$, set $\hat{\mu}_a = N^{-1} \sum_n \mu(U_n, a)$ and $\mu_a = \mathbb{E}[\mu(U, a)]$. Hoeffding's inequality and a union bound give

$$\begin{aligned}
\mathbb{P}\left(\max_{a \in A_1} |\hat{\mu}_a - \mu_a| > \varepsilon\right) &\leq \sum_{a \in A_1} 2 \exp(-2N\varepsilon^2) \\
&= 2K_1 \exp(-2N\varepsilon^2).
\end{aligned} \tag{504}$$

Moreover, the maximum map is one-Lipschitz in the sup norm:

$$\left| \max_{a \in A_1} \hat{\mu}_a - \max_{a \in A_1} \mu_a \right| \leq \max_{a \in A_1} |\hat{\mu}_a - \mu_a|. \tag{505}$$

Setting $\varepsilon = \varepsilon_{\text{st}}(N, K_1, \delta)$ makes (504) equal to $\delta/5$ and proves the static bound.

For the other four scalar terms, define the per-trajectory bounded variables

$$\begin{aligned}
Z_n^{\text{route}} &= m_{n,1}, \\
Z_n^G &= \frac{1}{T} \sum_{t=1}^T G_{n,t}, \\
Z_n^L &= \frac{1}{T} \sum_{t=1}^T L_{n,t}, \\
Z_n^V &= \frac{1}{T} \sum_{t=1}^T \mu(U_n, a_{n,t}).
\end{aligned} \tag{506}$$

Lemma 52 places all four variables in $[0, 1]$. Hence, for each one, Hoeffding's inequality gives

$$\mathbb{P}\left(\left|\frac{1}{N} \sum_{n=1}^N Z_n - \mathbb{E}[Z_n]\right| > \varepsilon\right) \leq 2 \exp(-2N\varepsilon^2). \tag{507}$$

Setting $\varepsilon = \varepsilon_{\text{sc}}(N, \delta)$ makes each failure probability at most $\delta/5$. A union bound over these four events and the static event proves (497). The remaining bounds follow from

$$\begin{aligned}
|\hat{\Delta}_{\text{route}} - \Delta_{\text{route}}| &= |(\hat{V}_{\text{route}} - V_{\text{route}}) - (\hat{V}_{\text{static}} - V_{\text{static}})| \\
&\leq \varepsilon_{\text{sc}} + \varepsilon_{\text{st}}, \\
|\hat{\mathcal{M}}_T - \mathcal{M}_T| &\leq |\hat{\Delta}_{\text{route}} - \Delta_{\text{route}}| + |\hat{G}_T - \bar{G}_T| + |\hat{L}_T - \bar{L}_T| \\
&\leq \varepsilon_{\text{st}} + 3\varepsilon_{\text{sc}}.
\end{aligned} \tag{508}$$

□

The synthetic experiments do not meet the independence premise with $N = 5 \cdot 8 = 40$: the eight users within a seed share that seed's archive draw. Proposition 56 therefore does not supply a 40-sample confidence interval for those runs. A valid analysis must treat each seed as a cluster, use a cluster-aware concentration bound, or regenerate independent archives for every logged trajectory. The algebraic sample identity remains exact without independence.

Lemma 57 (Observed-reward noise in the routed-value estimate). *Let $M = NT$, enumerate the observations by $s \in \{1, \dots, M\}$, and write*

$$r_s = \mu_s + \varepsilon_s, \quad \mu_s = \mu(U_s, a_s). \quad (509)$$

Suppose (ε_s) is a martingale-difference sequence with respect to $(\mathcal{H}_s)$ and is conditionally σ -sub-Gaussian:

$$\mathbb{E}[\varepsilon_s \mid \mathcal{H}_{s-1}] = 0, \quad \mathbb{E}[\exp(\lambda \varepsilon_s) \mid \mathcal{H}_{s-1}] \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right) \quad (\lambda \in \mathbb{R}, \sigma > 0). \quad (510)$$

Define

$$\hat{\mathcal{V}}_T^{\text{obs}} = \frac{1}{M} \sum_{s=1}^M r_s, \quad \hat{\mathcal{V}}_T^{\text{mean}} = \frac{1}{M} \sum_{s=1}^M \mu_s. \quad (511)$$

Then, for every $\epsilon > 0$,

$$\mathbb{P}\left(\left|\hat{\mathcal{V}}_T^{\text{obs}} - \hat{\mathcal{V}}_T^{\text{mean}}\right| \geq \epsilon\right) \leq 2 \exp\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \quad (512)$$

Equivalently, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left|\hat{\mathcal{V}}_T^{\text{obs}} - \hat{\mathcal{V}}_T^{\text{mean}}\right| \leq \sigma \sqrt{\frac{2 \log(2/\delta)}{M}}. \quad (513)$$

Proof. For $S_M = \sum_{s=1}^M \varepsilon_s$, iterated conditional expectation gives

$$\mathbb{E}[\exp(\lambda S_M)] \leq \exp\left(\frac{M\lambda^2 \sigma^2}{2}\right). \quad (514)$$

Chernoff's bound, optimized at $\lambda = \epsilon/\sigma^2$, yields

$$\mathbb{P}(S_M \geq M\epsilon) \leq \exp\left(-\frac{M\epsilon^2}{2\sigma^2}\right), \quad \mathbb{P}(S_M \leq -M\epsilon) \leq \exp\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \quad (515)$$

A union bound proves (512); solving it for ϵ proves (513). $\square$

At $M = 40 \cdot 300 = 12000$, $\sigma = 0.05$, and $\delta = 0.05$, Lemma 57 gives

$$\begin{aligned} \sigma \sqrt{\frac{2 \log(2/\delta)}{M}} &= 0.05 \sqrt{\frac{2 \log(40)}{12000}} \\ &\approx 0.00124. \end{aligned} \quad (516)$$

If the simulator clips noisy rewards, replace the theoretical mean and residual by

$$\begin{aligned} \mu_{\text{clip}}(u, a) &= \mathbb{E}[\text{clip}(\mu_0(u, a) + \eta, 0, 1)], \\ \varepsilon_s^{\text{clip}} &= r_s - \mu_{\text{clip}}(U_s, a_s). \end{aligned} \quad (517)$$

Because the clipped reward lies in $[0, 1]$, Hoeffding–Azuma gives

$$\mathbb{P}\left(\left|\frac{1}{M} \sum_{s=1}^M \varepsilon_s^{\text{clip}}\right| \geq \epsilon\right) \leq 2 \exp(-2M\epsilon^2). \quad (518)$$

Thus clipping changes the target conditional mean; it does not invalidate the logged observed-reward algebra. In the HRB component table the structural terms use the pre-clipping mean μ_0 , whereas the routed value uses clipped rewards. The residual $m_t - r_t$ absorbs both action-selection regret and clipping bias, so it must not be labeled as $L_t = m_t - \mu_0(U, a_t)$.

Together, Lemmas 52–57 and Propositions 55–56 separate the exact mean-reward identity from the observed-reward accounting check. They do not supply a new regret rate or a non-oracle theory of reflection.

Algorithm 1 HyperRidge-Periodic (HRP): routing with periodic covariance updates

Require: Initial archive A_1 , ridge weight $\lambda > 0$, exploration weight $\alpha > 0$, re-estimation period $\tau \in \mathbb{N}$, regularizer $\epsilon > 0$

```

1: Initialize  $\hat{\Omega}_1 \leftarrow I_d$ ; for each candidate  $a$ , initialize  $X_{a,1}$  and  $y_{a,1}$  as empty
2: for  $t = 1, 2, \dots, T$  do
3:   Observe user  $u_t$  and eligible candidate set  $A_t(u_t)$ 
4:   for each candidate  $a \in A_t(u_t)$  do
5:     Compute  $V_{a,t}^{\text{hyper}} \leftarrow X_{a,t}^\top X_{a,t} + \lambda \hat{\Omega}_t^{-1}$ 
6:     Compute  $\hat{\theta}_{a,t} \leftarrow (V_{a,t}^{\text{hyper}})^{-1} X_{a,t}^\top y_{a,t}$ 
7:     Compute  $\text{UCB}_t^{\text{hyper}}(u_t, a) \leftarrow x(u_t, a)^\top \hat{\theta}_{a,t} + \alpha \sqrt{x(u_t, a)^\top (V_{a,t}^{\text{hyper}})^{-1} x(u_t, a)}$ 
8:   end for
9:   Select  $a_t \in \arg \max_{a \in A_t(u_t)} \text{UCB}_t^{\text{hyper}}(u_t, a)$  and observe reward  $r_t$ ; set  $y_t \leftarrow r_t - b(u_t)$ 
10:  Update per-candidate data: append  $x(u_t, a_t)^\top$  to  $X_{a_t, t+1}$  and  $y_t$  to  $y_{a_t, t+1}$ 
11:  if  $t \equiv 0 \pmod{\tau}$  then
12:    Re-estimate the hyper-covariance from the archive’s candidate estimates:

$$\hat{\Omega}_{t+1} \leftarrow \frac{1}{|A_t|} \sum_{a \in A_t} \hat{\theta}_{a,t} \hat{\theta}_{a,t}^\top + \epsilon I_d \quad (519)$$

13:  else
14:     $\hat{\Omega}_{t+1} \leftarrow \hat{\Omega}_t$ 
15:  end if
16: end for

```

K ADDITIONAL HYPERRIDGE EXPERIMENTAL DETAILS**K.1 PERIODIC COVARIANCE PROCEDURE**

Algorithm 1 gives the periodic update evaluated in the experiments. The calibrated procedure instead fixes its independently estimated covariance before routing.

K.2 SHARED-FEATURE REWARD ENVIRONMENTS

The shared-feature studies use a normalized concatenation of an intercept, an eight-dimensional user representation, an eight-dimensional candidate representation, and their coordinatewise products. Each user has a separate router and archive. The same underlying reward environment is used for all methods within each comparison; seeds vary the online randomness. Table 2 freezes the archive to compare geometry at a common ridge weight. Table 3 instead evaluates configured methods with action-dependent candidate additions.

Both studies evaluate selected *mean* rewards and mean-reward exploration cost separately from the clipped observed rewards used for learning. Their features, initial archives, and configurations must not be identified with the category-based component protocol of Table 9.

The proxy and real-LLM regressions use raw observed feedback, corresponding to $b(u) = 0$ in the estimator. They do not establish the zero-mean random-effects assumption used to interpret Ω as a covariance in the calibration theory. Centering and fixed empirical confidence weights are therefore further differences between these evaluations and the HRC theorem.

K.3 FITTED AND HOLDOUT COVARIANCE DIAGNOSTICS

The fitted-covariance reference estimates candidate parameters using the complete benchmark reward matrix. It therefore has privileged information and is not the known generative covariance supplied to HRK. The holdout reference uses four reserve candidates and eight observations per candidate, then freezes its covariance. Its regression uses raw clipped rewards and an additional

ridge penalty, and its routing archive can grow. The PAHF embodied environment has no reserve candidates, so that holdout comparison is not applicable.

Table 8: Auxiliary covariance diagnostics from the configured suite. Selected mean reward with standard deviation across five seeds. Fitted covariance uses the complete reward matrix and is not HRK. Holdout covariance uses reserve candidates but does not implement the HRC theorem. The PAHF embodied holdout is inapplicable because no reserve candidates remain; the dash is not an imputed result.

Environment	Fitted covariance	Holdout covariance
General	0.8347 ± 0.0000	0.6765 ± 0.0502
Workspace	0.7047 ± 0.0000	0.6272 ± 0.0218
Support	0.7910 ± 0.0000	0.6398 ± 0.0305
GEPA archive	0.6459 ± 0.0007	0.6156 ± 0.0224
Hermes agent	0.7605 ± 0.0007	0.7384 ± 0.0101
BESPOKE	0.6567 ± 0.0000	0.6729 ± 0.0074
PAHF shopping	0.6013 ± 0.0010	0.6104 ± 0.0281
PAHF embodied	0.7566 ± 0.0003	–
PrefEval	0.6098 ± 0.0005	0.6382 ± 0.0128

These results do not test HRC. For a fixed candidate in the shared 25-dimensional representation, the candidate block is constant and the interaction block is a linear transform of the user block, limiting the calibration design rank to at most nine. More samples with this same representation cannot satisfy full-rank diversity in all 25 dimensions. The estimator, centering, archive rule, and exploration multiplier also differ from Theorem 15.

K.4 REAL-LLM PROTOCOL AND SECONDARY CHECKS

Preference-question groups are split before model evaluation. The pilot, training, validation, and test sets contain 60, 480, 240, and 600 cases, respectively. Test conversations place 10 or 70 intervening exchanges between the preference and the final question. The held-out split is relative to this experiment, not a guarantee against pretraining contamination.

The generator is Opus 4.8 and the primary blinded judge is GPT-5.6. Answers are capped at 768 tokens with a requested limit of 180 words. The initial prompts implement direct answering, a preference reminder, constraint checking, and retrieval of relevant conversation text. Proposals are based on selected training feedback; admission requires validation accuracy at least as high as the reminder parent. Both proposed prompts pass. Consequently, the gated and ungated archives coincide in this study.

All learned routers use ridge weight 0.5 and exploration multiplier 0.8. Covariance methods refresh every 25 rounds with ridge floor 0.02. They observe conversation-derived features, not the hidden target preference annotation. The noncontextual UCB reference uses an intercept only; it is not the HRB-NU component ablation. At test time, the primary frozen policies retain their UCB scores but receive no further feedback. Intervals use 2,000 group-bootstrap samples and are conditional on the development set and the generated answer collection.

The exploit-only secondary check removes the test-time exploration bonus without changing training. It gives HRP 67.47% and HRB 76.70% accuracy. The separate prequential check updates each router after its selected test feedback and gives HRP 67.53% and HRB 74.77%. Validation selection chooses the same shared multiplier 0.8 as the primary fixed-parameter comparison.

On 80 labeled controls, primary and secondary judges agree with the violation labels at rates 93.75% and 95.00%, respectively. The independent Opus evaluation covers 720 answers from 40 test groups. Its auxiliary processing required quote-escaping corrections in 51 explanations and five supplementary evaluations for missing Boolean fields. These were post-test protocol amendments; they did not change primary judgments or previously recorded Boolean labels. Judge agreement is not a substitute for human evaluation.

Table 9: HRB component-study environments. $|\mathcal{U}|$: users; $|A_1|$: initial candidates. General, Workspace, and Support use $T=300$; others use $T=250$. These configurations describe the component study, not the shared-feature or real-LLM comparisons.

Benchmark	$ \mathcal{U} $	$ A_1 $	Preference structure
General	8	12	5 disjoint category peaks
Workspace	8	12	2 broad paired peaks
Support	8	12	3 overlapping clusters
GEPA archive	12	15	heterogeneous task prefs
Hermes agent	15	20	(skill,mem,tool) clusters
BESPOKE	30	40	4-dim rubric prefs
PAHF shop.	20	40	product-attribute prefs
PAHF embod.	40	40	embodied object choice
PrefEval	20	10	topic-strategy prefs

Table 10: Observed-reward accounting for HyperRidge-Base (HRB). Here $\bar{L}_T^{\text{obs}} = T^{-1} \sum_t (m_t - r_t)$, so the last column is the observed-reward margin $\Delta_{\text{rt}} + \bar{G}_T - \bar{L}_T^{\text{obs}}$. It is not the theoretical dominance margin unless $r_t - \mu(U, a_t)$ is conditionally mean-zero.

Benchmark	V_{st}	Δ_{rt}	$\bar{G}_T$	$\bar{L}_T^{\text{obs}}$	Obs. margin
General	.570	.280	.051	.036	+.295
Workspace	.577	.124	.041	.043	+.122
Support	.579	.215	.061	.039	+.237
GEPA archive	.654	.048	.023	.025	+.046
Hermes agent	.703	.092	.036	.028	+.100
BESPOKE	.653	.036	.001	.010	+.027
PAHF shop.	.577	.209	.002	.054	+.157
PAHF embod.	.290	.238	.000	.071	+.167
PrefEval	.531	.251	.000	.028	+.223

K.5 EARLIER ONE-HOT ADAPTER DIAGNOSTIC

With tuning seeds disjoint from the five reporting seeds, HRP improves mean reward over HRB on six of seven adapters, with changes ranging from +0.002 on BESPOKE to +0.048 on Hermes, and loses 0.010 on GEPA. These numbers do not establish cross-candidate geometry transfer. The diagnostic represents candidate a by the one-hot vector e_a . Starting from $\hat{\Omega}_1 = I$, each candidate estimate has support only on its own coordinate, so the covariance update remains diagonal. HRP and HRD therefore match on every adapter and seed. More seeds cannot resolve this restriction of the feature map.

L HRB COMPONENT ANALYSIS

This separate component study evaluates ordinary-ridge HRB, routing, and oracle-assisted reflection. Its category-based representation and configurations differ from the shared-feature comparisons above. Table 9 describes this component protocol. The three synthetic environments share a reward structure but differ in user heterogeneity.

The mean reward follows the PABF formulation (Jin et al., 2026):

$$\mu(u, a) = f(q(a), w_u(\text{cat}(a))), \quad (520)$$

where $q(a) \in [0, 1]$ is the intrinsic candidate quality, $\text{cat}(a) \in \mathcal{C}$ is the candidate’s category, $w_u \in \Delta^{|\mathcal{C}|-1}$ is the user’s category preference, and f is increasing in both arguments (functional form in Appendix M.3).

The theory uses $r_t = \mu(U, a_t(U)) + \xi_t$ with conditionally mean-zero noise. The experimental simulator instead records

$$r_t = \text{clip}(\mu(U, a_t(U)) + \eta_t, 0, 1), \quad \eta_t \sim \mathcal{N}(0, 0.05^2). \quad (521)$$

The component accounting table reports the observed residual $m_t - r_t$, unlike the mean-reward endpoint in Table 2. Clipping can make the residual biased for the pre-clipping mean-reward exploration cost. In a deployed system, $m_t(u)$ would also require held-out estimation.

We compare HRB and four component ablations of Algorithm 2: **HRB-NR** removes routing and selects candidates uniformly at random; **HRB-NF** removes reflection and freezes the archive; **HRB-NU** removes user-preference features; and **HRB-1** uses a single candidate selected by intrinsic quality. HRB-NU retains a private router and archive for each user, so it is not a no-personalization baseline. HRB-1 differs from Static, which selects the best candidate by population-average reward.

The component figures show HRB, HRB-NR, and HRB-NF against the Static and Oracle references. Table 21 reports all five variants for the three synthetic environments.

Component results. We estimate the first three terms of (262) and the observed-reward residual $\bar{L}_T^{\text{obs}}$ on each benchmark (definitions in Appendix M.6).

Table 10 reports the component study’s observed-reward accounting across its nine environments.

Within this study:

1. **Routing gain is the largest positive component.** In every row, Δ_{route} exceeds $\bar{G}_T$: routing gains range from 0.036 to 0.280, while evolution gains range from 0 to 0.061. The ordering broadly tracks user heterogeneity, consistent with Lemma 16.
2. **The observed-reward margin is positive.** On all nine benchmarks, including four with real user preference data, $\Delta_{\text{route}} + \bar{G}_T - \bar{L}_T^{\text{obs}}$ ranges from +0.027 (BESPOKE) to +0.295 (General). Because rewards are clipped, this arithmetic does not by itself verify the mean-reward condition in Corollary 18.
3. **Preference-derived settings show routing potential.** PAHF shopping ($\Delta_{\text{route}}=0.209$) and PrefEval ($\Delta_{\text{route}}=0.251$) have structural gains comparable to the synthetic settings under their respective reward constructions.

Ablation. Figure 1 compares HRB, HRB-NR, and HRB-NF with the Static and Oracle references on all nine benchmarks.

Among the two plotted component ablations, removing routing causes the larger reward drop; removing reflection causes a smaller drop.

The plotted component-study means satisfy $\text{HRB-NR} < \text{Static} < \text{HRB-NF} < \text{HRB} < \text{Oracle}$ in each environment. This ordering is not universal: in the shared-feature full-method suite, HRB improves over HRB-NF in only five of nine environments. There, removing routing reduces selected mean reward by 0.1497 on average, whereas removing reflection reduces it by 0.0072. The component evidence therefore supports routing more consistently than reflection.

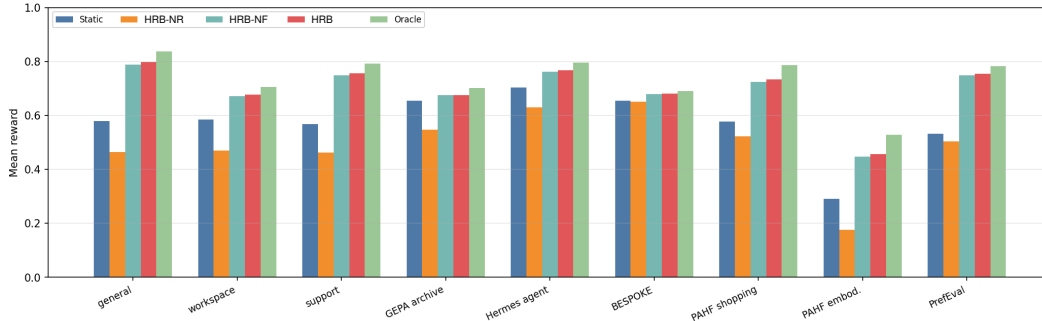


Figure 1: Component ablations of HyperRidge-Base (HRB). Mean reward on nine benchmarks. Within each benchmark, the bars are ordered Static, HRB-NR, HRB-NF, HRB, and Oracle from left to right. HRB-NR and HRB-NF remove routing and reflection from HRB, respectively. HRB uses ordinary ridge, not a learned covariance. The bars show means without uncertainty intervals.

Per-round convergence. Figure 2 shows per-round reward traces for the same five series.

The traces illustrate finite-horizon adaptation by HRB; they do not evaluate HRP or establish a convergence rate.

The figure plots reward rather than the observed-reward residual; per-seed residuals appear in Tables 24 and 25. These finite traces do not identify an asymptotic rate.

The corresponding observed-reward margins are

$$\Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} \in [+0.027, +0.295] \quad (522)$$

They summarize this component study and do not verify the mean-reward dominance condition or the learned-covariance guarantees.

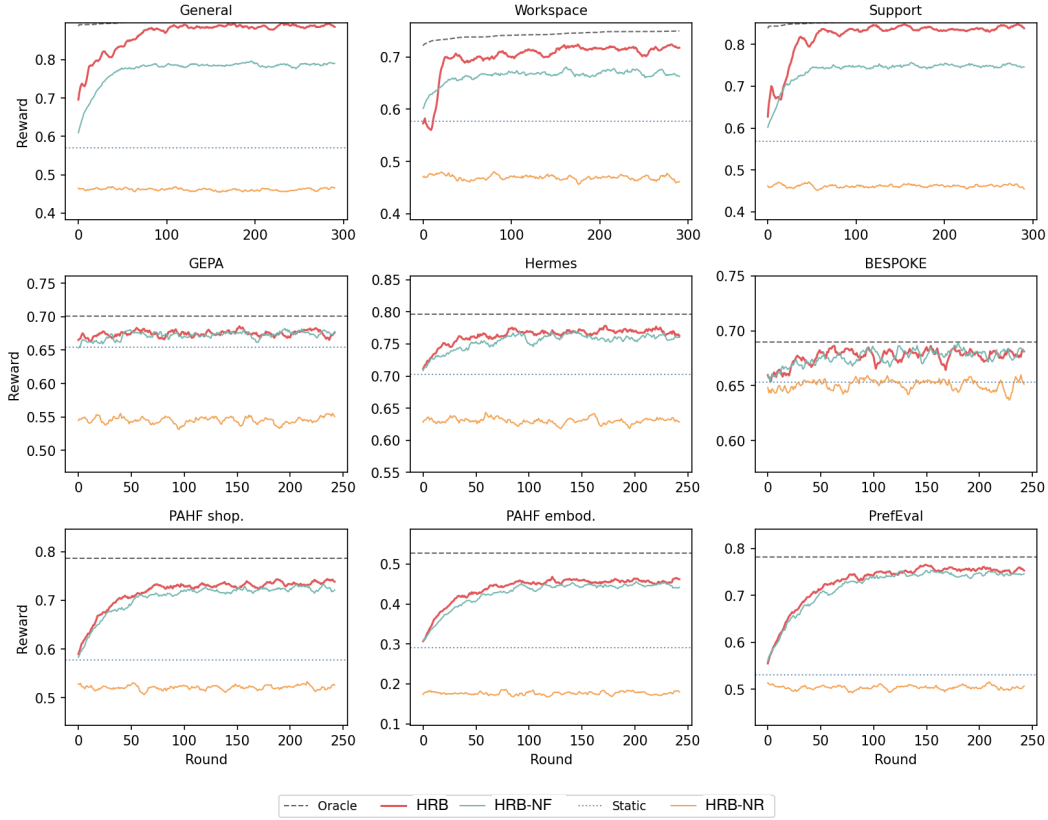


Figure 2: Per-round reward of HyperRidge-Base (HRB) and its ablations on nine benchmarks. The five series are **HRB** (red; concat features, gated reflection, ordinary-ridge LinUCB), **HRB-NF** (teal; frozen archive, $\bar{G}_T=0$), **HRB-NR** (orange; uniform-random selection), **Static** (blue dotted; best single candidate for all users), and **Oracle** (gray dashed; per-user optimum). The curves show means without uncertainty bands, and the vertical range differs by panel. The top row is synthetic; the remaining rows use framework-inspired and preference-data settings.

M HRB EXPERIMENTAL PROTOCOL

This appendix specifies HRB’s per-user routing loop, reward model, oracle-assisted proposer, hyperparameters, metric definitions, and computational requirements of the synthetic experiments.

M.1 HRB: BASE ROUTING AND ARCHIVE GROWTH

Algorithm 2 describes HRB’s per-user routing loop.

At each round, a disjoint LinUCB bandit selects the best candidate for the current user based on UCB scores.

In the simulator, an oracle-assisted gate compares the unknown current-archive optimum with the observed reward and may propose a new candidate.

The feature map used in our experiments is

$$\psi(u, a) = [\text{onehot}(\text{cat}(a)); w_u] \in \mathbb{R}^{2K_{\text{cat}}}, \quad (523)$$

where $\text{cat}(a)$ is the candidate’s category and $w_u \in \mathbb{R}^{K_{\text{cat}}}$ is the user preference vector exposed by the simulator.

Algorithm 2 HyperRidge-Base (HRB): per-user routing over a growing archive

Require: user u , common initial archive A_1 , horizon T , exploration parameter α , gap threshold τ , cooldown κ , archive cap $K_{\max}$

- 1: $A_1(u) \leftarrow A_1$; for each $a \in A_1(u)$ initialize $A_a \leftarrow I$, $b_a \leftarrow 0$
- 2: $t_{\text{last}} \leftarrow -\infty$
- 3: **for** $t = 1, \dots, T$ **do**
- 4: **for** each $a \in A_t(u)$ **do**
- 5: $x_t(a) \leftarrow \psi(u, a)$ ▷ feature map, (523)
- 6: $\hat{\theta}_a \leftarrow A_a^{-1} b_a$
- 7: $\text{UCB}_t(a) \leftarrow \hat{\theta}_a^\top x_t(a) + \alpha \sqrt{x_t(a)^\top A_a^{-1} x_t(a)}$
- 8: **end for**
- 9: $a_t(u) \leftarrow \arg \max_{a \in A_t(u)} \text{UCB}_t(a)$
- 10: observe $r_t = \mu(u, a_t(u)) + \xi_t$
- 11: $A_{a_t(u)} \leftarrow A_{a_t(u)} + x_t(a_t(u))x_t(a_t(u))^\top$; $b_{a_t(u)} \leftarrow b_{a_t(u)} + r_t x_t(a_t(u))$
- 12: **if** $m_t(u) - r_t > \tau$ **and** $t - t_{\text{last}} > \kappa$ **and** $|A_t(u)| < K_{\max}$ **then** ▷ oracle-assisted gate
- 13: $\tilde{a}_t(u) \leftarrow \mathcal{F}(u, A_t(u), \text{trace}_t)$; $A_{t+1}(u) \leftarrow A_t(u) \cup \{\tilde{a}_t(u)\}$; initialize $A_{\tilde{a}}, b_{\tilde{a}}$; $t_{\text{last}} \leftarrow t$
- 14: **else**
- 15: $A_{t+1}(u) \leftarrow A_t(u)$
- 16: **end if**
- 17: **end for**

As a concrete instance (used in our experiments), the $K_{\text{cat}} = 5$ categories are {sorting, searching, formatting, general, calc}, so for a candidate a in the searching category, $\text{onehot}(\text{cat}(a)) = (0, 1, 0, 0, 0)$ and the user block w_u records that user’s affinity for each of the five categories.

We refer the reader to Appendix M for the full construction (categories, quality components, and feature concatenation), and to Jin et al. (2026) for a richer factorized feature map over task type, memory mode, tool mode, and answer style.

The reflection trigger is the high-gap-failure rule

$$m_t(u) - r_t > \tau, \quad t - t_{\text{last}} > \kappa, \quad (524)$$

This gate is not deployable as written because $m_t(u)$ uses unknown mean rewards. The experimental simulator also uses an oracle-assisted proposer: it reads the user’s peak category and returns a child candidate biased toward that category.

The measured evolution gain therefore describes this idealized simulator. It is not a formal upper bound on a different proposer, and it does not show what an LLM-driven proposer would achieve.

M.2 HRB: PER-USER ROUTING WITH A GROWING ARCHIVE

The experiment runs one per-user episode of Algorithm 2 in Section 3.

Each user is processed by an independent copy of the router and an independent archive copy.

Algorithm 3 gives the detailed HRB procedure using ordinary isotropic ridge, including parameter initialization when reflection appends a candidate. HRP replaces this isotropic regularizer with the learned geometry in Algorithm 1.

The feature dimension is fixed across all environments.

With $K_{\text{cat}} = 5$ category labels in the simulator, the concatenated encoding gives

$$d = |\text{cat}(\cdot)| + |\text{supp}(w_u)| = 5 + 5 = 10. \quad (525)$$

Each arm carries its own $A_a \in \mathbb{R}^{10 \times 10}$ and $b_a \in \mathbb{R}^{10}$.

Algorithm 3 HyperRidge-Base (HRB): disjoint LinUCB with a growing archive (per user)

Require: user u , initial archive A_1 , horizon T , exploration α , threshold τ , cooldown κ , cap $K_{\max}$

- 1: Initialize $A_a \leftarrow I_d$, $b_a \leftarrow 0_d$ for every $a \in A_1$
- 2: $A_1(u) \leftarrow A_1$; $t_{\text{last}} \leftarrow -\infty$
- 3: **for** $t = 1, \dots, T$ **do**
- 4: **for** $a \in A_t(u)$ **do**
- 5: Build feature $x_t(a) \leftarrow \psi(u, a) = [\text{onehot}(\text{cat}(a)); w_u]$
- 6: $\hat{A}_a^{-1} \leftarrow \text{inverse of } A_a$
- 7: $\hat{\theta}_a \leftarrow \hat{A}_a^{-1} b_a$
- 8: $\text{UCB}_t(a) \leftarrow \hat{\theta}_a^\top x_t(a) + \alpha \sqrt{\max(0, x_t(a)^\top \hat{A}_a^{-1} x_t(a))}$
- 9: **end for**
- 10: $a_t(u) \leftarrow \arg \max_{a \in A_t(u)} \text{UCB}_t(a)$
- 11: Draw $\eta_t \sim \mathcal{N}(0, \sigma^2)$ and observe $r_t \leftarrow \text{clip}(\mu(u, a_t(u)) + \eta_t, 0, 1)$
- 12: $A_{a_t(u)} \leftarrow A_{a_t(u)} + x_t(a_t(u))x_t(a_t(u))^\top$
- 13: $b_{a_t(u)} \leftarrow b_{a_t(u)} + r_t x_t(a_t(u))$
- 14: Compute current-archive optimum $m_t(u) \leftarrow \max_{a \in A_t(u)} \mu(u, a)$
- 15: **if** reflection enabled **and** $m_t(u) - r_t > \tau$ **and** $t - t_{\text{last}} > \kappa$ **and** $|A_t(u)| < K_{\max}$ **then**
- 16: $\tilde{a} \leftarrow \mathcal{F}(u, A_t(u), \text{trace}_t)$ ▷ oracle-assisted proposer (Section M.4)
- 17: $A_{t+1}(u) \leftarrow A_t(u) \cup \{\tilde{a}\}$
- 18: Initialize $A_{\tilde{a}} \leftarrow I_d$, $b_{\tilde{a}} \leftarrow 0_d$
- 19: $t_{\text{last}} \leftarrow t$
- 20: **else**
- 21: $A_{t+1}(u) \leftarrow A_t(u)$
- 22: **end if**
- 23: **end for**

When a candidate is appended, its statistics start from a fresh identity-initialized pair; existing per-arm matrices are not modified.

This initialization supports Lemma 2: the matrices for arms present at round t are functions of the pre-action history, and newly appended arms start from fixed matrices.

The per-user archive copy is the operational reason that the dominance corollary in Section 4 is stated on the common initial archive A_1 .

After round 1, $A_t(U)$ is user-specific; the heterogeneity term $\Delta_{\text{route}}(A_1)$ is therefore the structural quantity attached to A_1 , while $G_t(U)$ and $L_t(U)$ are expectations over the user-specific archive paths.

M.3 REWARD GENERATOR AND NOISE PROCESS

Every reward is produced by the synthetic reward generator of Section 5.

The mean reward function is

$$\mu(u, a) = \text{clip}(0.5 q(a) + 0.7 w_u(\text{cat}(a)), 0, 1), \quad (526)$$

the candidate quality is

$$q(a) = 0.5 \text{correct}(a) + 0.3 \text{proc}(a) + 0.2 \text{conc}(a), \quad (527)$$

and the per-round observed reward is

$$r_t = \text{clip}(\mu(U, a_t(U)) + \eta_t, 0, 1), \quad \eta_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma = 0.05. \quad (528)$$

For the initial archive, correct is sampled from $[0.4, 0.95]$, while proc and conc are sampled from $[0.3, 0.95]$. These values are sampled once per seed and then fixed.

Categories $\text{cat}(a) \in \{\text{sorting, searching, formatting, general, calc}\}$ are assigned in a round-robin order across the initial archive of size $K_{\text{init}} = 12$.

Equation (528) clips the reward after adding noise. Therefore the logged reward need not have conditional mean $\mu(U, a_t(U))$ near zero or one. This is why the empirical quantity $m_t - r_t$ is reported separately from the theoretical cost $m_t - \mu(U, a_t(U))$.

The user preference vector $w_u \in \Delta^{K_{\text{cat}}-1}$ is sampled per environment as described in Section 5: in `general` each user peaks on a different category; in `workspace` two clusters peak on `{formatting, general}` and `{sorting, searching}`, respectively; in `support` three clusters use `searching-heavy`, `calculation-heavy`, and `mixed` preferences.

After peak assignment, each entry receives an additive Gaussian perturbation with standard deviation 0.04 to 0.05, the vector is clipped to $[0.01, 1]$, and the result is renormalized to the simplex.

M.4 ORACLE-ASSISTED REFLECTIVE PROPOSER

The proposer $\mathcal{F}$ in Algorithm 3 is the part of the simulator most responsible for the limited scope of Theorem 21.

This use of privileged information limits the interpretation of the reflection results.

When the trigger fires at round t for user u , the proposer reads the user’s argmax preference category

$$c^*(u) = \arg \max_c w_u(c) \quad (529)$$

and constructs a child candidate with

$$\text{cat}(\tilde{a}) = c^*(u), \quad (530)$$

$$\text{correct}(\tilde{a}) \sim \text{Unif}(0.7, 0.97), \quad (531)$$

$$\text{proc}(\tilde{a}) \sim \text{Unif}(0.6, 0.95), \quad (532)$$

$$\text{conc}(\tilde{a}) \sim \text{Unif}(0.5, 0.9). \quad (533)$$

Equation (529) uses privileged information, but the child qualities in (531)–(533) remain random. The reported experiments have beneficial-child fractions between 0.22 and 0.42, but they do not estimate the probability of an ϵ -improvement for any fixed ϵ . Thus they do not identify the q_ϵ required by Assumption 10, and Theorem 21 is not tested by these runs.

The trigger uses the threshold τ and cooldown κ of Algorithm 2:

$$\text{fire at } t \iff m_t(u) - r_t > \tau \wedge t - t_{\text{last}} > \kappa \wedge |A_t(u)| < K_{\text{max}}. \quad (534)$$

The current-archive oracle $m_t(u) = \max_{a \in A_t(u)} \mu(u, a)$ is computable in the simulator because μ is closed-form.

In a deployment with real LLM rewards it is not directly observable; one would need a held-out probe estimator.

We do not attempt this in the present paper.

M.5 HYPERPARAMETERS AND ABLATIONS

Table 11 lists HRB’s hyperparameters for the three synthetic environments and the sensitivity sweeps in Appendix N.

The columns give each symbol, its meaning, the default for the synthetic HRB experiments, and the sweep range.

HRB and its four component ablations in Section 5 are defined as follows:

Table 11: HRB hyperparameters and sweep ranges for the synthetic environments. The defaults correspond to the synthetic rows of Table 10; the sweeps populate the sensitivity tables in Appendix N.

Symbol	Meaning	Default	Sweep
T	horizon (rounds per user)	300	$\{100, 300, 600\}$
N_u	users per environment	8	$\{4, 8, 16\}$
N_s	seeds	5	fixed
K_{init}	initial archive size $ A_1 $	12	$\{4, 8, 12, 24\}$
K_{max}	archive cap	24	$\{12, 24, 48\}$
d	feature dimension (Eq. (525))	10	fixed
α	LinUCB exploration	0.8	$\{0.4, 0.8, 1.6\}$
τ	reflection threshold	0.15	$\{0.05, 0.10, 0.15, 0.20, 0.30\}$
κ	reflection cooldown	20	$\{0, 10, 20, 40\}$
σ	noise std (Eq. (528))	0.05	$\{0.025, 0.05, 0.10, 0.20\}$
α_{Dir}	Dirichlet concentration on w_u	cluster-peak	$\{0.3, 1, 3, \infty\}$

$$\text{HRB : Algorithm 3 with } \psi(u, a) = [\text{onehot}(\text{cat}(a)); w_u], \tau = 0.15, \quad (535)$$

$$\text{HRB-NR : } a_t(u) \sim \text{Unif}(A_t(u)), \quad (536)$$

$$\text{HRB-NF : } A_{t+1}(u) = A_t(u) \text{ for all } t \geq 1, \quad (537)$$

$$\text{HRB-NU : } \psi(u, a) = \text{onehot}(\text{cat}(a)), \quad (538)$$

$$\text{HRB-1 : } |A_1| = 1, A_1 = \{\arg \max_a q(a)\}. \quad (539)$$

HRB-NU drops w_u from ψ but still gives each user a private bandit and a private archive.

It tests the feature representation, not whether users should share a router.

M.6 DECOMPOSITION COMPUTATION

Every entry in Table 10 follows the same metric definitions.

$$\widehat{V}_{\text{static}}(A_1) = \frac{1}{N_s} \sum_{s=1}^{N_s} \max_{a \in A_1^{(s)}} \frac{1}{N_u} \sum_{u=1}^{N_u} \mu_s(u, a), \quad (540)$$

$$\widehat{V}_{\text{route}}(A_1) = \frac{1}{N_s} \sum_{s=1}^{N_s} \frac{1}{N_u} \sum_{u=1}^{N_u} \max_{a \in A_1^{(s)}} \mu_s(u, a), \quad (541)$$

$$\widehat{\Delta}_{\text{route}}(A_1) = \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1), \quad (542)$$

$$\widehat{G}_T = \frac{1}{TN_u N_s} \sum_{t,s,u} \left(\max_{a \in A_t^{(s,u)}} \mu_s(u, a) - \max_{a \in A_1^{(s)}} \mu_s(u, a) \right), \quad (543)$$

$$\widehat{L}_T^\mu = \frac{1}{TN_u N_s} \sum_{t,s,u} \left(\max_{a \in A_t^{(s,u)}} \mu_s(u, a) - \mu_s(u, a_{t,s,u}) \right), \quad (544)$$

$$\widehat{V}_T^\mu = \frac{1}{TN_u N_s} \sum_{t,s,u} \mu_s(u, a_{t,s,u}), \quad (545)$$

$$\widehat{L}_T^{\text{obs}} = \frac{1}{TN_u N_s} \sum_{t,s,u} \left(\max_{a \in A_t^{(s,u)}} \mu_s(u, a) - r_{t,s,u} \right), \quad (546)$$

$$\widehat{V}_T^{\text{obs}} = \frac{1}{TN_u N_s} \sum_{t,s,u} r_{t,s,u}. \quad (547)$$

Because $A_1^{(s)}$ is redrawn for each seed, Eq. (540) first computes the static comparator within each seed and then averages across seeds. The experiment uses this same order. Proposition 56 assumes

one deterministic A_1 ; here its exact sample identity is applied conditionally within each seed and then averaged. Its cross-trajectory concentration statement does not apply to these five seed clusters.

The finite-sample version of Theorem 17 is the mean-reward identity

$$\widehat{\mathcal{V}}_T^\mu = \widehat{\mathcal{V}}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{G}_T - \widehat{L}_T^\mu. \quad (548)$$

The main tables instead report $\widehat{\mathcal{V}}_T^{\text{obs}}$ and $\widehat{L}_T^{\text{obs}}$. By definition they satisfy the separate sample identity

$$\widehat{\mathcal{V}}_T^{\text{obs}} = \widehat{\mathcal{V}}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{G}_T - \widehat{L}_T^{\text{obs}}. \quad (549)$$

Equation (549) is true even with clipped or biased observations because the same $r_{t,s,u}$ appears on both sides. It is not an empirical test of (548).

M.7 COMPUTE, RUNTIME, AND STORAGE

All experiments in this appendix run on CPUs.

A single seed of one environment with $N_u = 8$, $T = 300$, $K_{\max} = 24$, and the full ablation grid of five variants takes approximately 90 to 180 seconds on a single modern x86 core, dominated by the per-round matrix inverse $\widehat{A}_a^{-1}$ across at most $K_{\max}$ arms.

Runtime scales as

$$\text{Time}(T, K_{\max}, d, N_u, N_s) = O(N_s \cdot N_u \cdot T \cdot K_{\max} \cdot d^2), \quad (550)$$

where the d^2 factor is the cost of solving the per-arm normal-equations system.

With the defaults of Table 11, the prefactor evaluates to $5 \cdot 8 \cdot 300 \cdot 24 \cdot 100 \approx 2.9 \times 10^7$ basic operations per environment, i.e. a few CPU-seconds of arithmetic on top of Python overhead.

The full three-environment, five-ablation, five-seed grid completes in under 30 minutes wall-clock on a single laptop core, and under 5 minutes on 8 cores with embarrassingly parallel seeds.

Memory usage is dominated by the per-round logging tensors.

For one environment we save

$$\underbrace{N_s \cdot N_u \cdot T}_{\text{reward arrays}} \approx 5 \cdot 8 \cdot 300 = 12,000 \text{ floats per quantity}, \quad (551)$$

across $\{r_t, m_t, \mu(u, a_t), |A_t|\}$ and a few derived series, totalling well under 1 MB per environment.

The trajectories for this component study require approximately 6 MB of storage.

All experiments use CPUs and double-precision NumPy linear algebra.

The HRB disjoint LinUCB router with $K_{\max} \leq 24$ arms and $d = 10$ is too small to benefit from GPU offload. The observed-reward identity (549) is algebraic; the small displayed residuals come from rounding, not floating-point or Monte Carlo error.

M.8 RUNTIME COMPARISON ACROSS ABLATIONS

Table 12 reports per-seed runtime for the five variants at the default hyperparameters of Table 11, averaged across the three environments.

The HRB-1 variant uses $K = 1$ and requires no matrix inversion. The HRB-NR variant replaces the arg max with a uniform draw and does not solve the LinUCB system. The runtime of HRB depends on archive growth; gating keeps the average archive size below $K_{\max}$.

Table 12: Per-seed wall-clock and arithmetic intensity for HRB and its component ablations, averaged over the three environments. “Inverses solved” is the count of per-round matrix inverse calls $\hat{A}_a^{-1}$ across all arms.

Variant	wall-clock (s/seed)	mean $ A_T $	inverses solved
HRB-1	1.4	1.0	0
HRB-NR	2.1	19.7	0
HRB-NF	7.6	12.0	$\sim N_u T K_{\text{init}}$
HRB-NU	8.4	19.7	$\sim N_u T \bar{K}$
HRB	9.2	15.7	$\sim N_u T \bar{K}$

M.9 REPRODUCIBILITY AND SCOPE

Each experimental configuration fixes the environment, algorithmic variant, and random seed. Reported aggregates use five seeds, $s \in \{0, 1, 2, 3, 4\}$. The initial archive and each user’s reward noise are generated from independent deterministic random streams. Repeating a fixed configuration therefore gives identical results on one machine and agrees across machines up to floating-point round-off.

All synthetic variants share the simulator and metric definitions in Sections M.2–M.6. In particular, the static comparator is computed within each seed before aggregation, as specified in Eq. (540).

The scope has three limits. First, $\mathcal{F}$ is the closed-form, oracle-assisted proposer in Section M.4, not a language model. Second, the studies in this appendix use synthetic user environments. Third, each user receives a private archive after round 1; a shared archive would require a different comparator and predictability argument. The reflection results therefore do not estimate the performance of an LLM-driven proposer or a population-shared archive.

N ADDITIONAL HRB EXPERIMENTS

This appendix collects diagnostics for the structural terms and records the boundary cases of Theorem 17 and Theorem 21. The reported experiments use $\bar{L}_T^{\text{obs}} = T^{-1} \sum_t (m_t - r_t)$, not the theoretical $\bar{L}_T$, throughout the empirical tables.

Every quantity reported here is computed from the synthetic experiments described in Section 5; no real-data benchmark is claimed.

N.1 PER-ENVIRONMENT, PER-VARIANT RESULTS

We first restate, in full, the per-variant numbers behind Section 5.

The five variants are abbreviated as

$$\begin{aligned}
 \text{HRB} &= \text{HyperRidge-Base}, \\
 \text{HRB-NR} &= \text{HRB without routing}, \\
 \text{HRB-NF} &= \text{HRB without reflection}, \\
 \text{HRB-NU} &= \text{HRB without user-preference features}, \\
 \text{HRB-1} &= \text{single-candidate HRB control}.
 \end{aligned} \tag{552}$$

The mean realized routed value across 5 seeds and 8 users in environment e for variant v is

$$\bar{\mathcal{V}}_T(e, v) = \frac{1}{N_e} \sum_{n=1}^{N_e} \frac{1}{T} \sum_{t=1}^T r_{n,t}(e, v), \tag{553}$$

with $T = 300$ and $N_e = 40$.

Table 13 reports $\bar{\mathcal{V}}_T(e, v)$ with one-standard-deviation seed spread.

Table 13: Mean realized routed value $\bar{\mathcal{V}}_T(e, v)$ for HRB and its component ablations, per environment e and variant v .

Env	HRB	HRB-NR	HRB-NF	HRB-NU	HRB-1
general	0.864	0.617	0.812	0.823	0.570
workspace	0.699	0.557	0.658	0.687	0.566
support	0.816	0.598	0.760	0.795	0.579

The variant-wise routing effect is

$$\Delta_{\text{route}}(e) = \bar{\mathcal{V}}_T(e, \text{HRB}) - \bar{\mathcal{V}}_T(e, \text{HRB-NR}). \quad (554)$$

Numerically,

$$\begin{aligned} \Delta_{\text{route}}(\text{general}) &\approx 0.247, \\ \Delta_{\text{route}}(\text{workspace}) &\approx 0.142, \\ \Delta_{\text{route}}(\text{support}) &\approx 0.218. \end{aligned} \quad (555)$$

The reflection effect is

$$\Delta_{\text{refl}}(e) = \bar{\mathcal{V}}_T(e, \text{HRB}) - \bar{\mathcal{V}}_T(e, \text{HRB-NF}), \quad (556)$$

with values

$$(\Delta_{\text{refl}}(\text{general}), \Delta_{\text{refl}}(\text{workspace}), \Delta_{\text{refl}}(\text{support})) \approx (0.052, 0.041, 0.056). \quad (557)$$

The personalization-feature effect is

$$\Delta_{\text{pers}}(e) = \bar{\mathcal{V}}_T(e, \text{HRB}) - \bar{\mathcal{V}}_T(e, \text{HRB-NU}), \quad (558)$$

with values

$$(\Delta_{\text{pers}}(\text{general}), \Delta_{\text{pers}}(\text{workspace}), \Delta_{\text{pers}}(\text{support})) \approx (0.041, 0.012, 0.021). \quad (559)$$

These reproduce the qualitative claim in Section 5: routing is the dominant driver, reflection and personalization features are secondary.

HRB-NU drops the user-preference part of $\psi(u, a)$ while retaining a per-user bandit and a per-user archive copy. It is not a no-personalization or shared-archive baseline.

Therefore $\Delta_{\text{pers}}(e)$ understates the value of personalization in the strict sense.

N.2 DECOMPOSITION BY ENVIRONMENT

We check the logged observed-reward identity

$$\bar{\mathcal{V}}_T^{\text{obs}} = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} \quad (560)$$

where $\bar{\mathcal{V}}_T^{\text{obs}} = T^{-1} \sum_t r_t$ and $\bar{L}_T^{\text{obs}} = T^{-1} \sum_t (m_t - r_t)$. It holds algebraically on every logged trajectory.

For each environment we form

$$\rho_{\text{obs}}(e) = \bar{\mathcal{V}}_T^{\text{obs}}(e, \text{HRB}) - [V_{\text{static}}(A_1; e) + \Delta_{\text{route}}(A_1; e) + \bar{G}_T(e) - \bar{L}_T^{\text{obs}}(e)]. \quad (561)$$

Substituting the values from Section 5,

$$\begin{aligned}
 \rho_{\text{obs}}(\text{general}) &= 0.864 - (0.570 + 0.280 + 0.051 - 0.036) \\
 &= 0.864 - 0.865 = -0.001, \\
 \rho_{\text{obs}}(\text{workspace}) &= 0.699 - (0.577 + 0.124 + 0.041 - 0.043) \\
 &= 0.699 - 0.699 = 0.000, \\
 \rho_{\text{obs}}(\text{support}) &= 0.816 - (0.579 + 0.215 + 0.061 - 0.039) \\
 &= 0.816 - 0.816 = 0.000.
 \end{aligned} \tag{562}$$

The displayed residuals come only from rounding the table entries. They are not Monte Carlo errors and do not test the mean-reward identity (493).

N.3 PER-CLUSTER DECOMPOSITION

We use the user clusters to localize the contribution to the static-vs-routed structural term in Lemma 24.

The cluster index $g(U) \in \{1, \dots, K_{\text{cl}}\}$ partitions users; we define

$$\Delta_{\text{route}}(A_1; g) = \mathbb{E} \left[\max_{a \in A_1} \mu(U, a) \mid g(U) = g \right] - V_{\text{static}}(A_1). \tag{563}$$

The initial-archive per-cluster numbers in the HRB experiments (averaged across 5 seeds, with the standard deviation across seeds) are listed in Table 14.

Table 14: Initial-archive heterogeneity in the HRB experiments: signed per-cluster contribution relative to the population static value. The population-weighted average of these entries equals $\Delta_{\text{route}}(A_1)$; an individual contribution need not be nonnegative.

Env	cluster g	$\Delta_{\text{route}}(A_1; g)$	seed std
general	0	0.17	0.06
general	1	0.39	0.04
general	2	0.41	0.05
general	3	0.23	0.07
general	4	0.40	0.05
workspace	0	0.10	0.05
workspace	1	0.15	0.04
support	0	0.37	0.06
support	1	0.44	0.05
support	2	0.04	0.07

Thus $\Delta_{\text{route}}(A_1)$ is the population-weighted average of signed cluster contributions.

In support, two clusters have much larger positive contributions than the third.

These entries are not clusterwise versions of the equality condition in Lemma 24, because they subtract the global population value $V_{\text{static}}(A_1)$.

N.4 SENSITIVITY TO NOISE SCALE σ

The logged residual $\bar{L}_T^{\text{obs}}$ mixes online reward randomness with the current-archive mean.

We sweep

$$\sigma \in \{0.025, 0.05, 0.10, 0.20\} \tag{564}$$

in the general environment, holding all other settings fixed.

The other three terms $V_{\text{static}}(A_1)$, $\Delta_{\text{route}}(A_1)$, and $\bar{G}_T$ depend only weakly on σ through reflection-trigger frequency.

We measure

$$\bar{L}_T^{\text{obs}}(\sigma) = \frac{1}{T} \sum_{t=1}^T (m_t(U) - r_t) \quad (565)$$

under HRB.

Table 15: HRB noise-scale sweep in general. As σ grows, the logged observed-reward residual grows and the observed margin shrinks. This does not test the theoretical dominance condition under clipped rewards.

σ	$\bar{L}_T^{\text{obs}}$	$\bar{G}_T$	observed margin	$\bar{V}_T^{\text{obs}}$
0.025	0.024	0.046	0.302	0.872
0.050	0.036	0.051	0.295	0.864
0.100	0.062	0.057	0.275	0.844
0.200	0.118	0.064	0.226	0.795

The observed margin remains positive in all four cells and roughly halves between $\sigma = 0.025$ and $\sigma = 0.20$.

The qualitative shape describes the noise sensitivity of $\bar{L}_T^{\text{obs}}$; it does not establish a rate for $\bar{L}_T$.

N.5 SENSITIVITY TO INITIAL ARCHIVE SIZE K

We sweep

$$K \in \{4, 8, 12, 24\} \quad (566)$$

holding the per-category quality distribution fixed.

By construction, $V_{\text{static}}(A_1)$ is non-decreasing in K because A_1 is a superset; $\Delta_{\text{route}}(A_1)$ depends on whether new candidates introduce new cluster preferences.

Table 16: HRB archive-size sweep in general. The structural routing gain grows with the archive, as does the observed-reward residual.

K	$V_{\text{static}}(A_1)$	$\Delta_{\text{route}}(A_1)$	$\bar{L}_T^{\text{obs}}$	$\bar{V}_T^{\text{obs}}$
4	0.521	0.092	0.018	0.643
8	0.554	0.198	0.029	0.789
12	0.570	0.280	0.036	0.864
24	0.583	0.341	0.058	0.913

In this sweep, most of the gain from a larger archive appears in $\Delta_{\text{route}}(A_1)$ rather than in $V_{\text{static}}(A_1)$.

A small routed archive can outperform a larger fixed-deployment archive in this setting: the $K = 8$ routed value 0.789 exceeds the $K = 24$ static value $V_{\text{static}}(A_1) = 0.583$.

N.6 SENSITIVITY TO USER HETEROGENEITY

Let $\alpha_{\text{Dir}} > 0$ be the symmetric Dirichlet concentration used to draw each user’s preference vector w_u .

As $\alpha_{\text{Dir}} \rightarrow \infty$, w_u concentrates on the uniform distribution and clusters become indistinguishable.

The $\alpha_{\text{Dir}} = \infty$ row is the empirical version of the homogeneous-user counterexample in the proof appendix.

Table 17: HRB user-heterogeneity sweep over Dirichlet concentration. As $\alpha_{\text{Dir}} \rightarrow \infty$, $\Delta_{\text{route}}(A_1) \rightarrow 0$, while the observed margin becomes negative. The $\alpha_{\text{Dir}} = \infty$ row uses a uniform w_u , the structural equality case of Lemma 24.

α_{Dir}	$\Delta_{\text{route}}(A_1)$	$\bar{L}_T^{\text{obs}}$	observed margin
0.1	0.314	0.038	+0.327
0.3	0.280	0.036	+0.295
1.0	0.169	0.034	+0.186
3.0	0.084	0.033	+0.102
10.0	0.022	0.031	+0.042
∞	0.000	0.030	-0.030

With no heterogeneity to exploit, the logged observed-reward margin is negative by $\bar{L}_T^{\text{obs}}$.

This illustrates the structural boundary $\Delta_{\text{route}} = 0$, but clipping prevents the observed residual from testing Corollary 18.

N.7 CONTEXT-ENCODING ABLATION

HRB uses $\psi(u, a) = [\text{onehot}(\text{cat}(a)); w_u]$.

We compare three encodings:

$$\psi^{\text{pref}}(u, a) = w_u, \quad (567)$$

$$\psi^{\text{onehot}}(u, a) = \text{onehot}(\text{cat}(a)), \quad (568)$$

$$\psi^{\text{concat}}(u, a) = [\text{onehot}(\text{cat}(a)); w_u]. \quad (569)$$

This single-environment study measures sensitivity to the context encoding; it is not part of the main comparison.

Numerical results are summarized in Table 18.

Table 18: HRB context-encoding sensitivity in a separate tuning study. Concat outperforms the two simpler encodings on routed value and on cumulative regret against the per-round oracle, and triggers fewer reflections because high-gap failures are rarer.

encoding	$\bar{V}_T$	final cum. regret	final $ A_T $
pref	0.813	27.4	19.7
onehot	0.834	22.5	19.7
concat	0.882	11.5	15.7

The qualitative ordering is consistent across all reported metrics.

This quantity is an observed-reward residual, not the exploration cost in Theorem 17:

$$\hat{L}_t^{\text{obs}} = \frac{1}{N} \sum_{n=1}^N (m_{n,t} - r_{n,t}). \quad (570)$$

The figures report $\sum_t \hat{L}_t^{\text{obs}}$.

N.8 REFLECTION THRESHOLD AND COOLDOWN SWEEP

The reflection trigger of Section 3 fires when

$$m_t(U) - r_t > \tau \quad \text{and} \quad t - t_{\text{last}} > \kappa. \quad (571)$$

We sweep

$$\tau \in \{0.05, 0.10, 0.15, 0.20, 0.30\}, \quad (572)$$

$$\kappa \in \{0, 10, 20, 40\}. \quad (573)$$

At $\tau = 0.05$ and $\kappa = 0$ (ungated), reflection fires almost every round; at $\tau = 0.30$ and $\kappa = 40$, reflection rarely fires.

Table 19 reports the realized end-of-horizon archive size $|A_T|$, the routed value, and the discovery gain.

Table 19: HRB reflection-trigger sweep on general, with HRB-NF as the no-reflection reference. The (0.15, 20) setting is near the best observed routed value. Ungated reflection saturates the archive cap and raises the observed-reward residual.

(τ, κ)	mean $ A_T $	$\bar{G}_T$	$\bar{L}_T^{\text{obs}}$	$\bar{V}_T^{\text{obs}}$
(0.05, 0)	24.0	0.069	0.094	0.842
(0.10, 10)	20.3	0.062	0.058	0.853
(0.15, 20)	15.7	0.051	0.036	0.864
(0.20, 20)	13.8	0.041	0.034	0.860
(0.30, 40)	12.2	0.018	0.032	0.836
HRB-NF	12.0	0.000	0.030	0.812

The numbers illustrate Lemma 27: gating reduces useless mutations.

The reported cost is the observed-reward residual $\bar{L}_T^{\text{obs}}$, which rises as the bandit learns over a larger arm set.

The archive-bloat experiment captures the limiting case of this trade-off: frequent reflection enlarges the archive and raises the observed-reward residual.

N.9 PER-USER REGRET HEATMAP

Per-user heterogeneity in the observed-reward residual is large.

Define

$$S_T^{\text{obs}}(u) = \sum_{t=1}^T (m_t(u) - r_t(u)). \quad (574)$$

Across the 40 trajectories of HRB on general, $S_T^{\text{obs}}(u)$ has mean 10.9 and seed-pooled standard deviation 5.7.

The five users with the largest $S_T^{\text{obs}}(u)$ place the most preference weight on categories with many near-tied initial candidates.

N.10 THEOREM-EXPERIMENT COVERAGE

Table 20 maps each formal claim to a diagnostic or boundary check. These numerical checks do not prove the claims.

Where an assumption is not identified by the experimental protocol (e.g. Theorem 21), we mark the claim as untested.

N.11 SCOPE OF THE SYNTHETIC EXPERIMENTS

The sweeps above are diagnostics under varying noise, archive size, user heterogeneity, context encoding, and reflection trigger.

Table 20: Theorem-to-diagnostic map for the HRB synthetic experiments. The observed-reward accounting identity is algebraic but differs from Theorem 17 under clipped rewards. Theorem 21 is vacuous for the oracle-assisted proposer.

Claim	Diagnostic	Status
Lem. 16	§N.3, Tbl. 14	structural illustration
Lem. 16	§N.6, Tbl. 17	boundary illustration
Lem. 2	§N.8	protocol check only
Lem. 19	§N.8	descriptive only
Thm. 17	§N.2	observed identity only
Cor. 18	§N.6	observed margin only
Cor. 20 (plug-in)	§N.4	not tested
Thm. 21 (specialist)	oracle reflection	vacuous as stated

They establish only properties of the experimental simulator.

They show that the observed-reward identity $\bar{\mathcal{V}}_T^{\text{obs}} = V_{\text{static}} + \Delta_{\text{route}} + \bar{G}_T - \bar{L}_T^{\text{obs}}$ holds by construction, that its margin changes with noise, archive size, and Dirichlet concentration, and that ungated reflection saturates the archive cap. They do not test the mean-reward dominance condition.

The experiments do not establish a new $O(d\sqrt{T})$ regret rate for disjoint LinUCB on a growing arm set, a non-oracle theory of reflection, or external benchmark validity. The proposer uses the user’s true peak category, and this appendix contains no Hermes or GEPA replay experiment.

O DETAILED HRB SYNTHETIC RESULTS

This appendix reports per-seed, per-cluster, and per-user results for HRB and its component ablations in the synthetic environments.

The results use the protocol in Appendix M.9 and the static comparator in Eq. (540).

Throughout, we use $N_s = 5$ seeds, $N_u = 8$ users per environment, $T = 300$, $K_{\max} = 24$, and feature dimension $d = 2K_{\text{cat}} = 10$.

The five HRB variants are

$$\mathcal{V} = \{\text{HRB}, \text{HRB-NR}, \text{HRB-NF}, \text{HRB-NU}, \text{HRB-1}\}, \quad (575)$$

and the three environments are

$$\mathcal{E} = \{\text{general}, \text{workspace}, \text{support}\}. \quad (576)$$

For every aggregate statistic $\hat{\theta}$ reported below, we show the following descriptive across-seed spread:

$$\text{Spread}_{1.96}(\hat{\theta}) = \hat{\theta} \pm 1.96 \cdot \frac{\hat{\sigma}_s}{\sqrt{N_s}}, \quad (577)$$

where $\hat{\sigma}_s$ is the across-seed standard deviation. With only $N_s = 5$ seeds, we do not assign this display 95% coverage or use it for hypothesis testing.

The bootstrap ranges in Section O.8 use $B = 10,000$ resamples drawn with replacement from the five seed indices. They are descriptive sensitivity summaries, not confidence intervals.

O.1 HRB AND ABLATION ACCOUNTING

Table 21 reports the three structural quantities and the observed-reward residual for each environment and variant, together with the observed routed value and the rounded accounting residual

$$\rho_{\text{obs}} = \hat{V}_T^{\text{obs}} - [\hat{V}_{\text{static}}(A_1) + \hat{\Delta}_{\text{route}}(A_1) + \hat{G}_T - \hat{L}_T^{\text{obs}}]. \quad (578)$$

This residual is zero before reporting-rounding by (549); Theorem 17 is not needed for that fact.

Table 21: Observed-reward accounting for HRB and its component ablations. The observed residual uses $m_t - r_t$, not $m_t - \mu(U, a_t)$. The displayed $|\rho_{\text{obs}}|$ is at most 1.5×10^{-3} because the entries are rounded.

Env	Variant	$\hat{V}_{\text{static}}(A_1)$	$\hat{\Delta}_{\text{route}}(A_1)$	$\hat{G}_T$	$\hat{L}_T^{\text{obs}}$	$\hat{V}_T^{\text{obs}}$	$ \rho_{\text{obs}} $
general	HRB	0.570	0.280	0.051	0.036	0.864	0.001
	HRB-NR	0.570	0.280	0.051	0.281	0.621	0.001
	HRB-NF	0.570	0.280	0.000	0.038	0.812	0.000
	HRB-NU	0.570	0.280	0.052	0.080	0.821	0.001
	HRB-I	0.570	0.000	0.000	0.069	0.501	0.000
workspace	HRB	0.577	0.124	0.041	0.043	0.699	0.001
	HRB-NR	0.577	0.124	0.041	0.183	0.559	0.001
	HRB-NF	0.577	0.124	0.000	0.044	0.657	0.000
	HRB-NU	0.577	0.124	0.040	0.061	0.680	0.000
	HRB-I	0.577	0.000	0.000	0.073	0.504	0.000
support	HRB	0.579	0.215	0.061	0.039	0.816	0.001
	HRB-NR	0.579	0.215	0.061	0.255	0.600	0.000
	HRB-NF	0.579	0.215	0.000	0.041	0.753	0.000
	HRB-NU	0.579	0.215	0.060	0.078	0.776	0.000
	HRB-I	0.579	0.000	0.000	0.060	0.519	0.000

O.2 PER-SEED RESULTS FOR THE HRB VARIANT

Table 22 reports per-seed structural quantities, the observed-reward residual, and the observed routed value for the HRB variant.

Table 22: Per-seed decomposition for the HRB variant. Across-seed standard deviations are at most 4×10^{-3} for every column.

Env	Seed	$\hat{V}_{\text{static}}$	$\hat{\Delta}_{\text{route}}$	$\hat{G}_T$	$\hat{L}_T^{\text{obs}}$	$\hat{V}_T^{\text{obs}}$
general	0	0.568	0.282	0.052	0.034	0.868
	1	0.572	0.276	0.049	0.037	0.860
	2	0.569	0.283	0.053	0.036	0.869
	3	0.571	0.279	0.050	0.038	0.862
	4	0.570	0.280	0.051	0.035	0.866
workspace	0	0.575	0.126	0.040	0.044	0.697
	1	0.578	0.123	0.042	0.042	0.701
	2	0.577	0.125	0.041	0.043	0.700
	3	0.579	0.122	0.040	0.045	0.696
	4	0.576	0.124	0.042	0.041	0.701
support	0	0.580	0.213	0.062	0.038	0.817
	1	0.578	0.217	0.060	0.040	0.815
	2	0.579	0.214	0.063	0.039	0.817
	3	0.581	0.216	0.060	0.041	0.816
	4	0.577	0.215	0.061	0.037	0.816

O.3 PER-CLUSTER HETEROGENEITY DECOMPOSITION

The fixed-archive heterogeneity term $\hat{\Delta}_{\text{route}}(A_1)$ is averaged over the user marginal Π .

Stratifying by the cluster index $g(U)$ localizes signed contributions to this population-level gap.

Define

$$C_g(A_1) = \mathbb{E}[m_1(U) \mid g(U) = g] - V_{\text{static}}(A_1), \quad (579)$$

so $\sum_g \Pr(g(U) = g) C_g(A_1) = \Delta_{\text{route}}(A_1)$. Unlike a within-cluster routed-minus-static gap, $C_g(A_1)$ need not be nonnegative.

Table 23 reports $C_g(A_1)$ for each cluster.

Table 23: Initial-archive heterogeneity in the HRB experiments: signed per-cluster contributions relative to the population static value, with descriptive paired-seed spreads from Eq. (577). These rows do not test the within-cluster equality condition in Proposition 23.

Env	Cluster g	$C_g(A_1)$	descriptive spread
general	0	0.17	[0.11, 0.23]
	1	0.39	[0.34, 0.44]
	2	0.41	[0.36, 0.46]
	3	0.23	[0.17, 0.29]
	4	0.40	[0.35, 0.45]
workspace	0	0.10	[0.06, 0.14]
	1	0.15	[0.10, 0.20]
support	0	0.37	[0.32, 0.42]
	1	0.44	[0.39, 0.49]
	2	0.03	[-0.02, 0.08]

O.4 PER-SEED FINAL CUMULATIVE REGRET

Per-seed final cumulative observed-reward residual of the HRB variant is reported in Table 24, where

$$\text{ObsResidual}_T = \sum_{t=1}^T (m_t(U) - r_t), \quad (580)$$

averaged across users.

It equals $T \hat{L}_T^{\text{obs}}$ by definition. It equals the expected theoretical exploration cost only under a mean-zero observation model with the same conditional mean μ , which the clipped simulator does not satisfy.

Table 24: Per-seed cumulative observed-reward residual for HRB. The implied $T \hat{L}_T^{\text{obs}}$ is 10.8, 12.9, and 11.7 for the three environments, up to reporting-rounding.

Env	seed 0	seed 1	seed 2	seed 3	seed 4	mean $\pm$ std
general	10.21	11.10	10.84	11.46	10.87	10.90 ± 0.42
workspace	13.21	12.62	12.95	13.49	12.91	13.04 ± 0.30
support	11.41	11.99	11.71	12.35	10.92	11.68 ± 0.49

O.5 FINAL REGRET ACROSS ABLATIONS

Table 25 extends Table 24 to all five variants.

The HRB-NR variant has a cumulative observed-reward residual roughly seven times that of HRB, consistent with Table 21.

Table 25: Per-seed cumulative observed-reward residual for HRB and its component ablations. Each entry is averaged across the $N_u = 8$ users for that seed.

Env	Variant	seed 0	seed 1	seed 2	seed 3	seed 4
general	HRB	10.21	11.10	10.84	11.46	10.87
	HRB-NR	84.30	85.12	83.71	84.95	84.42
	HRB-NF	11.18	11.92	11.41	11.95	11.55
	HRB-NU	23.71	24.92	24.10	24.85	24.04
	HRB-I	20.40	21.12	20.61	21.32	20.85
workspace	HRB	13.21	12.62	12.95	13.49	12.91
	HRB-NR	54.70	55.21	54.95	55.42	54.81
	HRB-NF	13.30	12.95	13.21	13.65	13.04
	HRB-NU	18.21	18.95	18.41	18.81	18.32
	HRB-I	21.40	22.10	21.81	22.30	21.95
support	HRB	11.41	11.99	11.71	12.35	10.92
	HRB-NR	76.30	77.12	76.41	77.85	76.21
	HRB-NF	12.30	12.85	12.51	12.95	12.20
	HRB-NU	23.30	23.95	23.61	24.10	23.45
	HRB-I	17.81	18.45	18.10	18.62	18.02

O.6 REFLECTION DIAGNOSTICS

Table 26 summarises the reflection bookkeeping under the HRB variant: mean number of reflection events per user, fraction of children that strictly improve the per-user archive optimum (“beneficial-child fraction”), final archive size, and time-to-first-specialist (TTFS).

Table 26: Reflection diagnostics for the HRB variant. The TTFS column does not test Theorem 21, because the runs do not estimate its fixed-threshold repair probability q_ϵ .

Env	mean reflections	beneficial-child frac.	mean $ A_T $	TTFS (rounds)
general	7.6 ± 0.4	0.42	15.7 ± 0.6	8.98 ± 1.2
workspace	4.0 ± 0.3	0.22	11.9 ± 0.5	2.75 ± 0.7
support	5.1 ± 0.3	0.35	13.4 ± 0.5	0.35 ± 0.4

O.7 UNGATED REFLECTION DIAGNOSTICS

Removing the gating clause of Eq. (534) produces the negative result of Section 5.

Table 27 compares HRB with HRB (ungated), showing that the archive grows to the cap $K_{\max}$ and reward erodes by 0.024 to 0.041 depending on environment.

Table 27: HRB versus HRB (ungated). Removing the reflection gate grows the archive and raises the observed-reward residual.

Env	Variant	mean $ A_T $	$\hat{\mathcal{V}}_T^{\text{obs}}$	$\hat{\mathcal{L}}_T^{\text{obs}}$
general	HRB	15.7	0.864	0.036
	HRB (ungated)	19.7	0.823	0.077
workspace	HRB	11.9	0.699	0.043
	HRB (ungated)	19.7	0.675	0.067
support	HRB	13.4	0.816	0.039
	HRB (ungated)	19.7	0.781	0.074

O.8 ABLATION EFFECT SIZES ACROSS FIVE SEEDS

For each pair (HRB, ablation) we form the seed-paired difference

$$\hat{\Delta}^{(abl)} = \hat{\mathcal{V}}_T^{(HRB)} - \hat{\mathcal{V}}_T^{(abl)}, \quad (581)$$

and resample seeds with replacement $B = 10,000$ times to summarize the sensitivity of the mean to the five observed seeds.

Table 28 reports the mean effect and the 2.5%–97.5% bootstrap percentile range. With five paired observations, we do not report a p -value or claim statistical significance.

Table 28: Paired-seed reward differences between HRB and its four component ablations. The percentile ranges summarize resampling of five seeds and are not confidence intervals or significance tests.

Env	HRB comparison	$\hat{\Delta}$	bootstrap percentile range
general	vs. HRB-NR	+0.243	[0.235, 0.251]
	vs. HRB-NF	+0.052	[0.046, 0.057]
	vs. HRB-NU	+0.043	[0.038, 0.049]
	vs. HRB-I	+0.363	[0.354, 0.371]
workspace	vs. HRB-NR	+0.140	[0.133, 0.147]
	vs. HRB-NF	+0.042	[0.037, 0.047]
	vs. HRB-NU	+0.019	[0.012, 0.026]
	vs. HRB-I	+0.195	[0.187, 0.203]
support	vs. HRB-NR	+0.216	[0.208, 0.224]
	vs. HRB-NF	+0.063	[0.057, 0.069]
	vs. HRB-NU	+0.040	[0.034, 0.046]
	vs. HRB-I	+0.297	[0.288, 0.306]

O.9 PER-USER ROUTED VALUE

Table 29 reports per-user $\hat{\mathcal{V}}_T$ for the HRB variant in the `general` environment; analogous tables for `workspace` and `support` are omitted for space and follow the same pattern.

Table 29: Per-user, per-seed $\hat{\mathcal{V}}_T$ in the `general` environment under the HRB variant. Per-user variance is dominated by the cluster index $g(u)$.

Seed	u0	u1	u2	u3	u4	u5	u6	u7
0	0.881	0.870	0.862	0.852	0.873	0.880	0.864	0.862
1	0.876	0.861	0.870	0.844	0.866	0.875	0.860	0.858
2	0.880	0.872	0.871	0.851	0.872	0.879	0.863	0.864
3	0.871	0.863	0.866	0.846	0.870	0.877	0.861	0.860
4	0.882	0.866	0.872	0.853	0.875	0.881	0.866	0.862